

RTL Design Sherpa

RAPIDS Micro-Architecture Specification 0.4

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Table of Contents

1 Rapids Mas Index....33
2 Document Information....34
2.1 Revision History....34
2.2 References....35
2.2.1 Related Documents....35
2.3 Terminology....35
2.4 Document Conventions....37
3 Architecture Overview....38
3.1 Overview....38
3.1.1 Key Features....38
3.1.2 What Is Shared with RAPIDS Beats....38
3.2 Block Diagram....39
3.2.1 Figure 1.1.1: RAPIDS Architecture Block Diagram....39
3.3 Data Flow Overview....40
3.3.1 Sink Path (Network to Memory)....40
3.3.2 Figure 1.1.2: Sink Path Data Flow....40
3.3.3 Source Path (Memory to Network)....40
3.3.4 Figure 1.1.3: Source Path Data Flow....40
3.4 Byte-Granular Design Points....41
3.5 Chapter Map....42
4 Top-Level Port List....43
4.1 Parameters....43

4.2 Port List (324 ports)..... 44
4.2.1 Clock and Reset (shared - single clock/reset for the whole wrapper)..... 45
4.3 SOURCE HALF (u_src) - shared-infrastructure ports (src_ prefixed)..... 45
4.3.1 APB Programming Interface..... 45
4.3.2 Per-channel configuration..... 45
4.3.3 Scheduler Configuration (global)..... 45
4.3.4 Descriptor Engine Configuration (global)..... 46
4.3.5 Control Engine Configuration (Phase 2, global)..... 46
4.3.6 Descriptor AXI Monitor Configuration..... 47
4.3.7 Network-plane monitor input (rapids TASK-024)..... 48
4.3.8 Status..... 48
4.3.9 Descriptor AXI Monitor Status..... 48
4.3.10 Descriptor AXI Monitor perf window (feeds SRC_.MON.DAXMON_PERF_*)..... 49
4.3.11 Descriptor Fetch AXI Master..... 50
4.3.12 Control Read AXI Master (32-bit) [Phase 2]..... 51
4.3.13 Control Write AXI Master (32-bit) [Phase 2]..... 52
4.4 SOURCE HALF (u_src) - direction-unique ports (no prefix)..... 53
4.4.1 AXI Transfer Configuration (source-only)..... 53
4.4.2 Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id 53
4.4.3 AXI4 Master - Data Read (Memory -> Source SRAM)..... 54
4.5 SINK HALF (u_snk) - shared-infrastructure ports (snk_ prefixed)..... 55
4.5.1 APB Programming Interface..... 55

4.5.2 Per-channel configuration....55
4.5.3 Scheduler Configuration (global)....55
4.5.4 Descriptor Engine Configuration (global)....56
4.5.5 Control Engine Configuration (Phase 2, global)....56
4.5.6 Descriptor AXI Monitor Configuration....56
4.5.7 Network-plane monitor input (rapids TASK-024)....57
4.5.8 Status....58
4.5.9 Descriptor AXI Monitor Status....58
4.5.10 Descriptor AXI Monitor perf window (feeds SNK_.MON.DAXMON_PERF_*)....59
4.5.11 Descriptor Fetch AXI Master....59
4.5.12 Control Read AXI Master (32-bit) [Phase 2]....60
4.5.13 Control Write AXI Master (32-bit) [Phase 2]....61
4.6 Monitor Bus (SINGLE aggregated stream for the whole core)....63
4.6.1 Monitor Bus....63
4.7 SINK HALF (u_snk) - direction-unique ports (no prefix)....63
4.7.1 AXI Transfer Configuration (sink-only)....63
4.7.2 Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id...63
4.7.3 AXI4 Master - Data Write (Sink SRAM -> Memory)....64
4.8 Debug Interface (per half, src_dbg_ / snk_dbg_ prefixed)....65
4.8.1 Source half debug....65
4.8.2 Sink half debug....66
4.8.3 Active-channel sideband for per-channel bus instrumentation....66
5 Clocks and Reset....67

5.1 Clock Domain....67
5.1.1 Clock Specifications....67
5.1.2 Clock Distribution....67
5.2 Reset Specification....67
5.2.1 Reset Signal....68
5.2.2 Reset Timing Diagram....68
5.2.3 Figure 1.3.1: Reset Timing....68
5.3 Reset Behavior....69
5.3.1 On Reset Assert....69
5.3.2 Reset Sequence....69
5.4 Configuration During Reset....69
5.5 Reset Macros....70
5.6 Power-On Reset Considerations....70
6 Scheduler....72
6.1 Overview....72
6.1.1 Key Features....72
6.1.2 Block Diagram....73
6.1.3 Figure 2.1.1: Scheduler Byte Math....73
6.2 Parameters....73
6.3 Port List....74
6.3.1 Clock, Reset and Configuration....74
6.3.2 Status....75
6.3.3 Descriptor Engine Interface....75
6.3.4 Data Read Interface (to AXI Read Engine)....76

6.3.5 Data Write Interface (to AXI Write Engine)....76
6.3.6 Completion Interface....77
6.3.7 Control Engine Interface....77
6.3.8 Monitor Bus Interface....78
6.4 Descriptor Format....78
6.5 Byte-to-Beat Math....79
6.6 Address Advance....80
6.7 Packet Records....81
6.8 Finite State Machine....81
6.9 Operation....82
6.9.1 Data Request Outputs....82
6.9.2 Write Completion....83
6.9.3 Control Descriptors....83
6.10 Error Handling....83
6.11 Recoverable Write-Progress Timeout....83
6.12 MonBus Events....84
6.13 Waveforms....84
7 Descriptor Engine Specification....86
7.1 Overview....86
7.1.1 Key Features....86
7.1.2 Block Diagram....86
7.1.3 Figure 2.2.1: Descriptor Engine Block Diagram....86
7.2 Parameters....87
7.3 Port List....87

7.3.1 Clock and Reset..... 87
7.3.2 APB Programming Interface..... 87
7.3.3 Configuration Interface..... 88
7.3.4 AXI4 Read Master Interface..... 88
7.3.5 Scheduler Interface..... 89
7.4 FSM States..... 90
7.4.1 Figure 2.2.2: Descriptor Engine FSM..... 90
7.5 Operation..... 91
7.5.1 Descriptor Chain Processing..... 91
7.5.2 Figure 2.2.3: Descriptor Chain Timing..... 91
7.6 Address Range Checking..... 91
7.7 Prefetch and Autonomous Chaining..... 92
7.7.1 Chaining Decision..... 92
7.7.2 Deferred (Throttle-Blocked) Chain Fetch..... 92
8 AXI Read Engine..... 94
8.1 Overview..... 94
8.1.1 Key Features..... 94
8.1.2 Block Diagram..... 95
8.1.3 Figure 2.3.1: AXI Read Engine Block Diagram..... 95
8.2 Parameters..... 95
8.2.1 Derived Constants..... 96
8.3 Port List..... 97
8.3.1 Clock, Reset and Configuration..... 97
8.3.2 Scheduler Interface..... 97

8.3.3 SRAM Allocation and Fill Interface....98
8.3.4 AXI4 Read Master Interface....98
8.3.5 Debug....99
8.4 Operation....99
8.4.1 Burst Length Cap (rapids BUG-009)....99
8.4.2 Byte-Granular Burst Sizing....100
8.4.3 Streaming Pipeline....100
8.4.4 Figure 2.3.2: Burst Segmentation with a Mid-Beat Start....100
8.4.5 Done Strobe....101
8.5 Error Handling....101
8.5.1 Channel Reset....101
8.6 Differences from RAPIDS Beats....102
9 AXI Write Engine....103
9.1 Overview....103
9.1.1 Key Features....103
9.1.2 Block Diagram....104
9.1.3 Figure 2.4.1: AXI Write Engine Block Diagram....104
9.2 Parameters....104
9.2.1 Derived Constants....105
9.3 Port List....106
9.3.1 Clock, Reset and Configuration....106
9.3.2 Scheduler Interface....106
9.3.3 SRAM Reservation and Drain Interface....107
9.3.4 AXI4 Write Master Interface....108

9.3.5 Debug and Status..... 109
9.4 Operation..... 109
9.4.1 Burst Length Cap (rapids BUG-009)..... 109
9.4.2 Byte-Granular Burst Sizing..... 109
9.4.3 Write Strobes..... 110
9.4.4 Streaming Pipeline..... 110
9.4.5 Two Completion Strobes..... 110
9.5 Error Handling..... 111
9.5.1 Channel Reset..... 111
9.6 Differences from RAPIDS Beats..... 112
10 Beats Alloc Control Specification..... 113
10.1 Overview..... 113
10.1.1 Key Features..... 113
10.1.2 Block Diagram..... 114
10.1.3 Figure 2.5.1: Beats Alloc Control Block Diagram..... 114
10.2 Concept: Virtual FIFO..... 114
10.2.1 Use Case: Pre-Allocation for AXI Reads..... 114
10.3 Parameters..... 115
10.4 Port List..... 115
10.4.1 Clock and Reset..... 115
10.4.2 Write Interface (Allocation Requests)..... 115
10.4.3 Read Interface (Data Written)..... 116
10.4.4 Status Outputs..... 116
10.5 Operation..... 116

10.5.1 Pointer Arithmetic....116
10.5.2 Timing Diagram....117
10.5.3 Figure 2.5.2: Allocation and Release Timing....117
10.6 Integration Example....117
11 Beats Drain Control Specification....119
11.1 Overview....119
11.1.1 Key Features....119
11.1.2 Block Diagram....120
11.1.3 Figure 2.6.1: Beats Drain Control Block Diagram....120
11.2 Concept: Data Availability Tracking....120
11.2.1 Complementary Roles....120
11.3 Parameters....121
11.4 Port List....121
11.4.1 Clock and Reset....121
11.4.2 Write Interface (Data Written to SRAM)....121
11.4.3 Read Interface (Drain Requests)....121
11.4.4 Status Outputs....122
11.5 Operation....122
11.5.1 Pointer Arithmetic....122
11.5.2 Timing Diagram....123
11.5.3 Figure 2.6.2: Data Arrival and Drain Timing....123
11.6 Integration Example....123
11.7 Relationship: alloc_ctrl vs drain_ctrl....124
12 Latency Bridge (beats) Specification....125

12.1 Overview....125
12.1.1 Key Features....125
12.1.2 Block Diagram....126
12.1.3 Figure 2.7.1: Latency Bridge Block Diagram....126
12.2 Concept: bridging a registered read....126
12.3 Parameters....127
12.4 Port List....127
12.4.1 Clock and Reset....127
12.4.2 Upstream Interface (from the registered FIFO)....127
12.4.3 Downstream Interface (to the consumer)....127
12.4.4 Status and Debug....128
12.5 Operation....129
12.5.1 Figure 2.7.2: Latency Bridge Timing (consumer always ready)....129
12.6 Integration Context....130
12.7 Design Considerations....130
13 Control-Read Engine Specification....132
13.1 Overview....132
13.1.1 Key Features....132
13.1.2 Block Diagram....133
13.1.3 Figure 2.8.1: Control-Read Engine Block Diagram....133
13.2 Parameters....133
13.3 Port List....134
13.3.1 Clock and Reset....134
13.3.2 Scheduler Interface....134

13.3.3 Configuration Interface....135
13.3.4 AXI4 Read Master Interface....135
13.3.5 MonBus Interface....136
13.4 FSM States....137
13.4.1 Figure 2.8.2: Control-Read Engine FSM....137
13.5 Operation....138
13.5.1 Masked Compare....138
13.5.2 Retry Budget and Pacing....138
13.5.3 Null Address Fast-Path....138
13.5.4 AXI Read Transaction....138
13.5.5 Channel Reset....139
13.6 MonBus Reporting....139
13.7 Descriptor Field Mapping....140
14 Control-Write Engine Specification....141
14.1 Overview....141
14.1.1 Key Features....141
14.1.2 Block Diagram....142
14.1.3 Figure 2.9.1: Control-Write Engine Block Diagram....142
14.2 Parameters....142
14.3 Port List....143
14.3.1 Clock and Reset....143
14.3.2 Configuration Interface....143
14.3.3 Scheduler Interface....143
14.3.4 AXI4 Write Master Interface....144

14.3.5 MonBus Interface....145
14.4 FSM States....145
14.4.1 Figure 2.9.2: Control-Write Engine FSM....145
14.5 Operation....147
14.5.1 Doorbell Write Sequence....147
14.5.2 Figure 2.9.3: Control-Write Doorbell Timing....147
14.5.3 Request Handshake and Skid Buffer....148
14.5.4 Address Validation....148
14.5.5 Error Reporting....148
14.5.6 Channel Reset....148
14.6 Contrast with the Control-Read Engine....149
15 Scheduler Group....151
15.1 Overview....151
15.1.1 Key Features....151
15.1.2 Block Diagram....152
15.1.3 Figure 3.1.1: Scheduler Group Block Diagram....152
15.2 Parameters....152
15.3 Port List....153
15.3.1 Clock, Reset and Kick-Off....153
15.3.2 Configuration Interface....154
15.3.3 Status....155
15.3.4 Descriptor AXI Master Interface....155
15.3.5 Control Read and Control Write AXI Interfaces....156
15.3.6 Scheduler Data Interface....156

15.3.7 Packet Record Interface....157
15.3.8 MonBus Interface....158
15.4 Internal Architecture....158
15.4.1 Descriptor Path....158
15.4.2 Control Engines....159
15.4.3 MonBus Aggregation....159
15.5 Differences from RAPIDS Beats....159
16 Scheduler Group Array....160
16.1 Overview....160
16.1.1 Key Features....160
16.1.2 Block Diagram....161
16.1.3 Figure 3.2.1: Scheduler Group Array Block Diagram....161
16.2 Parameters....161
16.3 Port List....162
16.3.1 Clock, Reset and Kick-Off....162
16.3.2 Global Scheduler and Descriptor Engine Configuration....163
16.3.3 Descriptor Monitor Configuration and Status....164
16.3.4 Descriptor-AXI Perf Window....165
16.3.5 Status....166
16.3.6 Shared AXI Masters....166
16.3.7 Per-Channel Scheduler Data Interface....166
16.3.8 Per-Channel Packet Record Interface....167
16.3.9 MonBus Interface....168
16.4 Shared Master Arbitration....168

16.4.1 Figure 3.2.2: Shared Master Arbitration....169
16.5 MonBus Aggregation....169
16.6 Differences from RAPIDS Beats....170
17 Sink Data Path....171
17.1 Overview....171
17.1.1 Key Features....171
17.1.2 Figure 3.3.1: Sink Data Path Block Diagram....172
17.2 Parameters....172
17.3 Interfaces....173
17.3.1 Fill Interface....173
17.3.2 Configuration....174
17.3.3 Scheduler Interface....174
17.3.4 AXI4 Write Master....175
17.4 Byte Enable Path....175
17.4.1 Channel Reset....175
17.5 Differences from RAPIDS Beats....175
18 Sink Data Path with AXIS (Ingress)....177
18.1 Overview....177
18.1.1 Key Features....177
18.1.2 Figure 3.4.1: Ingress Shifter....178
18.2 Parameters....178
18.3 Interfaces....179
18.3.1 AXI-Stream Slave....179
18.3.2 Scheduler Interface....179

18.3.3 Other Ports....180
18.4 Ingress Operation....180
18.4.1 Packet Records....180
18.4.2 Placement....180
18.4.3 Flush and Completion....181
18.4.4 Length Check....181
18.4.5 Channel Reset....181
18.4.6 Allocation....182
18.5 Differences from RAPIDS Beats....183
19 Sink SRAM Controller Specification....184
19.1 Overview....184
19.1.1 Key Features....184
19.1.2 Block Diagram....185
19.1.3 Figure 3.5.1: Sink SRAM Controller Block Diagram....185
19.2 Parameters....185
19.3 Port List....186
19.3.1 Clock and Reset....186
19.3.2 Fill Interface (ID-Selected)....186
19.3.3 Arbitrated Drain Interface (to AXI Write Engine)....186
19.3.4 Debug Interface....187
19.4 Drain Selection....187
19.4.1 Figure 3.5.2: Drain Selection by the Consumer....187
19.5 Integration with AXI Write Engine....188
20 Source Data Path Specification....190

20.1 Overview....190
20.1.1 Key Features....190
20.1.2 Block Diagram....191
20.1.3 Figure 3.6.1: Source Data Path Block Diagram....191
20.2 Parameters....191
20.3 Data Flow....192
20.3.1 Figure 3.6.2: Source Path Data Flow Sequence....192
20.4 Port List....193
20.4.1 Clock and Reset....193
20.4.2 Configuration....193
20.4.3 Scheduler Interface....193
20.4.4 AXI Read Master Interface....193
20.4.5 Drain Interface (Output to Network)....194
20.5 Timing Diagram....195
20.5.1 Figure 3.6.3: Source Path Transfer Timing....195
20.6 Integration Example....195
21 Source Data Path with AXIS (Egress)....197
21.1 Overview....197
21.1.1 Key Features....197
21.1.2 Figure 3.7.1: Egress Shifter....198
21.2 Parameters....198
21.3 Interfaces....199
21.3.1 Configuration....199
21.3.2 Scheduler Interface....199

21.3.3 AXI-Stream Master....200
21.3.4 Other Ports....201
21.4 Egress Operation....201
21.4.1 Reservation and Drain....201
21.4.2 Pop....201
21.4.3 Shift and Hold....201
21.4.4 Strobe and Last....202
21.4.5 Waveform 3.7: Egress Re-pack....202
21.5 Channel Reset....202
21.6 Differences from RAPIDS Beats....203
22 Source SRAM Controller Specification....205
22.1 Overview....205
22.1.1 Key Features....205
22.1.2 Block Diagram....206
22.1.3 Figure 3.8.1: Source SRAM Controller Block Diagram....206
22.2 Parameters....206
22.3 Port List....207
22.3.1 Clock and Reset....207
22.3.2 Fill Interface (ID-Selected, from AXI Read Engine)....207
22.3.3 Arbitrated Drain Interface (to Network)....207
22.3.4 Debug Interface....208
22.4 Drain Selection....208
22.4.1 Figure 3.8.2: Drain Selection by the Consumer....208
22.5 Comparison: Sink vs Source SRAM Controller....209

22.6 Integration Example....209
23 RAPIDS Core, Sink Half and Source Half....211
23.1 Overview....211
23.1.1 Figure 3.9.1: Core Block Diagram....211
23.2 Byte-Granular Content....212
23.2.1 Packet Record Nets....212
23.2.2 Port Renames....212
23.2.3 Data Ports....212
23.3 Parameters....213
23.3.1 Channel Reset Routing....214
23.4 Differences from RAPIDS Beats....215
24 RAPIDS Registers Specification....216
24.1 Overview....216
24.2 Base Register Map (0x100-0x3FF)....216
24.2.1 Key Register Fields....218
24.3 Monitor Register Map (rapids_mon_regs @ 0x1000)....219
25 RAPIDS Config Block Specification....221
25.1 Overview....221
25.2 Mapping Groups....221
25.2.1 Base Configuration....221
25.2.2 Monitor Configuration (hwif_out.MON.*)....222
25.2.3 Performance and Observation....222
26 RAPIDS Top....224
26.1 Overview....224

26.1.1 Figure 3.12.1: Top Hierarchy....225
26.2 APB Address Map....225
26.3 Parameters....226
26.4 Differences from RAPIDS Beats....227
27 Network Planes Specification....228
27.1 Overview....228
27.2 Generated Tops....229
27.3 CXS Planes....230
27.3.1 Figure 3.13.1: CXS Network Planes....230
27.3.2 Receive (sink) plane....230
27.3.3 Transmit (source) plane....230
27.3.4 Per-stream framing....231
27.3.5 Parameters....231
27.3.6 Monitoring....231
27.4 Proposed Changes....232
27.5 Verification....232
28 AXI4 Interface Specification....233
28.1 Overview....233
28.2 Byte Address to Beat Mapping....233
28.3 Burst Rules....234
28.4 Write Strobes....234
28.5 Limitations....234
29 AXI-Stream Interface Specification....236
29.1 Overview....236

29.2 Packet Contract..... 236
29.3 Source Master Signals..... 237
29.4 Sink Slave Signals..... 237
29.5 Length Check..... 238
30 MonBus Interface Specification..... 239
30.1 Overview..... 239
30.2 MonBus Signal Interface..... 239
30.2.1 Signal Table..... 239
30.2.2 Handshaking Protocol..... 239
30.2.3 Timing Diagram..... 240
30.2.4 Figure 4.3.1: MonBus Packet Transfer Timing..... 240
30.3 MonBus Packet Format..... 240
30.3.1 64-bit Packet Structure..... 240
30.3.2 Figure 4.3.2: MonBus Packet Format..... 240
30.4 Packet Types..... 241
30.4.1 Type Code Definitions..... 241
30.4.2 Payload Formats by Type..... 242
30.5 MonBus Source Aggregation..... 243
30.5.1 RAPIDS Core MonBus Sources..... 243
30.5.2 Aggregation Hierarchy..... 244
30.5.3 Figure 4.3.3: MonBus Aggregation Tree..... 244
30.6 Integration Guidelines..... 244
30.6.1 Consumer Requirements..... 244
30.6.2 Recommended Consumer Patterns..... 244

30.7 Timing Requirements.....245
31 Packet Record Interface Specification.....246
31.1 Overview.....246
31.1.1 Figure 4.4.1: Scheduler Byte Math.....246
31.2 Signals.....247
31.3 Protocol.....247
31.4 Unused Direction.....248
31.5 Consumer Behavior.....248
32 CXS Interface Specification.....249
32.1 Overview.....249
32.2 Link Profile.....249
32.3 Signal Table.....250
32.4 Credit Flow.....251
32.5 Link Activation.....251
32.6 Backpressure.....252
32.7 Proposed Changes.....252
33 RAPIDS Beats Architecture MAS.....253
33.1 Document Organization.....253
33.1.1 Front Matter.....253
33.1.2 Chapter 1: Overview.....253
33.1.3 Chapter 2: FUB (Functional Unit Blocks).....253
33.1.4 Chapter 3: Macro (Integration Blocks).....253
33.1.5 Chapter 4: Interfaces.....254
33.2 Quick Reference.....254

33.2.1 FUB Modules (fub_beats/)..... 254
33.2.2 Macro Modules (macro_beats/)..... 255
33.3 Architecture Comparison: RAPIDS vs STREAM..... 256
33.4 Clock and Reset Summary..... 257
33.4.1 Clock Domains..... 257
33.4.2 Reset Signals..... 257
33.5 Interface Summary..... 258
33.5.1 External Interfaces..... 258
33.5.2 Internal Buses..... 258
33.6 Related Documentation..... 258
33.7 Specification Conventions..... 259
33.7.1 Signal Naming..... 259
33.7.2 Parameter Naming..... 259
33.7.3 Figure and Table Numbering..... 259

List of Figures

Figure 1.1.1: RAPIDS Architecture Block Diagram....39
Figure 1.1.2: Sink Path Data Flow....40
Figure 1.1.3: Source Path Data Flow....40
Figure 1.3.1: Reset Timing....68
Figure 2.1.1: Scheduler Byte Math....72
Figure 2.2.1: Descriptor Engine Block Diagram....85
Figure 2.2.2: Descriptor Engine FSM....89
Figure 2.2.3: Descriptor Chain Timing....90
Figure 2.3.1: AXI Read Engine Block Diagram....94
Figure 2.3.2: Burst Segmentation with a Mid-Beat Start....99
Figure 2.4.1: AXI Write Engine Block Diagram....103
Figure 2.5.1: Beats Alloc Control Block Diagram....113
Figure 2.5.2: Allocation and Release Timing....116
Figure 2.6.1: Beats Drain Control Block Diagram....119
Figure 2.6.2: Data Arrival and Drain Timing....122
Figure 2.7.1: Latency Bridge Block Diagram....125
Figure 2.7.2: Latency Bridge Timing (consumer always ready)....128
Figure 2.8.1: Control-Read Engine Block Diagram....132
Figure 2.8.2: Control-Read Engine FSM....136
Figure 2.9.1: Control-Write Engine Block Diagram....141
Figure 2.9.2: Control-Write Engine FSM....144
Figure 2.9.3: Control-Write Doorbell Timing....146
Figure 3.1.1: Scheduler Group Block Diagram....151
Figure 3.2.1: Scheduler Group Array Block Diagram....160
Figure 3.2.2: Shared Master Arbitration....168
Figure 3.3.1: Sink Data Path Block Diagram....170
Figure 3.4.1: Ingress Shifter....177
Figure 3.5.1: Sink SRAM Controller Block Diagram....184
Figure 3.5.2: Drain Selection by the Consumer....186
Figure 3.6.1: Source Data Path Block Diagram....190
Figure 3.6.2: Source Path Data Flow Sequence....191
Figure 3.6.3: Source Path Transfer Timing....194
Figure 3.7.1: Egress Shifter....197
Figure 3.8.1: Source SRAM Controller Block Diagram....205
Figure 3.8.2: Drain Selection by the Consumer....207
Figure 3.9.1: Core Block Diagram....210

Figure 3.12.1: Top Hierarchy.....224
Figure 3.13.1: CXS Network Planes.....229
Figure 4.3.1: MonBus Packet Transfer Timing.....239
Figure 4.3.2: MonBus Packet Format.....239
Figure 4.3.3: MonBus Aggregation Tree.....243
Figure 4.4.1: Scheduler Byte Math.....245

List of Tables

Table 1: Table 0.1: Revision History....34
Table 2: Table 0.2: Related Documents and Specifications....35
Table 3: Table 1.1.1: Block Roles....39
Table 4: Table 1.1.2: Design Points....41
Table 5: Table 1.2.1: Parameters (from rapids_core_beats.sv)....43
Table 6: Table 1.2.2: Clock and Reset (shared - single clock/reset for the whole wrapper)....45
Table 7: Table 1.2.3: APB Programming Interface....45
Table 8: Table 1.2.4: Per-channel configuration....45
Table 9: Table 1.2.5: Scheduler Configuration (global)....45
Table 10: Table 1.2.6: Descriptor Engine Configuration (global)....46
Table 11: Table 1.2.7: Control Engine Configuration (Phase 2, global)....46
Table 12: Table 1.2.8: Descriptor AXI Monitor Configuration....47
Table 13: Table 1.2.9: Network-plane monitor input (rapids TASK-024)....48
Table 14: Table 1.2.10: Status....48
Table 15: Table 1.2.11: Descriptor AXI Monitor Status....48
Table 16: Table 1.2.12: Descriptor AXI Monitor perf window (feeds SRC_.MON.DAXMON_PERF_*)....49
Table 17: Table 1.2.13: Descriptor Fetch AXI Master....50
Table 18: Table 1.2.14: Control Read AXI Master (32-bit) [Phase 2]....51
Table 19: Table 1.2.15: Control Write AXI Master (32-bit) [Phase 2]....52
Table 20: Table 1.2.16: AXI Transfer Configuration (source-only)....53
Table 21: Table 1.2.17: Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id....53
Table 22: Table 1.2.18: AXI4 Master - Data Read (Memory -> Source SRAM)....54
Table 23: Table 1.2.19: APB Programming Interface....55
Table 24: Table 1.2.20: Per-channel configuration....55
Table 25: Table 1.2.21: Scheduler Configuration (global)....55
Table 26: Table 1.2.22: Descriptor Engine Configuration (global)....56
Table 27: Table 1.2.23: Control Engine Configuration (Phase 2, global)....56
Table 28: Table 1.2.24: Descriptor AXI Monitor Configuration....56
Table 29: Table 1.2.25: Network-plane monitor input (rapids TASK-024)....58
Table 30: Table 1.2.26: Status....58
Table 31: Table 1.2.27: Descriptor AXI Monitor Status....58
Table 32: Table 1.2.28: Descriptor AXI Monitor perf window (feeds SNK_.MON.DAXMON_PERF_*)....59

Table 33: Table 1.2.29: Descriptor Fetch AXI Master....59
Table 34: Table 1.2.30: Control Read AXI Master (32-bit) [Phase 2]....60
Table 35: Table 1.2.31: Control Write AXI Master (32-bit) [Phase 2]....61
Table 36: Table 1.2.32: Monitor Bus....63
Table 37: Table 1.2.33: AXI Transfer Configuration (sink-only)....63
Table 38: Table 1.2.34: Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id....63
Table 39: Table 1.2.35: AXI4 Master - Data Write (Sink SRAM -> Memory)....64
Table 40: Table 1.2.36: Source half debug....65
Table 41: Table 1.2.37: Sink half debug....66
Table 42: Table 1.2.38: Active-channel sideband for per-channel bus instrumentation....66
Table 43: Table 1.3.1: Clock Specifications....67
Table 44: Table 1.3.2: Reset Specifications....67
Table 45: Table 1.3.3: Reset States....69
Table 46: Table 1.3.4: Configuration Timing Requirements....69
Table 47: Table 2.1.1: Scheduler Parameters....72
Table 48: Table 2.1.2: Clock, Reset and Configuration Ports....73
Table 49: Table 2.1.3: Status Ports....74
Table 50: Table 2.1.4: Descriptor Engine Interface....74
Table 51: Table 2.1.5: Data Read Interface....75
Table 52: Table 2.1.6: Data Write Interface....75
Table 53: Table 2.1.7: Completion and Error Interface....76
Table 54: Table 2.1.8: Control Engine Interface....76
Table 55: Table 2.1.9: Monitor Bus Interface....77
Table 56: Table 2.1.10: Descriptor Layout....77
Table 57: Table 2.1.11: Beat Totals for a 32-byte Beat....78
Table 58: Table 2.1.12: State Encoding....80
Table 59: Table 2.1.13: Transitions....81
Table 60: Table 2.1.14: MonBus Events....83
Table 61: Table 2.2.2: Clock and Reset....86
Table 62: Table 2.2.3: APB Programming Interface....86
Table 63: Table 2.2.4: Configuration Interface....87
Table 64: Table 2.2.5: AXI4 Read Master Interface....87
Table 65: Table 2.2.6: Scheduler Interface....88
Table 66: Table 2.3.1: AXI Read Engine Parameters....94
Table 67: Table 2.3.2: AXI Read Engine Derived Constants....95
Table 68: Table 2.3.3: Clock, Reset and Configuration....96

Table 69: Table 2.3.4: Scheduler Interface....96
Table 70: Table 2.3.5: SRAM Allocation and Fill Interface....97
Table 71: Table 2.3.6: AXI4 Read Master Interface....97
Table 72: Table 2.3.7: Debug Outputs....98
Table 73: Table 2.3.8: Burst Segmentation Example....99
Table 74: Table 2.3.9: Error Handling....100
Table 75: Table 2.3.10: Read Engine Delta....101
Table 76: Table 2.4.1: AXI Write Engine Parameters....103
Table 77: Table 2.4.2: AXI Write Engine Derived Constants....104
Table 78: Table 2.4.3: Clock, Reset and Configuration....105
Table 79: Table 2.4.4: Scheduler Interface....105
Table 80: Table 2.4.5: SRAM Drain Interface....106
Table 81: Table 2.4.6: AXI4 Write Master Interface....107
Table 82: Table 2.4.7: Debug and Status Outputs....108
Table 83: Table 2.4.8: Completion Strobes....109
Table 84: Table 2.4.9: Error Handling....110
Table 85: Table 2.4.10: Write Engine Delta....111
Table 86: Table 2.5.2: Clock and Reset....114
Table 87: Table 2.5.3: Write Interface....114
Table 88: Table 2.5.4: Read Interface....115
Table 89: Table 2.5.5: Status Outputs....115
Table 90: Table 2.6.2: Clock and Reset....120
Table 91: Table 2.6.3: Write Interface....120
Table 92: Table 2.6.4: Read Interface....120
Table 93: Table 2.6.5: Status Outputs....121
Table 94: Table 2.6.6: alloc_ctrl vs drain_ctrl Comparison....123
Table 95: Table 2.7.2: Clock and Reset....126
Table 96: Table 2.7.3: Upstream Interface....126
Table 97: Table 2.7.4: Downstream Interface....126
Table 98: Table 2.7.5: Status and Debug....127
Table 99: Table 2.7.6: Skid Depth Selection....129
Table 100: Table 2.8.2: Clock and Reset....133
Table 101: Table 2.8.3: Scheduler Interface....133
Table 102: Table 2.8.4: Configuration Interface....134
Table 103: Table 2.8.5: AXI4 Read Master Interface....134
Table 104: Table 2.8.6: MonBus Interface....135
Table 105: Table 2.8.7: MonBus Events....138
Table 106: Table 2.8.8: CTRL_READ Descriptor Field Mapping....139

Table 107: Table 2.9.2: Clock and Reset....142
Table 108: Table 2.9.3: Configuration Interface....142
Table 109: Table 2.9.4: Scheduler Interface....142
Table 110: Table 2.9.5: AXI4 Write Master Interface....143
Table 111: Table 2.9.6: MonBus Interface....144
Table 112: Table 3.1.1: Scheduler Group Parameters....151
Table 113: Table 3.1.2: Clock, Reset and Kick-Off....152
Table 114: Table 3.1.3: Configuration Interface....153
Table 115: Table 3.1.4: Status Interface....154
Table 116: Table 3.1.5: Descriptor AXI Master Interface....154
Table 117: Table 3.1.6: Scheduler Data Interface....155
Table 118: Table 3.1.7: Packet Record Interface....156
Table 119: Table 3.1.8: MonBus Interface....157
Table 120: Table 3.1.9: Scheduler Group Delta....158
Table 121: Table 3.2.1: Scheduler Group Array Parameters....160
Table 122: Table 3.2.2: Clock, Reset and Kick-Off....161
Table 123: Table 3.2.3: Global Configuration....162
Table 124: Table 3.2.4: Descriptor Monitor Configuration and Status....163
Table 125: Table 3.2.5: Descriptor-AXI Perf Window....164
Table 126: Table 3.2.6: Status....165
Table 127: Table 3.2.7: Per-Channel Scheduler Data Interface....165
Table 128: Table 3.2.8: Per-Channel Packet Record Interface....166
Table 129: Table 3.2.9: MonBus Interface....167
Table 130: Table 3.2.10: MonBus Source Assignment....168
Table 131: Table 3.2.11: Array Delta....169
Table 132: Table 3.3.1: Sink Data Path Parameters....171
Table 133: Table 3.3.2: Fill Interface....172
Table 134: Table 3.3.3: Configuration....172
Table 135: Table 3.3.4: Scheduler Interface....173
Table 136: Table 3.3.5: Sink Data Path Delta....174
Table 137: Table 3.4.1: Additional Parameters....177
Table 138: Table 3.4.2: AXI-Stream Slave....178
Table 139: Table 3.4.3: Packet Record Ports....178
Table 140: Table 3.4.4: Channel Reset Actions....180
Table 141: Table 3.4.5: Ingress Delta....182
Table 142: Table 3.5.2: Clock and Reset....185
Table 143: Table 3.5.3: Fill Interface (ID-Selected)....185
Table 144: Table 3.5.4: Arbitrated Drain Interface....185

Table 145: Table 3.5.5: Debug Interface....186
Table 146: Table 3.6.2: Clock and Reset....192
Table 147: Table 3.6.3: Configuration....192
Table 148: Table 3.6.4: Scheduler Interface....192
Table 149: Table 3.6.5: AXI Read Master Interface....192
Table 150: Table 3.6.6: Drain Interface....193
Table 151: Table 3.7.1: Parameters....197
Table 152: Table 3.7.2: Configuration....198
Table 153: Table 3.7.3: Scheduler Interface....198
Table 154: Table 3.7.4: AXI-Stream Master....199
Table 155: Table 3.7.5: Egress Shift Rules....200
Table 156: Table 3.7.6: Channel Reset Actions....202
Table 157: Table 3.7.7: Egress Delta....202
Table 158: Table 3.8.2: Clock and Reset....206
Table 159: Table 3.8.3: Fill Interface (ID-Selected)....206
Table 160: Table 3.8.4: Arbitrated Drain Interface....206
Table 161: Table 3.8.5: Debug Interface....207
Table 162: Table 3.8.6: Sink vs Source Controller Comparison....208
Table 163: Table 3.9.1: Packet Record Nets per Half....211
Table 164: Table 3.9.2: Core Port Renames....211
Table 165: Table 3.9.3: Data Ports....211
Table 166: Table 3.9.4: Core Parameters....212
Table 167: Table 3.9.5: Channel Reset Routing....213
Table 168: Table 3.9.6: Core Delta....214
Table 169: Table 3.10.1: Base Register Map....215
Table 170: Table 3.10.2: Monitor Register Map (base 0x1000)....218
Table 171: Table 3.11.1: Monitor Register-to-Config Mapping....221
Table 172: Table 3.12.1: APB Address Map....224
Table 173: Table 3.12.2: Top Parameters....225
Table 174: Table 3.12.3: Top Delta....226
Table 175: Table 3.13.1: RAPIDS Network Planes....227
Table 176: Table 3.13.2: Generated Tops....228
Table 177: Table 3.13.3: CXS Plane Parameters....230
Table 178: Table 3.13.4: CXS Top Test Cells....231
Table 179: Table 4.1.1: Address Split....232
Table 180: Table 4.1.2: Burst Rules....233
Table 181: Table 4.1.3: Byte-Granular Write Signal....233
Table 182: Table 4.2.1: Packet Contract....235

Table 183: Table 4.2.2: Source Master Signals....236
Table 184: Table 4.2.3: Sink Slave Signals....236
Table 185: Table 4.3.1: MonBus Interface Signals....238
Table 186: Table 4.3.2: MonBus Packet Fields....239
Table 187: Table 4.3.3: MonBus Packet Types....240
Table 188: Table 4.3.4: Descriptor Packet Payload....241
Table 189: Table 4.3.5: Scheduler Packet Payload....241
Table 190: Table 4.3.6: AXI Event Packet Payload....241
Table 191: Table 4.3.7: Error Packet Payload....241
Table 192: Table 4.3.8: MonBus Source Assignment....242
Table 193: Table 4.3.9: MonBus Timing Parameters....244
Table 194: Table 4.4.1: Packet Record Signals....246
Table 195: Table 4.4.2: Record Rules....246
Table 196: Table 4.4.1: CXS Link Profile....248
Table 197: Table 4.4.2: CXS Interface Signals....249
Table 198: Table 4.4.3: Link Activation Sequence....250
Table 199: FUB Module Summary....253
Table 200: Macro Module Summary....254
Table 201: RAPIDS Beats vs STREAM Comparison....255
Table 202: Clock Domains....256
Table 203: Reset Signals....256
Table 204: External Interfaces....257
Table 205: Internal Buses....257

List of Waveforms

Waveform 2.1: Byte Descriptor Start.....83
Waveform 3.7: Egress Re-pack.....201

1 Rapids Mas Index

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2 Document Information

This document describes RAPIDS, the byte-granular Rapid AXI Programmable In-band Descriptor System. RAPIDS provides network-to-memory and memory-to-network data transfer using descriptor-based DMA with 8 channels. A descriptor length is a byte count, source and destination addresses are byte addresses, and the sink write master drives byte enables. The earlier “Beats” architecture, which counted everything in beats, was the stepping stone to this design and is documented separately in the RAPIDS Beats MAS.

2.1 Revision History

Table 0.1: Revision History

Version	Date	Author	Description
0.4	2026-10-16	RTL Design Sherpa	Shared Sections 3.13 and 4.4 point at the routing and payload-type proposal in the tile’s TILE_SPEC Section 11
0.3	2026-10-16	RTL Design Sherpa	Network planes (rapids TASK-024/025): `rapids_top` / `rapids_cxs_top` generated from `rapids_top_base`; links the shared Network Planes chapter and CXS Interface Specification; block diagrams updated for the planes; the CXS ID mapping is marked provisional.
0.2	2026-09-30	RTL Design Sherpa	Channel reset recovers every error class: `cfg_channel_reset` routed into the AXI read and write engines, both data paths and the SRAM wrappers; scheduler clears sticky flags on the reset.

Version	Date	Author	Description
0.1	2026-09-30	RTL Design Sherpa	First byte-granular MAS. Rewrites the chapters that changed from RAPIDS Beats and links the rest.

2.2 References

2.2.1 Related Documents

Table 0.2: Related Documents and Specifications

Source	Title	Version
RTL Design Sherpa	RAPIDS Beats MAS (shared chapters)	0.7
RTL Design Sherpa	RAPIDS HAS	0.2
RTL Design Sherpa	RAPIDS Product Requirements Document	1.0
ARM	AMBA AXI and ACE Protocol Specification	IHI0022H
ARM	AMBA AXI-Stream Protocol Specification	IHI0051A
ARM	AMBA CXS Protocol Specification	IHI0079D

2.3 Terminology

Beat

One data transfer of an AXI burst or an AXI-Stream transfer. One beat is `BYTE_LANES` bytes, where `BYTE_LANES = DATA_WIDTH/8`. The board design uses 256 bits, so a beat is 32 bytes.

Byte offset

The low `OFF_W` bits of a byte address, where `OFF_W = $clog2(BYTE_LANES)`. It is the lane at which a transfer begins inside its first memory beat.

Memory beat

A beat as it appears on the AXI side, aligned to `BYTE_LANES`. A packet that starts at a non-zero offset occupies one more memory beat than a stream of the same length would.

Stream beat

A beat as it appears on AXI-Stream, packed from lane 0 with no offset.

Packet record

The pair `{bytes, offset}` that the scheduler issues once per DATA descriptor, so the AXIS shifter knows how the memory-side and stream-side layouts relate.

Spill

The bytes of a stream beat that did not fit in the current memory beat during ingress shifting. They are held for the next memory beat.

Channel

One of 8 independent DMA channels. Each has its own descriptor chain, scheduler and SRAM allocation.

Descriptor

A 256-bit structure holding the source address, destination address, length in bytes, next pointer, and control flags.

TYPE=EXT

The descriptor type that describes a row/column transfer. It stays beat-aligned by design: addresses are aligned and lengths are multiples of the beat size.

DMA

Direct Memory Access. Data movement without processor involvement.

MonBus

The 64-bit monitor bus that carries event packets out of the design.

2.4 Document Conventions

- Shared chapters are linked from the RAPIDS Beats MAS and marked “(shared with RAPIDS Beats)” in the index. Module names in those chapters carry a `_beats` suffix that this design drops.
- Requirement priorities follow the RAPIDS PRD.
- No emojis appear in this document.

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3 Architecture Overview

Module: `rapids_core.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

3.1 Overview

RAPIDS moves data between AXI-Stream and system memory in units of bytes. A descriptor's length is a byte count in the existing 32-bit field, and its source and destination are byte addresses. A linear descriptor may start at any byte on either side. The design keeps the streaming, no-FSM data path of RAPIDS Beats and adds three things around it: byte-to-beat math in the scheduler, byte enables through the sink SRAM to the write master, and a shifter at each AXI-Stream edge that converts between packed stream beats and memory-aligned beats.

3.1.1 Key Features

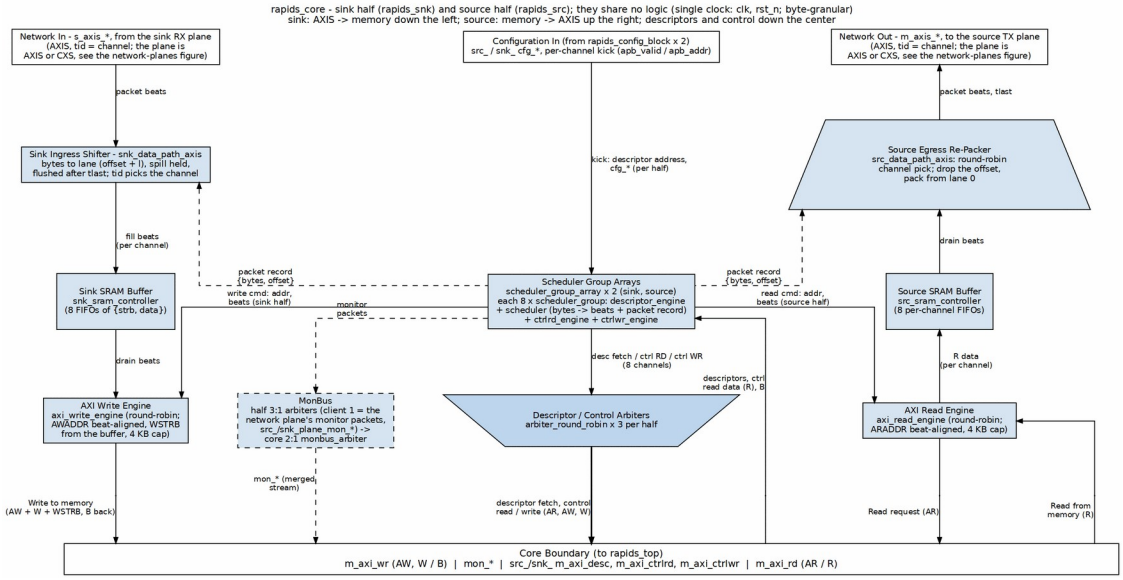
- **8 Independent Channels:** Each channel has its own descriptor chain and SRAM allocation.
- **Separate Data Paths:** Sink (network to memory) and source (memory to network).
- **Byte Granularity:** Length in bytes, byte addresses, any starting offset for linear descriptors.
- **Byte Enables:** The sink SRAM entry is `{strb, data}`; the write engine drives `WSTRB` from it.
- **Packet Records:** The scheduler issues one `{bytes, offset}` record per DATA descriptor to the AXIS shifter.
- **One AXIS Packet per Descriptor:** The source path emits exactly one packet, with `TLAST` on the final byte.
- **Safe Bursts:** AXI bursts never cross a 4 KB boundary; `AxADDR` is beat-aligned.
- **MonBus Integration:** Same monitor packets as RAPIDS Beats.

3.1.2 What Is Shared with RAPIDS Beats

The descriptor engine, the control-read and control-write engines, the SRAM controllers, the register block and config block, and the MonBus specification are unchanged. Those chapters are linked from the RAPIDS Beats MAS.

3.2 Block Diagram

3.2.1 Figure 1.1.1: RAPIDS Architecture Block Diagram



RAPIDS core block diagram

Source: 04_rapids_core.dot

Source: `rapids_core.sv`, `rapids_snk.sv`, `rapids_src.sv`

Table 1.1.1: Block Roles

Block	Role
scheduler	Loads a descriptor, computes beats from offset and length, issues the packet record, tracks completion
scheduler_group_array	Eight scheduler groups sharing one descriptor AXI master; per-channel packet-record ports
snk_data_path_axis	Converts packed AXIS beats into memory-aligned beats with byte enables; checks the packet length
snk_data_path	SRAM of `{strb, data}` entries plus the AXI write engine
src_data_path	SRAM of data entries plus the AXI read engine; unchanged from Beats

Block	Role
src_data_path_axis	Converts memory-aligned beats into packed stream beats with a partial last beat

3.3 Data Flow Overview

3.3.1 Sink Path (Network to Memory)

3.3.2 Figure 1.1.2: Sink Path Data Flow

```
1. Scheduler loads a descriptor, computes beats_total, pulses
sched_wr_pkt_valid
        |   (record {bytes, offset} enters the per-channel record
queue)
        v
2. s_axis packet arrives; tready is held low until the channel's
record is known
        |
        v
3. Ingress shifter: stream beat << (offset*8), OR the spill of the
previous beat
        |   low DATA_WIDTH bits = this memory beat, high bits become
the new spill
        v
4. Fill logic writes {strb, data} into the sink SRAM
(snk_sram_controller)
        |
        v
5. axi_write_engine: aligned AWADDR, capped burst, WSTRB from the SRAM
entry
        |
        v
6. System memory; B response commits the beats and completes the
descriptor
```

3.3.3 Source Path (Memory to Network)

3.3.4 Figure 1.1.3: Source Path Data Flow

```
1. Scheduler loads a descriptor, computes beats_total, pulses
sched_rd_pkt_valid
        |
        v
2. axi_read_engine: aligned ARADDR, capped burst, beats into the
source SRAM
```

```
        |
        v
3. src_data_path_axis reserves drain blocks (round-robin,
cfg_drain_size beats)
        |
        v
4. Egress shifter: memory beat >> (offset*8), OR the next beat's low
lanes
        |   the first pop only primes the hold; the last beat is
partial
        v
5. m_axis: TDATA packed from lane 0, TSTRB contiguous, TLAST on the
final byte
```

3.4 Byte-Granular Design Points

Table 1.1.2: Design Points

Point	Rule
Length unit	Bytes, in the existing 32-bit descriptor field
Beats moved	`(offset + length + BYTE_LANES - 1) >> OFF_W`, zero when length is zero
First burst	May begin mid-beat; the address it presents is beat-aligned and the offset is carried by the record
Later bursts	Beat-aligned
4 KB rule	A burst is truncated at the next 4 KB boundary
Sink lanes not written	`WSTRB` bit is 0
Source last beat	Partial; `TSTRB` marks the valid low lanes, `TLAST` set
TYPE=EXT	Beat-aligned by design (permanent): aligned addresses, beat-multiple lengths

The 4 KB rule and the byte math are specified in Scheduler and AXI Read Engine.

3.5 Chapter Map

Topic	Chapter
Scheduler byte math and record issue	Scheduler
Record queue interface	Packet Record Interface
Ingress shifter	Sink Data Path AXIS
Egress shifter	Source Data Path AXIS
Everything else	Linked from the RAPIDS Beats MAS via the index

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4 Top-Level Port List

Module: `rapids_core_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2026-10-16

4.1 Parameters

Every parameter of `rapids_core_beats`, as declared. The short aliases at the end are what the port widths below are written in (`NC`, `AW`, `DW`, `IW`, `SCW`, `CIW`, `SW`).

Table 1.2.1: Parameters (from `rapids_core_beats.sv`)

Parameter	Type	Default	Notes
`NUM_CHANNELS`	int	`8`	-
`CHAN_WIDTH`	int	`$clog2(NUM_CHANNELS)`	-
`ADDR_WIDTH`	int	`64`	-
`DATA_WIDTH`	int	`512`	-
`AXI_ID_WIDTH`	int	`8`	-
`SRAM_DEPTH`	int	`512`	-
`SEG_COUNT_WIDTH`	int	`$clog2(SRAM_DEPTH) + 1`	-
`PIPELINE`	int	`1`	write-engine pipelining; 1 is the perf design point (see rapids_beats_top)
`AR_MAX_OUTSTANDING`	int	`8`	-
`AW_MAX_OUTSTANDING`	int	`8`	-
`W_PHASE_FIFO_DEPTH`	int	`64`	-
`B_PHASE_FIFO_DEPTH`	int	`16`	-
`AXIS_ID_WIDTH`	int	`8`	-
`AXIS_DEST_WIDTH`	int	`4`	-

Parameter	Type	Default	Notes
AXIS_USER_WIDTH	int	1	-
DESC_MON_BASE_AGENT_ID	int	16	0x10 - Descriptor Engines (16-23)
SCHED_MON_BASE_AGENT_ID	int	48	0x30 - Schedulers (48-55)
DESC_AXI_MON_AGENT_ID	int	8	0x08 - Descriptor AXI Master Monitor
MON_UNIT_ID	int	1	0x1
MON_MAX_TRANSACTIONS	int	16	-
USE_AXI_MONITORS	int	1	-
USE_ROW_COL_MAJOR_ADDRESSING	int	1	-
GEN_MON	bit	1'b1	-
NC	int	NUM_CHANNELS	-
AW	int	ADDR_WIDTH	-
DW	int	DATA_WIDTH	-
IW	int	AXI_ID_WIDTH	-
SD	int	SRAM_DEPTH	-
SCW	int	SEG_COUNT_WIDTH	-
CIW	int	(NC > 1) ? $clog2(NC) : 1	-
SW	int	DW / 8	-

4.2 Port List (324 ports)

The module is two independent halves. Ports both halves carry are prefixed `src_` / `snk_`; the direction-unique ports (the source's AXI read master and AXIS egress, the sink's AXI write master and AXIS ingress) carry no prefix. Sections follow the declaration order in the RTL, so a reader can walk this page and the module side by side. Widths use the parameter aliases.

4.2.1 Clock and Reset (shared - single clock/reset for the whole wrapper)

Table 1.2.2: Clock and Reset (shared - single clock/reset for the whole wrapper)

Signal	Direction	Width	Description
`clk`	input	1	-
`rst_n`	input	1	-

4.3 SOURCE HALF (u_src) - shared-infrastructure ports (src_ prefixed)

4.3.1 APB Programming Interface

Table 1.2.3: APB Programming Interface

Signal	Direction	Width	Description
`src_apb_valid`	input	NC	-
`src_apb_ready`	output	NC	-
`src_apb_addr`	input	NC x AW	-

4.3.2 Per-channel configuration

Table 1.2.4: Per-channel configuration

Signal	Direction	Width	Description
`src_cfg_channel_enable`	input	NC	-
`src_cfg_channel_reset`	input	NC	-

4.3.3 Scheduler Configuration (global)

Table 1.2.5: Scheduler Configuration (global)

Signal	Direction	Width	Description
`src_cfg_sched_enable`	input	1	-
`src_cfg_sched_timeout_cycles`	input	32	-
`src_cfg_sched_timeout_limit`	input	8	-
`src_cfg_sched_timeout_enab`	input	1	-

Signal	Direction	Width	Description
le			
src_cfg_sched_err_enable	input	1	-
src_cfg_sched_compl_enable	input	1	-
src_cfg_sched_perf_enable	input	1	-

4.3.4 Descriptor Engine Configuration (global)

Table 1.2.6: Descriptor Engine Configuration (global)

Signal	Direction	Width	Description
src_cfg_desceng_enable	input	1	-
src_cfg_desceng_prefetch	input	1	-
src_cfg_desceng_fifo_thresh	input	4	-
src_cfg_desceng_addr0_base	input	AW	-
src_cfg_desceng_addr0_limit	input	AW	-
src_cfg_desceng_addr1_base	input	AW	-
src_cfg_desceng_addr1_limit	input	AW	-

4.3.5 Control Engine Configuration (Phase 2, global)

Table 1.2.7: Control Engine Configuration (Phase 2, global)

Signal	Direction	Width	Description
src_cfg_ctrlrd_max_try	input	9	-
src_tick_1us	input	1	-

4.3.6 Descriptor AXI Monitor Configuration

Table 1.2.8: Descriptor AXI Monitor Configuration

Signal	Direction	Width	Description
src_cfg_desc_mon_enable	input	1	-
src_cfg_desc_mon_err_enable	input	1	-
src_cfg_desc_mon_perf_enable	input	1	-
src_cfg_desc_mon_perf_run	input	1	-
src_cfg_desc_mon_timeout_enable	input	1	-
src_cfg_desc_mon_timeout_cycles	input	32	-
src_cfg_desc_mon_latency_thresh	input	32	-
src_cfg_desc_mon_pkt_mask	input	16	-
src_cfg_desc_mon_err_select	input	4	-
src_cfg_desc_mon_err_mask	input	8	-
src_cfg_desc_mon_timeout_mask	input	8	-
src_cfg_desc_mon_compl_mask	input	8	-
src_cfg_desc_mon_thresh_mask	input	8	-
src_cfg_desc_mon_perf_mask	input	8	-
src_cfg_desc_mon_addr_mask	input	8	-

Signal	Direction	Width	Description
src_cfg_desc_mon_debug_mask	input	8	-

4.3.7 Network-plane monitor input (rapids TASK-024)

The network endpoint and its monitor-lite live in the plane (rtl/plane/) in the top. Its monbus packets enter here as client 1 of this half's monbus arbiter; the plane's monitor configuration and status come from this half's register block in the top.

Table 1.2.9: Network-plane monitor input (rapids TASK-024)

Signal	Direction	Width	Description
src_plane_mon_valid	input	1	-
src_plane_mon_ready	output	1	-
src_plane_mon_packet	input	128	monitor_packet_t
src_plane_mon_timestamp	input	64	monbus_timestamp_t

4.3.8 Status

Table 1.2.10: Status

Signal	Direction	Width	Description
src_system_idle	output	1	-
src_descriptor_engine_idle	output	NC	-
src_scheduler_idle	output	NC	-
src_scheduler_state	output	NC x 7	-
src_sched_error	output	NC	-

4.3.9 Descriptor AXI Monitor Status

Table 1.2.11: Descriptor AXI Monitor Status

Signal	Direction	Width	Description
src_cfg_sts_desc_mon_busy	output	1	-

Signal	Direction	Width	Description
src_cfg_sts_desc_mon_active_txns	output	8	-
src_cfg_sts_desc_mon_error_count	output	16	-
src_cfg_sts_desc_mon_txn_count	output	32	-
src_cfg_sts_desc_mon_conflict_error	output	1	-

4.3.10 Descriptor AXI Monitor perf window (feeds SRC_.MON.DAXMON_PERF_*).

Table 1.2.12: Descriptor AXI Monitor perf window (feeds SRC_.MON.DAXMON_PERF_*).

Signal	Direction	Width	Description
src_sts_desc_mon_win_active	output	1	-
src_sts_desc_mon_win_cycles	output	32	-
src_sts_desc_mon_prod_cycles	output	32	-
src_sts_desc_mon_bp_cycles	output	32	-
src_sts_desc_mon_starv_cycles	output	32	-
src_sts_desc_mon_idle_cycles	output	32	-
src_sts_desc_mon_beat_count	output	32	-
src_sts_desc_mon_byte_count	output	64	-

Signal	Direction	Width	Description
src_sts_desc_mon_burst_count	output	32	-

4.3.11 Descriptor Fetch AXI Master

Table 1.2.13: Descriptor Fetch AXI Master

Signal	Direction	Width	Description
src_m_axi_desc_arvalid	output	1	-
src_m_axi_desc_arready	input	1	-
src_m_axi_desc_araddr	output	AW	-
src_m_axi_desc_arlen	output	8	-
src_m_axi_desc_arsize	output	3	-
src_m_axi_desc_arburst	output	2	-
src_m_axi_desc_arid	output	IW	-
src_m_axi_desc_arlock	output	1	-
src_m_axi_desc_arcache	output	4	-
src_m_axi_desc_arprot	output	3	-
src_m_axi_desc_arqos	output	4	-
src_m_axi_desc_arregion	output	4	-
src_m_axi_desc_rvalid	input	1	-
src_m_axi_desc_rready	output	1	-
src_m_axi_desc_rdata	input	256	-
src_m_axi_desc_rresp	input	2	-

Signal	Direction	Width	Description
src_m_axi_desc_rlast	input	1	-
src_m_axi_desc_rid	input	IW	-

4.3.12 Control Read AXI Master (32-bit) [Phase 2]

Table 1.2.14: Control Read AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
src_m_axi_ctrlrd_arvalid	output	1	-
src_m_axi_ctrlrd_arready	input	1	-
src_m_axi_ctrlrd_araddr	output	AW	-
src_m_axi_ctrlrd_arlen	output	8	-
src_m_axi_ctrlrd_arsize	output	3	-
src_m_axi_ctrlrd_arburst	output	2	-
src_m_axi_ctrlrd_arid	output	IW	-
src_m_axi_ctrlrd_arlock	output	1	-
src_m_axi_ctrlrd_arcache	output	4	-
src_m_axi_ctrlrd_arprot	output	3	-
src_m_axi_ctrlrd_arqos	output	4	-
src_m_axi_ctrlrd_arregion	output	4	-
src_m_axi_ctrlrd_rvalid	input	1	-
src_m_axi_ctrlrd_rready	output	1	-
src_m_axi_ctrlrd_rdata	input	32	-
src_m_axi_ctrlrd_rresp	input	2	-

Signal	Direction	Width	Description
src_m_axi_ctrlrd_rlast	input	1	-
src_m_axi_ctrlrd_rid	input	IW	-

4.3.13 Control Write AXI Master (32-bit) [Phase 2]

Table 1.2.15: Control Write AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
src_m_axi_ctrlwr_awvalid	output	1	-
src_m_axi_ctrlwr_awready	input	1	-
src_m_axi_ctrlwr_awaddr	output	AW	-
src_m_axi_ctrlwr_awlen	output	8	-
src_m_axi_ctrlwr_awsize	output	3	-
src_m_axi_ctrlwr_awburst	output	2	-
src_m_axi_ctrlwr_awid	output	IW	-
src_m_axi_ctrlwr_awlock	output	1	-
src_m_axi_ctrlwr_awcache	output	4	-
src_m_axi_ctrlwr_awprot	output	3	-
src_m_axi_ctrlwr_awqos	output	4	-
src_m_axi_ctrlwr_awregion	output	4	-
src_m_axi_ctrlwr_wvalid	output	1	-
src_m_axi_ctrlwr_wready	input	1	-
src_m_axi_ctrlwr_wdata	output	32	-
src_m_axi_ctrlwr_wstrb	output	4	-

Signal	Direction	Width	Description
src_m_axi_ctrlwr_wlast	output	1	-
src_m_axi_ctrlwr_bvalid	input	1	-
src_m_axi_ctrlwr_bready	output	1	-
src_m_axi_ctrlwr_bid	input	IW	-
src_m_axi_ctrlwr_bresp	input	2	-

4.4 SOURCE HALF (u_src) - direction-unique ports (no prefix)

4.4.1 AXI Transfer Configuration (source-only)

Table 1.2.16: AXI Transfer Configuration (source-only)

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	-
cfg_drain_size	input	8	source: beats drained per AXIS packet

4.4.2 Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id

Table 1.2.17: Source Path - AXIS Master Interface (SRAM -> Network); tid = channel id

Signal	Direction	Width	Description
m_axis_tdata	output	DW	-
m_axis_tstrb	output	SW	-
m_axis_tlast	output	1	-
m_axis_tid	output	AXIS_ID_WIDTH	-
m_axis_tdest	output	AXIS_DEST_WIDTH	-
m_axis_tuser	output	AXIS_USER_WIDTH	-
m_axis_tvalid	output	1	-

Signal	Direction	Width	Description
m_axis_tready	input	1	-

4.4.3 AXI4 Master - Data Read (Memory -> Source SRAM)

Table 1.2.18: AXI4 Master - Data Read (Memory -> Source SRAM)

Signal	Direction	Width	Description
m_axi_rd_arid	output	IW	-
m_axi_rd_araddr	output	AW	-
m_axi_rd_arlen	output	8	-
m_axi_rd_arsize	output	3	-
m_axi_rd_arburst	output	2	-
m_axi_rd_arvalid	output	1	-
m_axi_rd_arready	input	1	-
m_axi_rd_rid	input	IW	-
m_axi_rd_rdata	input	DW	-
m_axi_rd_rresp	input	2	-
m_axi_rd_rlast	input	1	-
m_axi_rd_rvalid	input	1	-
m_axi_rd_rready	output	1	-

4.5 SINK HALF (u_snk) - shared-infrastructure ports (snk_ prefixed)

4.5.1 APB Programming Interface

Table 1.2.19: APB Programming Interface

Signal	Direction	Width	Description
snk_apb_valid	input	NC	-
snk_apb_ready	output	NC	-
snk_apb_addr	input	NC x AW	-

4.5.2 Per-channel configuration

Table 1.2.20: Per-channel configuration

Signal	Direction	Width	Description
snk_cfg_channel_enable	input	NC	-
snk_cfg_channel_reset	input	NC	-

4.5.3 Scheduler Configuration (global)

Table 1.2.21: Scheduler Configuration (global)

Signal	Direction	Width	Description
snk_cfg_sched_enable	input	1	-
snk_cfg_sched_timeout_cycles	input	32	-
snk_cfg_sched_timeout_limit	input	8	-
snk_cfg_sched_timeout_enable	input	1	-
snk_cfg_sched_err_enable	input	1	-
snk_cfg_sched_compl_enable	input	1	-
snk_cfg_sched_perf_enable	input	1	-

4.5.4 Descriptor Engine Configuration (global)

Table 1.2.22: Descriptor Engine Configuration (global)

Signal	Direction	Width	Description
snk_cfg_desceng_enable	input	1	-
snk_cfg_desceng_prefetch	input	1	-
snk_cfg_desceng_fifo_thresh	input	4	-
snk_cfg_desceng_addr0_base	input	AW	-
snk_cfg_desceng_addr0_limit	input	AW	-
snk_cfg_desceng_addr1_base	input	AW	-
snk_cfg_desceng_addr1_limit	input	AW	-

4.5.5 Control Engine Configuration (Phase 2, global)

Table 1.2.23: Control Engine Configuration (Phase 2, global)

Signal	Direction	Width	Description
snk_cfg_ctrlrd_max_try	input	9	-
snk_tick_1us	input	1	-

4.5.6 Descriptor AXI Monitor Configuration

Table 1.2.24: Descriptor AXI Monitor Configuration

Signal	Direction	Width	Description
snk_cfg_desc_mon_enable	input	1	-
snk_cfg_desc_mon_err_enable	input	1	-
snk_cfg_desc_mon_perf_enable	input	1	-
snk_cfg_desc_	input	1	-

Signal	Direction	Width	Description
mon_perf_run			
snk_cfg_desc_mon_timeout_enable	input	1	-
snk_cfg_desc_mon_timeout_cycles	input	32	-
snk_cfg_desc_mon_latency_thresh	input	32	-
snk_cfg_desc_mon_pkt_mask	input	16	-
snk_cfg_desc_mon_err_select	input	4	-
snk_cfg_desc_mon_err_mask	input	8	-
snk_cfg_desc_mon_timeout_mask	input	8	-
snk_cfg_desc_mon_compl_mask	input	8	-
snk_cfg_desc_mon_thresh_mask	input	8	-
snk_cfg_desc_mon_perf_mask	input	8	-
snk_cfg_desc_mon_addr_mask	input	8	-
snk_cfg_desc_mon_debug_mask	input	8	-

4.5.7 Network-plane monitor input (rapids TASK-024)

The network endpoint and its monitor-lite live in the plane (`rtl/plane/`) in the top. Its monbus packets enter here as client 1 of this half's monbus arbiter; the plane's monitor configuration and status come from this half's register block in the top.

Table 1.2.25: Network-plane monitor input (rapids TASK-024)

Signal	Direction	Width	Description
snk_plane_mon_valid	input	1	-
snk_plane_mon_ready	output	1	-
snk_plane_mon_packet	input	128	monitor_packet_t
snk_plane_mon_timestamp	input	64	monbus_timestamp_t

4.5.8 Status

Table 1.2.26: Status

Signal	Direction	Width	Description
snk_system_idle	output	1	-
snk_descriptor_engine_idle	output	NC	-
snk_scheduler_idle	output	NC	-
snk_scheduler_state	output	NC x 7	-
snk_sched_error	output	NC	-

4.5.9 Descriptor AXI Monitor Status

Table 1.2.27: Descriptor AXI Monitor Status

Signal	Direction	Width	Description
snk_cfg_sts_desc_mon_busy	output	1	-
snk_cfg_sts_desc_mon_active_txns	output	8	-
snk_cfg_sts_desc_mon_error_count	output	16	-
snk_cfg_sts_desc_mon_txn_count	output	32	-
snk_cfg_sts_d	output	1	-

Signal	Direction	Width	Description
esc_mon_conflict_error			

4.5.10 Descriptor AXI Monitor perf window (feeds SNK_.MON.DAXMON_PERF_*).

Table 1.2.28: Descriptor AXI Monitor perf window (feeds SNK_.MON.DAXMON_PERF_*).

Signal	Direction	Width	Description
snk_sts_desc_mon_win_active	output	1	-
snk_sts_desc_mon_win_cycles	output	32	-
snk_sts_desc_mon_prod_cycles	output	32	-
snk_sts_desc_mon_bp_cycles	output	32	-
snk_sts_desc_mon_starv_cycles	output	32	-
snk_sts_desc_mon_idle_cycles	output	32	-
snk_sts_desc_mon_beat_count	output	32	-
snk_sts_desc_mon_byte_count	output	64	-
snk_sts_desc_mon_burst_count	output	32	-

4.5.11 Descriptor Fetch AXI Master

Table 1.2.29: Descriptor Fetch AXI Master

Signal	Direction	Width	Description
snk_m_axi_desc_arvalid	output	1	-

Signal	Direction	Width	Description
snk_m_axi_desc_arready	input	1	-
snk_m_axi_desc_araddr	output	AW	-
snk_m_axi_desc_arlen	output	8	-
snk_m_axi_desc_arsize	output	3	-
snk_m_axi_desc_arburst	output	2	-
snk_m_axi_desc_arid	output	IW	-
snk_m_axi_desc_arlock	output	1	-
snk_m_axi_desc_arcache	output	4	-
snk_m_axi_desc_arprot	output	3	-
snk_m_axi_desc_arqos	output	4	-
snk_m_axi_desc_arregion	output	4	-
snk_m_axi_desc_rvalid	input	1	-
snk_m_axi_desc_rready	output	1	-
snk_m_axi_desc_rdata	input	256	-
snk_m_axi_desc_rresp	input	2	-
snk_m_axi_desc_rlast	input	1	-
snk_m_axi_desc_rid	input	IW	-

4.5.12 Control Read AXI Master (32-bit) [Phase 2]

Table 1.2.30: Control Read AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
snk_m_axi_ctrlrd_arvalid	output	1	-

Signal	Direction	Width	Description
snk_m_axi_ctrlrd_arready	input	1	-
snk_m_axi_ctrlrd_araddr	output	AW	-
snk_m_axi_ctrlrd_arlen	output	8	-
snk_m_axi_ctrlrd_arsize	output	3	-
snk_m_axi_ctrlrd_arburst	output	2	-
snk_m_axi_ctrlrd_arid	output	IW	-
snk_m_axi_ctrlrd_arlock	output	1	-
snk_m_axi_ctrlrd_arcache	output	4	-
snk_m_axi_ctrlrd_arprot	output	3	-
snk_m_axi_ctrlrd_arqos	output	4	-
snk_m_axi_ctrlrd_arregion	output	4	-
snk_m_axi_ctrlrd_rvalid	input	1	-
snk_m_axi_ctrlrd_rready	output	1	-
snk_m_axi_ctrlrd_rdata	input	32	-
snk_m_axi_ctrlrd_rresp	input	2	-
snk_m_axi_ctrlrd_rlast	input	1	-
snk_m_axi_ctrlrd_rid	input	IW	-

4.5.13 Control Write AXI Master (32-bit) [Phase 2]

Table 1.2.31: Control Write AXI Master (32-bit) [Phase 2]

Signal	Direction	Width	Description
snk_m_axi_ctrlwr_awvalid	output	1	-

Signal	Direction	Width	Description
snk_m_axi_ctrlwr_awready	input	1	-
snk_m_axi_ctrlwr_awaddr	output	AW	-
snk_m_axi_ctrlwr_awlen	output	8	-
snk_m_axi_ctrlwr_awsize	output	3	-
snk_m_axi_ctrlwr_awburst	output	2	-
snk_m_axi_ctrlwr_awid	output	IW	-
snk_m_axi_ctrlwr_awlock	output	1	-
snk_m_axi_ctrlwr_awcache	output	4	-
snk_m_axi_ctrlwr_awprot	output	3	-
snk_m_axi_ctrlwr_awqos	output	4	-
snk_m_axi_ctrlwr_awregion	output	4	-
snk_m_axi_ctrlwr_wvalid	output	1	-
snk_m_axi_ctrlwr_wready	input	1	-
snk_m_axi_ctrlwr_wdata	output	32	-
snk_m_axi_ctrlwr_wstrb	output	4	-
snk_m_axi_ctrlwr_wlast	output	1	-
snk_m_axi_ctrlwr_bvalid	input	1	-
snk_m_axi_ctrlwr_bready	output	1	-
snk_m_axi_ctrlwr_bid	input	IW	-
snk_m_axi_ctrlwr_bresp	input	2	-

4.6 Monitor Bus (SINGLE aggregated stream for the whole core)

The two halves' monitor outputs are merged through a top-level monbus_arbiter, so the core exposes exactly one monitor stream.

4.6.1 Monitor Bus

Table 1.2.32: Monitor Bus

Signal	Direction	Width	Description
mon_valid	output	1	-
mon_ready	input	1	-
mon_packet	output	monitor_common_pkg::monitor_packet_t	-
mon_timestamp	output	monitor_common_pkg::monbus_timestamp_t	-

4.7 SINK HALF (u_snk) - direction-unique ports (no prefix)

4.7.1 AXI Transfer Configuration (sink-only)

Table 1.2.33: AXI Transfer Configuration (sink-only)

Signal	Direction	Width	Description
cfg_axi_wr_xfer_beats	input	8	-
cfg_alloc_size	input	8	sink: SRAM alloc size per AXIS fill

4.7.2 Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id

Table 1.2.34: Sink Path - AXIS Slave Interface (Network -> SRAM); tid = channel id

Signal	Direction	Width	Description
s_axis_tdata	input	DW	-
s_axis_tstrb	input	SW	-
s_axis_tlast	input	1	-
s_axis_tid	input	AXIS_ID_WIDT	-

Signal	Direction	Width	Description
		H	
s_axis_tdest	input	AXIS_DEST_WIDTH	-
s_axis_tuser	input	AXIS_USER_WIDTH	-
s_axis_tvalid	input	1	-
s_axis_tready	output	1	-

4.7.3 AXI4 Master - Data Write (Sink SRAM -> Memory)

Table 1.2.35: AXI4 Master - Data Write (Sink SRAM -> Memory)

Signal	Direction	Width	Description
m_axi_wr_awid	output	IW	-
m_axi_wr_awaddr	output	AW	-
m_axi_wr_awlen	output	8	-
m_axi_wr_awsize	output	3	-
m_axi_wr_awburst	output	2	-
m_axi_wr_awlock	output	1	-
m_axi_wr_awcache	output	4	-
m_axi_wr_awprot	output	3	-
m_axi_wr_awqos	output	4	-
m_axi_wr_awregion	output	4	-
m_axi_wr_awvalid	output	1	-
m_axi_wr_awready	input	1	-
m_axi_wr_wdata	output	DW	-
m_axi_wr_wstrb	output	(DW/8)	-

Signal	Direction	Width	Description
m_axi_wr_wlast	output	1	-
m_axi_wr_wvalid	output	1	-
m_axi_wr_wready	input	1	-
m_axi_wr_bid	input	IW	-
m_axi_wr_bresp	input	2	-
m_axi_wr_bvalid	input	1	-
m_axi_wr_bready	output	1	-

4.8 Debug Interface (per half, src_dbg_ / snk_dbg_ prefixed)

4.8.1 Source half debug

Table 1.2.36: Source half debug

Signal	Direction	Width	Description
src_dbg_rd_all_complete	output	NC	-
src_dbg_r_beats_rcvd	output	32	-
src_dbg_sram_writes	output	32	-
src_dbg_arb_request	output	NC	-
src_dbg_src_sram_bridge_pending	output	NC	-
src_dbg_src_sram_bridge_out_valid	output	NC	-
src_dbg_axis_beats_sent	output	32	-
src_dbg_axis_packets_sent	output	32	-

4.8.2 Sink half debug

Table 1.2.37: Sink half debug

Signal	Direction	Width	Description
snk_dbg_snk_sram_bridge_pending	output	NC	-
snk_dbg_snk_sram_bridge_out_valid	output	NC	-
snk_dbg_axis_beats_received	output	32	-
snk_dbg_axis_packets_received	output	32	-

4.8.3 Active-channel sideband for per-channel bus instrumentation

(axi_bus_meter). The W bus carries no wid, so the write engine's channel index must travel out of band to reach the meter at the top.

Table 1.2.38: Active-channel sideband for per-channel bus instrumentation

Signal	Direction	Width	Description
snk_active_channel_id	output	CIW	-
snk_active_channel_valid	output	1	-

Verified: every row above is a declared port of `rapids_core_beats` and every declared port has a row (324 of 324), generated from the module declaration on 2026-09-28. Regenerate rather than hand-edit when the interface changes.

5 Clocks and Reset

Module: `rapids_core_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

5.1 Clock Domain

RAPIDS Beats operates in a single clock domain for simplified design and maximum throughput.

5.1.1 Clock Specifications

Table 1.3.1: Clock Specifications

Parameter	Specification
Clock Signal	`clk` (also `axi_aclk` in some submodules)
Frequency Range	100 - 500 MHz
Duty Cycle	45% - 55%
Clock Jitter	< 100 ps peak-to-peak

5.1.2 Clock Distribution

```
clk (one clock domain; rst_n active-low, async assert)
├── scheduler_group_array_beats (one per half)
│   └── descriptor_engine_beats, scheduler_beats, ctrlrd_engine,
ctrlwr_engine (x8)
├── snk_data_path_axis_beats
│   └── snk_sram_controller_beats, axi_write_engine_beats
└── src_data_path_axis_beats
    └── axi_read_engine_beats, src_sram_controller_beats
```

5.2 Reset Specification

RAPIDS uses asynchronous assert, synchronous deassert reset methodology.

5.2.1 Reset Signal

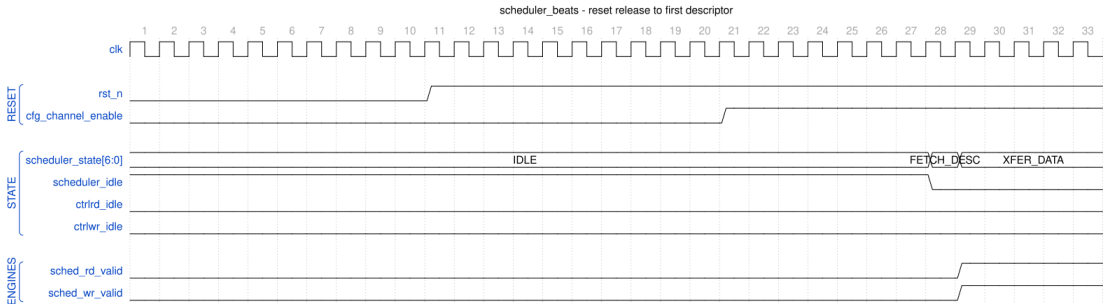
Table 1.3.2: Reset Specifications

Parameter	Specification

Reset Signal	`rst_n` (also `axi_aresetn` in some submodules)
Polarity	Active-low
Assert	Asynchronous (immediate)
Deassert	Synchronous (on rising clock edge)
Minimum Duration	10 clock cycles

5.2.2 Reset Timing Diagram

5.2.3 Figure 1.3.1: Reset Timing



scheduler_beats - reset release to first descriptor

Source: scheduler_reset.json, captured from `dv/tests/fub_beats/test_scheduler_beats.py` (basic transfer, the same run as Figure 2.1.3) with `WAVES=1`.

Reading it: `rst_n` is held low for ten cycles and released on a clock edge. Every state register is already at its reset value while reset is asserted – the FSM sits in `IDLE`, `scheduler_idle` is high, both control engines report idle and neither engine request is raised. Nothing changes at the release edge itself; the scheduler stays in `IDLE` until the testbench raises `cfg_channel_enable` ten cycles later and a descriptor becomes available, at which point it steps through `FETCH_DESC` to `XFER_DATA` and raises the read and write requests together.

5.3 Reset Behavior

5.3.1 On Reset Assert

All modules initialize to known states:

Table 1.3.3: Reset States

Component	Reset State
Scheduler FSM	IDLE
Descriptor Engine	IDLE, FIFOs empty
AXI Read Engine	Idle, no outstanding transactions
AXI Write Engine	Idle, no outstanding transactions
SRAM Pointers	All zeros
Alloc/Drain Controllers	Empty, full space available
MonBus	No packets pending

5.3.2 Reset Sequence

```
1. External reset asserted (rst_n = 0)
   - All flip-flops async reset
   - All outputs driven to safe states
   - No AXI transactions

2. System stabilization (minimum 10 cycles)
   - Clock running and stable
   - Configuration registers programmed (optional)

3. Reset deasserted (rst_n = 1, sync to clk rising edge)
   - FSMs begin operation
   - Ready to accept descriptors/data
```

5.4 Configuration During Reset

Some configuration signals MUST be stable before reset deassertion:

Table 1.3.4: Configuration Timing Requirements

Signal	Requirement
`src_cfg_channel_enable` / `snk_cfg_channel_enable`	Stable before rst_n=1 (one per half)
`src_cfg_sched_timeout_cycles` / `snk_cfg_sched_timeout_cyc`	Stable before rst_n=1 (one per half)

Signal	Requirement
les	
cfg_axi_rd_xfer_beats	Stable before rst_n=1
cfg_axi_wr_xfer_beats	Stable before rst_n=1

5.5 Reset Macros

RAPIDS uses standardized reset macros from `reset_defs.svh`:

```
`include "reset_defs.svh"

// Standard reset pattern
`ALWAYS_FF_RST(clk, rst_n,
    if (`RST_ASSERTED(rst_n)) begin
        r_state <= IDLE;
        r_counter <= '0;
    end else begin
        r_state <= w_next_state;
        r_counter <= w_next_counter;
    end
)
```

5.6 Power-On Reset Considerations

- External reset should be held for minimum 10 clock cycles after clock is stable
- Configuration registers should be programmed before or during reset
- No AXI transactions until reset is deasserted and channels are enabled

---

Last Updated: 2025-01-10

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6 Scheduler

Module: `scheduler.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/` **Status:** Implemented **Last Updated:** 2026-09-30

6.1 Overview

The scheduler coordinates one channel. It accepts a 256-bit descriptor from the descriptor engine, turns the descriptor's byte length and byte addresses into beat counts for the AXI engines, tells the data paths how the packet is laid out, and tracks progress until every source beat is read and every destination beat is committed.

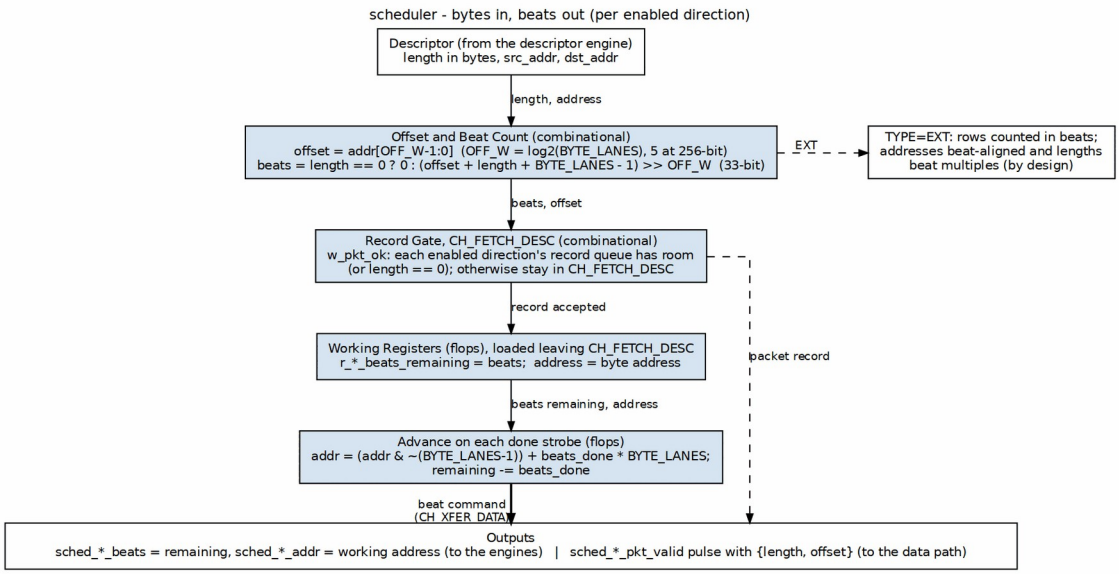
Compared with the RAPIDS Beats scheduler, the state machine, the timeout logic, the control-descriptor path and the MonBus events are the same. Three things change: the beat math, the address advance, and the packet record.

6.1.1 Key Features

- **Byte length:** the descriptor's 32-bit length is a byte count.
- **Byte addresses:** the offset of a transfer inside its first beat is `addr[OFF_W-1:0]`.
- **Beat math:** beats moved are derived from offset and length, in a 33-bit intermediate.
- **Packet record:** one `{bytes, offset}` pulse per DATA descriptor, per enabled direction, held back while the data path cannot take it.
- **Concurrent read and write:** both directions run inside one XFER_DATA state.
- **Recoverable write-progress timeout:** strikes escalate to a sticky error only when a limit is set.
- **Control descriptors:** CTRL_READ and CTRL_WRITE drive the control engines, unchanged.
- **Directional halves:** `EN_READ` and `EN_WRITE` select the source or sink half.

6.1.2 Block Diagram

6.1.3 Figure 2.1.1: Scheduler Byte Math



Scheduler byte math

Source: `assets/graphviz/03_scheduler_byte_math.dot`

6.2 Parameters

Table 2.1.1: Scheduler Parameters

Parameter	Type	Default	Description
`CHANNEL_ID`	int	0	Channel index of this scheduler
`NUM_CHANNELS`	int	8	Channel count
`CHAN_WIDTH`	int	`$clog2(NUM_CHANNELS)`	Channel field width
`ADDR_WIDTH`	int	64	Address width
`DATA_WIDTH`	int	512	Data width in bits; the board design uses 256
`MON_AGENT_ID`	logic [15:0]	16’h0040	MonBus agent ID
`MON_UNIT_ID`	logic [7:0]	8’h01	MonBus unit ID

Parameter	Type	Default	Description
MON_CHANNEL_ID	logic [8:0]	9'h000	MonBus base channel ID
DESC_WIDTH	int	256	Descriptor width; fixed at 256 (elaboration fails otherwise)
EN_READ	bit	1	Enable the read direction for DATA descriptors
EN_WRITE	bit	1	Enable the write direction for DATA descriptors
USE_ROW_COL_MAJOR_ADDRESSING	int	1	Enable TYPE=EXT run-base generators
BYTE_LANES	int	DATA_WIDTH/8	Bytes per beat
OFF_W	int	(BYTE_LANES>1) ? $clog2(BYTE_LANES) : 1	Offset field width

BYTE_LANES and OFF_W are the two parameters that do not exist in the Beats scheduler. A directional half sets exactly one of EN_READ and EN_WRITE: the source half reads only, the sink half writes only.

6.3 Port List

6.3.1 Clock, Reset and Configuration

Table 2.1.2: Clock, Reset and Configuration Ports

Signal	Dir	Width	Description
clk	in	1	Clock
rst_n	in	1	Active-low reset
cfg_channel_enable	in	1	Channel enable; IDLE leaves only when set
cfg_channel_reset	in	1	Channel reset; forces IDLE and clears sticky state
cfg_sched_timeout_cycles	in	32	Write-progress timeout window in cycles

Signal	Dir	Width	Description
cfg_sched_timeout_limit	in	8	Consecutive windows before escalation; 0 = never
cfg_sched_timeout_enable	in	1	Enable timeout detection

6.3.2 Status

Table 2.1.3: Status Ports

Signal	Dir	Width	Description
scheduler_idle	out	1	High in CH_IDLE with no channel reset pending
scheduler_state	out	7	One-hot state (see Table 2.1.12)
sched_error	out	1	Sticky, high in CH_ERROR
dbg_descriptor_error	out	1	Latched fatal descriptor-level error
dbg_read_error_sticky	out	1	Latched read-engine error
dbg_write_error_sticky	out	1	Latched write-engine error
dbg_timeout_expired	out	1	Live: the timeout window has elapsed

6.3.3 Descriptor Engine Interface

Table 2.1.4: Descriptor Engine Interface

Signal	Dir	Width	Description
descriptor_valid	in	1	A descriptor is available
descriptor_ready	out	1	High in CH_IDLE and CH_NEXT_DESC
descriptor_packet	in	256	Descriptor (layout in Table 2.1.10)
descriptor_ext_packet	in	256	Extended chunk; used only when TYPE=EXT
descriptor_error	in	1	Descriptor fetch or parse error

6.3.4 Data Read Interface (to AXI Read Engine)

Table 2.1.5: Data Read Interface

Signal	Dir	Width	Description
sched_rd_valid	out	1	Read request
sched_rd_addr	out	ADDR_WIDTH	Source byte address (working register)
sched_rd_beats	out	32	Beats remaining to read
sched_rd_pkt_valid	out	1	Packet record pulse to the source egress
sched_rd_pkt_ready	in	1	Source egress record queue has room
sched_rd_pkt_bytes	out	32	Descriptor length in bytes
sched_rd_pkt_offset	out	OFF_W	Source byte offset inside the first beat

6.3.5 Data Write Interface (to AXI Write Engine)

Table 2.1.6: Data Write Interface

Signal	Dir	Width	Description
sched_wr_valid	out	1	Write request
sched_wr_ready	in	1	Write engine can take the channel
sched_wr_addr	out	ADDR_WIDTH	Destination byte address (working register)
sched_wr_beats	out	32	Beats remaining to issue
sched_wr_pkt_valid	out	1	Packet record pulse to the sink ingress
sched_wr_pkt_ready	in	1	Sink ingress record queue has room
sched_wr_pkt_bytes	out	32	Descriptor length in bytes
sched_wr_pkt_offset	out	OFF_W	Destination byte offset inside the first beat

6.3.6 Completion Interface

Table 2.1.7: Completion and Error Interface

Signal	Dir	Width	Description
sched_rd_done_strobe	in	1	A read burst completed
sched_rd_beats_done	in	32	Beats in that burst
sched_wr_done_strobe	in	1	A write burst was issued (AW handshake)
sched_wr_beats_done	in	32	Beats in that burst
sched_wr_commit_strobe	in	1	A write burst was committed (B response)
sched_wr_commit_beats	in	32	Beats in that burst
sched_rd_error	in	1	Read engine error
sched_wr_error	in	1	Write engine error, including the sink packet length error

6.3.7 Control Engine Interface

Table 2.1.8: Control Engine Interface

Signal	Dir	Width	Description
ctrlrd_valid	out	1	CTRL_READ request
ctrlrd_ready	in	1	Engine accepted the request
ctrlrd_addr	out	ADDR_WIDTH	Poll address (descriptor source slot)
ctrlrd_data	out	32	Expected value (dst_addr[31:0])
ctrlrd_mask	out	32	Compare mask (dst_addr[63:32])
ctrlrd_error	in	1	Engine error (retries exhausted or AXI error)
ctrlrd_idle	in	1	Engine idle
ctrlwr_valid	out	1	CTRL_WRITE request
ctrlwr_ready	in	1	Engine accepted the request

Signal	Dir	Width	Description
ctrlwr_addr	out	ADDR_WIDTH	Doorbell address (descriptor source slot)
ctrlwr_data	out	32	Doorbell data (dst_addr[31:0])
ctrlwr_error	in	1	Engine error
ctrlwr_idle	in	1	Engine idle

6.3.8 Monitor Bus Interface

Table 2.1.9: Monitor Bus Interface

Signal	Dir	Width	Description
i_mon_time	in	timestamp	Time base for packets
mon_valid	out	1	Packet valid
mon_ready	in	1	Consumer ready
mon_packet	out	128	Monitor packet
mon_timestamp	out	64	Side-band timestamp

The control interface tables do not change from RAPIDS Beats. The packet-record ports in Tables 2.1.5 and 2.1.6 are the new ports.

6.4 Descriptor Format

Table 2.1.10: Descriptor Layout

Bits	Field	Notes
[63:0]	src_addr	Byte address
[127:64]	dst_addr	Byte address
[159:128]	length	Bytes
[191:160]	next_descriptor_ptr	Chain pointer
[192]	valid	Must be 1
[193]	gen_irq	Emit an IRQ event

Bits	Field	Notes
		on completion
[194]	`last`	End of chain
[195]	`error`	Error flag
[199:196]	`channel_id`	
[207:200]	`priority`	
[209:208]	`opcode`	DATA = 0, CTRL_READ = 1, CTRL_WRITE = 2
[212:210]	`type`	LEGACY = 0, EXT = 1

For CTRL_READ and CTRL_WRITE the address and data slots are reinterpreted as in Tables 2.1.8; the length field does not describe a byte count and no packet record is issued.

6.5 Byte-to-Beat Math

For each enabled direction the scheduler forms the offset from the low `OFF_W` bits of that side’s address and derives the number of memory beats the transfer touches.

```
w_rd_off        = src_addr[OFF_W-1:0]
w_wr_off        = dst_addr[OFF_W-1:0]
w_rd_span       = 33'(w_rd_off) + 33'(length) + 33'(BYTE_LANES-1)
w_wr_span       = 33'(w_wr_off) + 33'(length) + 33'(BYTE_LANES-1)
w_rd_beats_total = (length == 0) ? 0 : 32'(w_rd_span >> OFF_W)
w_wr_beats_total = (length == 0) ? 0 : 32'(w_wr_span >> OFF_W)
```

The span is 33 bits wide so a length near 4 GiB does not overflow. The read and write beat totals are computed separately because the two addresses can have different offsets; a 60-byte transfer read from offset 5 and written at offset 0 touches three source beats and two destination beats on a 32-byte design.

Table 2.1.11: Beat Totals for a 32-byte Beat

Length (bytes)	Offset	Beats
0	any	0
1	0	1
32	0	1
32	1	2
60	5	3

Length (bytes)	Offset	Beats
60	0	2
64	0	2

The values load in CH_FETCH_DESC:

- `r_beats_remaining` loads `EN_READ ? rd_total : wr_total`.
- `r_read_beats_remaining` loads `EN_READ ? rd_total : 0`.
- `r_write_beats_remaining` and `r_write_beats_to_commit` load `EN_WRITE ? wr_total : 0`.

A disabled direction therefore loads zero, so its valid never asserts and its completion term is true from the start. Completion collapses onto the enabled direction.

For TYPE=EXT the run counters load `min(run_size, beats_total)` instead of the whole total, and `stream_run_addr_gen.cfg_total_beats` takes the beats total. TYPE=EXT is beat-aligned by design: addresses are aligned and lengths are beat multiples, so the offset is zero and the math reduces to the Beats result.

6.6 Address Advance

The working address registers start at the descriptor's byte addresses. On each done strobe the address moves forward by the beats reported:

```
r_src_addr <= {r_src_addr[AW-1:OFF_W], {OFF_W{1'b0}}} +
(AW'(sched_rd_beats_done) << OFF_W)   // on sched_rd_done_strobe
r_dst_addr <= {r_dst_addr[AW-1:OFF_W], {OFF_W{1'b0}}} +
(AW'(sched_wr_beats_done) << OFF_W)   // on sched_wr_done_strobe
```

The low bits are cleared before the add. Only the first burst of a descriptor starts inside a beat; every later burst starts on a beat boundary, so aligning down first is correct for both. The write address advances on the AW handshake (issue), not on the B response. The B response decrements `r_write_beats_to_commit` instead.

The scheduler reports the working address to the engines unmodified. The engines align it (see AXI Read Engine); the offset reaches the data path through the packet record, not through the address bus.

6.7 Packet Records

A DATA descriptor with a non-zero length causes one record pulse per enabled direction, on the cycle the scheduler leaves CH_FETCH_DESC:

```
w_pkt_ok = !w_is_data || length == 0 ||
           ((!EN_READ  || sched_rd_pkt_ready) &&
            (!EN_WRITE || sched_wr_pkt_ready))

sched_{rd,wr}_pkt_valid = w_state_fetch_desc && r_descriptor.valid &&
w_is_data &&
                          EN_{READ,WRITE} && w_pkt_ok && (length != 0)
sched_{rd,wr}_pkt_bytes  = r_descriptor.length
sched_{rd,wr}_pkt_offset = w_{rd,wr}_off
```

Both directions share `w_pkt_ok`, so each enabled direction sees exactly one pulse. `w_pkt_ok` also gates the FETCH_DESC to XFER_DATA transition: while a record queue is full, the scheduler holds in CH_FETCH_DESC and the pulse does not fire. Control descriptors and zero-length transfers carry no record. See Packet Record Interface.

6.8 Finite State Machine

The state is one-hot and is `rapids_pkg::channel_state_t`.

Table 2.1.12: State Encoding

State	Encoding	Description
CH_IDLE	7'b0000001	Waiting for a descriptor and `cfg_channel_enable`
CH_FETCH_DESC	7'b0000010	Descriptor latched; working registers and counters load; record pulses
CH_XFER_DATA	7'b0000100	Read and write engines run concurrently; control ops issue
CH_COMPLETE	7'b0001000	Descriptor done; chain decision
CH_NEXT_DESC	7'b0010000	Waiting for the chained descriptor
CH_ERROR	7'b0100000	Sticky; only channel reset leaves it
CH_RESERVED	7'b1000000	Unused

Transition priority is: registered channel reset to IDLE, then a hard error or timeout escalation to ERROR, then the normal case.

Table 2.1.13: Transitions

From	To	Condition
IDLE	FETCH_DESC	`descriptor_valid && cfg_channel_enable`
FETCH_DESC	XFER_DATA	`r_descriptor.valid && w_pkt_ok`
FETCH_DESC	FETCH_DESC	`r_descriptor.valid` and `w_pkt_ok` false (record queue full)
FETCH_DESC	ERROR	descriptor `valid` bit clear
XFER_DATA	COMPLETE	`w_exec_complete`
COMPLETE	NEXT_DESC	`next_descriptor_ptr != 0 && !last`
COMPLETE	IDLE	otherwise
NEXT_DESC	FETCH_DESC	`descriptor_valid`
ERROR	ERROR	always; channel reset is the only exit
any other	ERROR	default

`w_exec_complete` is `w_transfer_complete` for a DATA descriptor and `w_ctrl_complete` for a control descriptor. `w_transfer_complete` is the read remaining count reaching zero together with the commit remaining count reaching zero. `w_ctrl_complete` is the control request having been accepted and the control engine having returned to idle.

`scheduler_idle` is `CH_IDLE && !r_channel_reset_active`. `sched_error` is the ERROR state.

6.9 Operation

6.9.1 Data Request Outputs

`sched_rd_valid` is high in XFER_DATA for a DATA descriptor while `!w_read_complete`, `!w_sched_rd_completing_this_cycle` and, for TYPE=EXT, `!w_rd_need_base`. `sched_rd_beats` is the read remaining count, or for EXT the current run's remainder. `sched_wr_valid` is high in XFER_DATA for a DATA descriptor while `r_write_beats_remaining != 0`, the commit count has not reached zero, `!w_sched_wr_completing_this_cycle` and `!w_wr_need_base`.

The look-ahead terms drop the valid on the same cycle the last done strobe arrives. Without them the engine would see a still-asserted valid for one more cycle and issue an extra burst.

6.9.2 Write Completion

The write direction has two counters. `r_write_beats_remaining` decrements on the AW done strobe and drives burst sizing and the destination address. `r_write_beats_to_commit` decrements on the B commit strobe and gates `w_write_complete`. The channel therefore completes only after every issued burst has been acknowledged.

6.9.3 Control Descriptors

CTRL_READ and CTRL_WRITE do not use the data engines. `ctrlrd_valid` and `ctrlwr_valid` assert in XFER_DATA until the engine accepts (`r_ctrl_issued`), and completion is the engine returning to idle after acceptance.

6.10 Error Handling

`w_hard_error` is the OR of:

- `descriptor_error`, `sched_rd_error`, `sched_wr_error`
- the sticky read and write error flags
- `ctrlrd_error` while the descriptor is CTRL_READ
- `ctrlwr_error` while the descriptor is CTRL_WRITE

A hard error, or a timeout escalation, moves the FSM to CH_ERROR. The sink packet length check reports through `sched_wr_error` (see Sink Data Path AXIS), so a stream packet whose byte count differs from the descriptor length is a hard error for that channel. Sticky read and write flags and the descriptor error clear in CH_IDLE and also on the registered channel reset, in every state. Clearing on the reset matters: the data engines and the data paths clear their own error levels on the same channel reset, so the scheduler no longer re-enters CH_ERROR from a stale level when it reaches CH_IDLE. A channel reset therefore recovers a data read or write response error and a sink packet length error, and the channel accepts a good descriptor afterwards. The reset also clears the descriptor-loaded flag, the remaining beat counts and the control issued flag, so nothing of the failed descriptor carries into the next one.

6.11 Recoverable Write-Progress Timeout

The timeout counter increments while `sched_wr_valid && !sched_wr_ready`. It resets on `sched_wr_done_strobe` or `sched_wr_commit_strobe`, and also when the condition is false. When the counter reaches `cfg_sched_timeout_cycles` with `cfg_sched_timeout_enable` set, `w_timeout_expired` pulses and the counter re-arms.

Each expiry adds one strike, in an 8-bit counter that saturates. Strikes clear on write progress, in CH_IDLE, and on channel reset. Escalation is:

```
w_timeout_escalate = (cfg_sched_timeout_limit != 0) && (strikes >=
cfg_sched_timeout_limit)
```

A limit of zero never escalates: the timeout is then purely informational.

6.12 MonBus Events

Packets are `PktTypeCompletion` from `PROTOCOL_CORE` unless noted.

Table 2.1.14: MonBus Events

State	Event	Data
CH_FETCH_DESC	`RAPIDS_EVENT_DESC_START`	descriptor length (bytes)
CH_COMPLETE	`RAPIDS_EVENT_IRQ` when `gen_irq`, else `RAPIDS_EVENT_DESC_COMPLETE`	
CH_ERROR	`RAPIDS_EVENT_ERROR` (`PktTypeError`), once per error episode	`{29'h0, wr_sticky, rd_sticky, 33'h0}`

There are no events during XFER_DATA. The only difference from RAPIDS Beats is that the DESC_START data field is a byte count.

6.13 Waveforms

6.13.0.1 Waveform 2.1: Byte Descriptor Start

The beat-count arithmetic and the record pulse are combinational functions of the latched descriptor, so the start of a descriptor is short:

```
cycle                   : 0     1       2       3
state                   : IDLE  FETCH   XFER    XFER
descriptor_valid        : 1     0       0       0
sched_wr_pkt_ready      : 1     1       1       1
sched_wr_pkt_valid      : 0     1       0       0     (one pulse,
the last FETCH cycle)
r_write_beats_remaining : 0     0       N       N     (loads at the
```

```
end of FETCH)
sched_wr_valid           : 0     0        1        1
```

If `sched_wr_pkt_ready` is low in FETCH, the state holds and the pulse is delayed until it rises.

---

7 Descriptor Engine Specification

Module: `descriptor_engine_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

7.1 Overview

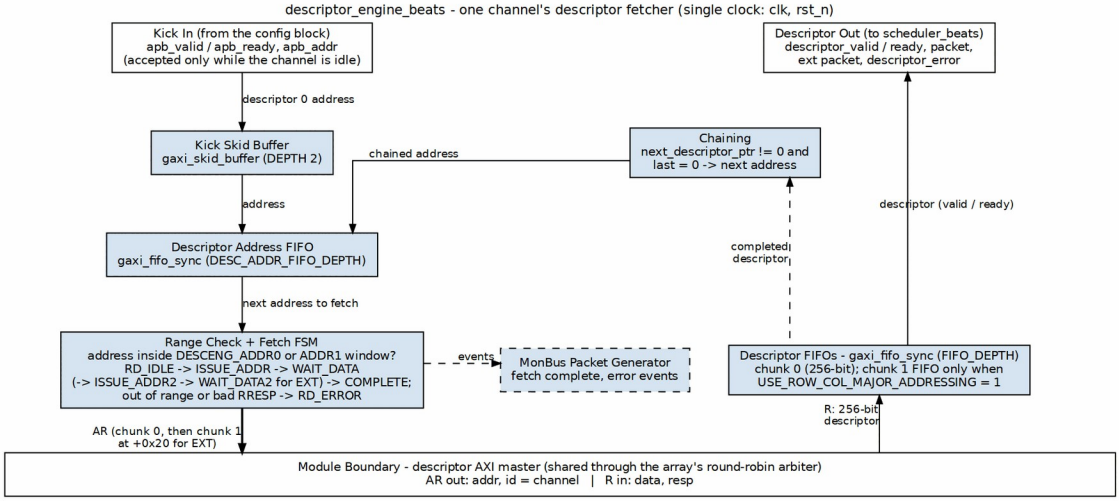
The Descriptor Engine fetches and manages 256-bit descriptors from memory via AXI4. It maintains a prefetch FIFO to hide memory latency and provides parsed descriptors to the scheduler.

7.1.1 Key Features

- **AXI4 Descriptor Fetch:** Single-beat 256-bit reads
- **Prefetch FIFO:** Configurable depth for latency hiding
- **Address Range Checking:** Validates descriptor addresses
- **Error Detection:** Address violations, AXI errors
- **MonBus Integration:** Fetch events and error reporting

7.1.2 Block Diagram

7.1.3 Figure 2.2.1: Descriptor Engine Block Diagram



Descriptor engine block diagram

Source: 02_descriptor_engine.dot

7.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 8;                  // Total channels
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int FIFO_DEPTH = 4;                    // Prefetch FIFO
depth

// Monitor Bus Parameters
parameter logic [7:0] MON_AGENT_ID = 8'h10;      // Descriptor Engine
Agent ID
parameter logic [3:0] MON_UNIT_ID = 4'h1;        // Unit identifier
```

: Table 2.2.1: Descriptor Engine Parameters

7.3 Port List

7.3.1 Clock and Reset

Table 2.2.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

7.3.2 APB Programming Interface

Table 2.2.3: APB Programming Interface

Signal	Direction	Width	Description
`apb_valid`	input	1	Kick-off valid (start descriptor chain)
`apb_ready`	output	1	Ready to accept kick-off
`apb_addr`	input	AW	First descriptor address

7.3.3 Configuration Interface

Table 2.2.4: Configuration Interface

Signal	Direction	Width	Description
`cfg_prefetch_enable`	input	1	Enable prefetch chaining (0 = on-demand, ~1 ahead)
`cfg_fifo_threshold`	input	4	Prefetch depth (descriptors buffered ahead when enabled)
`cfg_addr0_base`	input	AW	Valid address range 0 base
`cfg_addr0_limit`	input	AW	Valid address range 0 limit
`cfg_addr1_base`	input	AW	Valid address range 1 base
`cfg_addr1_limit`	input	AW	Valid address range 1 limit

7.3.4 AXI4 Read Master Interface

Table 2.2.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
`ar_valid`	output	1	Address valid
`ar_ready`	input	1	Address ready
`ar_addr`	output	ADDR_WIDTH	Descriptor fetch address
`ar_len`	output	8	Burst length - 1
`ar_size`	output	3	Burst size (log2 bytes)
`ar_burst`	output	2	Burst type (INCR)
`ar_id`	output	AXI_ID_WIDTH	Transaction ID
`ar_lock`	output	1	Lock type
`ar_cache`	output	4	Cache

Signal	Direction	Width	Description
			attributes
ar_prot	output	3	Protection attributes
ar_qos	output	4	Quality of service
ar_region	output	4	Region identifier
r_valid	input	1	Data valid
r_ready	output	1	Data ready
r_data	input	256	Descriptor payload (fixed 256-bit)
r_resp	input	2	Read response
r_last	input	1	Last beat
r_id	input	AXI_ID_WIDTH	Response ID

7.3.5 Scheduler Interface

Table 2.2.6: Scheduler Interface

Signal	Direction	Width	Description
descriptor_valid	output	1	Descriptor valid
descriptor_ready	input	1	Scheduler ready
descriptor_packet	output	256	Parsed descriptor (fixed 256-bit)
descriptor_ext_packet	output	256	Extended chunk 1 (0 when unused)
descriptor_error	output	1	Error flag
descriptor_eos	output	1	End of Stream
descriptor_eol	output	1	End of Line
descriptor_eod	output	1	End of Data

Signal	Direction	Width	Description
`descriptor_type`	output	2	Packet type
`channel_idle`	input	1	Scheduler idle (enables APB kick-off)
`descriptor_engine_idle`	output	1	Engine idle

7.4 FSM States

7.4.1 Figure 2.2.2: Descriptor Engine FSM

States and transitions of the `descriptor_engine_beats` fetch FSM.

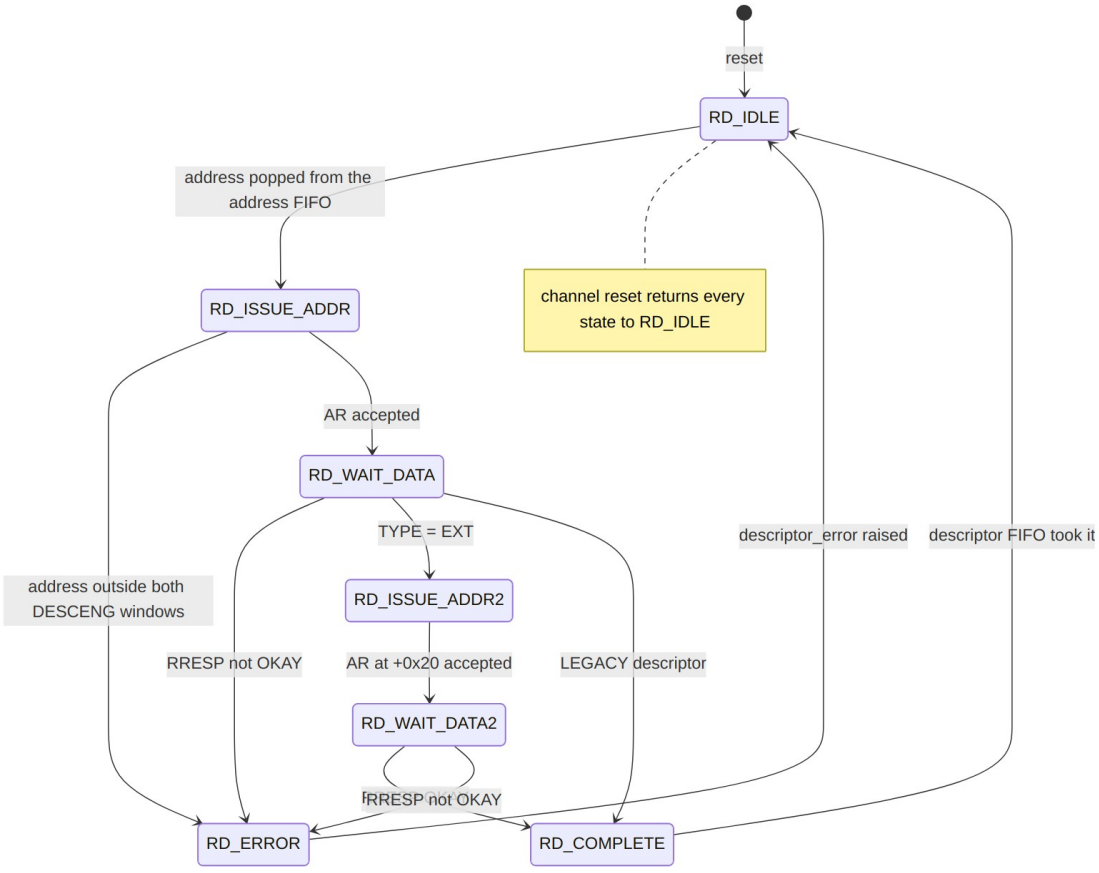
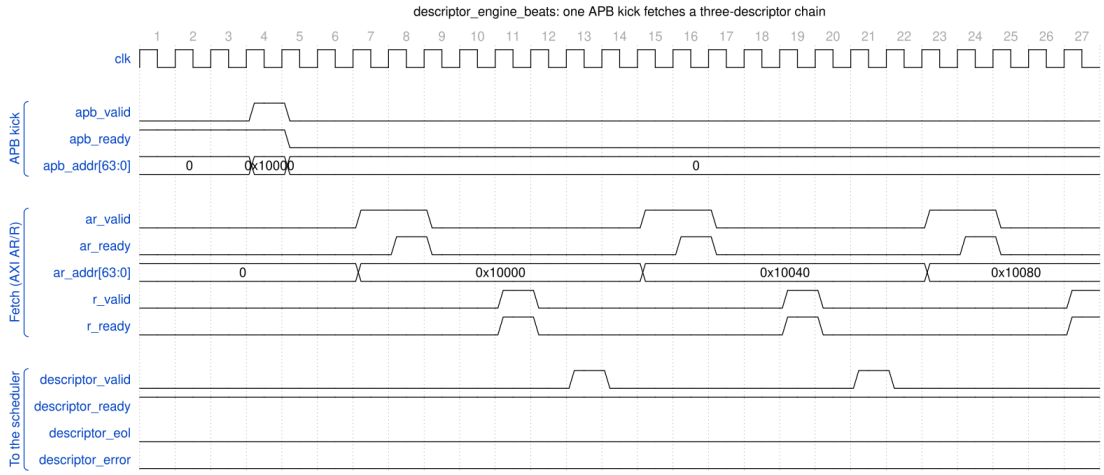


Diagram 1

7.5 Operation

7.5.1 Descriptor Chain Processing

7.5.2 Figure 2.2.3: Descriptor Chain Timing



Descriptor Engine - three-descriptor chain

Source: descriptor_engine_chain.json, captured from `dv/tests/fub_beats/test_descriptor_engine_beats.py` (`descriptor_chaining`, 32 channels, seed 7) with `WAVES=1`.

Reading it: one `apb_valid` kick carries the head address (0x10000). The engine issues an AXI read for it (`ar_valid`, `ar_addr`), the data returns on `r_valid`, and the parsed descriptor is presented on `descriptor_valid` to a ready scheduler. Because the descriptor's `next_ptr` points at 0x10040 and `last` is clear, the engine fetches the next one on its own – and the one after that at 0x10080 – with no further APB traffic; `descriptor_eol` marks the last of the chain. Each fetch costs the same AR -> R -> present latency, so a chain streams at one descriptor per fetch round trip.

7.6 Address Range Checking

Descriptors must fall within configured valid address ranges:

```
Valid if: (addr >= cfg_addr0_base && addr <= cfg_addr0_limit)
       || (addr >= cfg_addr1_base && addr <= cfg_addr1_limit)
```

Invalid addresses generate an error and halt the descriptor chain.

7.7 Prefetch and Autonomous Chaining

`cfg_prefetch_enable` and `cfg_fifo_threshold` are functional prefetch controls (previously declared-but-unused, which left chaining permanently greedy). They throttle how far ahead the engine fetches chained descriptors by comparing the output descriptor FIFO occupancy (`w_desc_fifo_count`) against a computed `w_prefetch_limit`:

```
if      (!cfg_prefetch_enable)      w_prefetch_limit = 1;   // on-
demand
else if (cfg_fifo_threshold == 0)   w_prefetch_limit = 1;   // guard
(0 would stall)
else                                w_prefetch_limit =
cfg_fifo_threshold;

w_prefetch_allows = (w_desc_fifo_count < w_prefetch_limit);
```

- **Prefetch OFF (`cfg_prefetch_enable = 0`):** limit forced to 1, so a chained fetch is only allowed when the output FIFO has drained – the engine stays roughly one descriptor ahead of the scheduler (on-demand).
- **Prefetch ON:** the engine buffers up to `cfg_fifo_threshold` descriptors ahead.

7.7.1 Chaining Decision

A chain fetch issues immediately when all of the following hold:

```
w_should_chain = w_chain_eligible     // valid non-last descriptor, in
range, no error
              && w_desc_committed      // current descriptor accepted
into output FIFO
              && w_prefetch_allows     // occupancy below prefetch
limit
              && w_desc_addr_fifo_wr_ready;  // address FIFO has a
slot
```

7.7.2 Deferred (Throttle-Blocked) Chain Fetch

If a descriptor commits and is chain-eligible but the fetch cannot issue this cycle (throttled by `w_prefetch_allows`, or the address FIFO is full), the owed fetch is recorded rather than dropped:

- `r_chain_pending` is set and the next-descriptor pointer is held in `r_pending_chain_addr`.
- The deferred fetch is reissued (`w_pending_push_fire`) once the output FIFO drains enough that `w_prefetch_allows` is true again and the address FIFO has a slot, clearing `r_chain_pending`.

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8 AXI Read Engine

Module: `axi_read_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-09-30

8.1 Overview

The AXI read engine turns per-channel read requests from the schedulers into AXI4 read bursts and routes the returning data to the SRAM controller. It is a streaming pipeline with no state machine. The channel identity travels in the AXI ID.

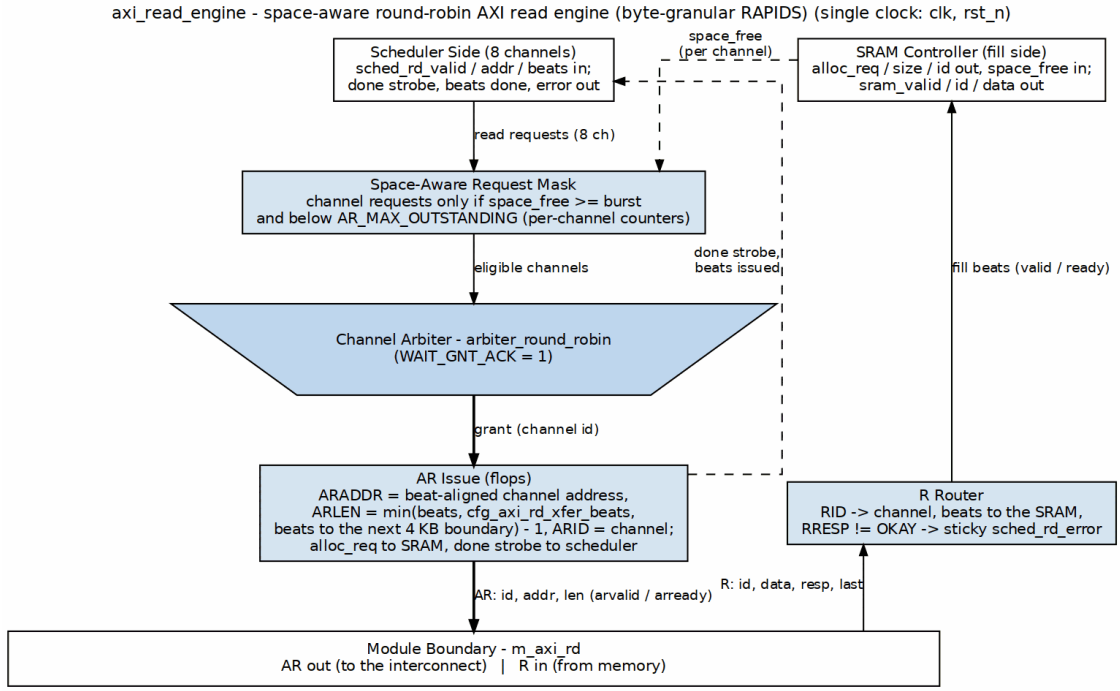
The engine is beat based, as in RAPIDS Beats. The scheduler has already converted the descriptor's byte length into a beat count. Two things change for byte-granular RAPIDS: the engine must cope with a channel address that is not beat aligned, and it must never let a burst cross a 4 KB boundary.

8.1.1 Key Features

- **Streaming pipeline:** no FSM, round-robin arbitration across channels.
- **Aligned AR address:** `ARADDR` is the channel address rounded down to a beat boundary. The lanes below the offset are read and ignored by the data path.
- **4 KB burst cap:** the burst length is limited so that no burst crosses a 4 KB boundary, whatever the start address.
- **Configurable burst length:** `cfg_axi_rd_xfer_beats`, clamped to what the buffer can hold (rapids BUG-009).
- **Space-checked:** a channel is only granted when the SRAM controller has room for the whole burst.
- **Sticky error:** SLVERR or DECERR on R sets `sched_rd_error` for the channel.

8.1.2 Block Diagram

8.1.3 Figure 2.3.1: AXI Read Engine Block Diagram



AXI read engine block diagram

Source: 05_axi_read_engine.dot

8.2 Parameters

Table 2.3.1: AXI Read Engine Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels. Channel id is carried in the AXI ID.
ADDR_WIDTH	64	AXI address width.
DATA_WIDTH	512	AXI data width in bits. The Genesys 2 build uses 256.
ID_WIDTH	8	AXI ID width.
SEG_COUNT_WIDTH	8	Width of the per-channel free-space count from the SRAM

Parameter	Default	Description
		controller. Sets the burst clamp.
PIPELINE	1	1 allows the next AR before the previous burst's data has returned. 0 waits for the burst to complete.
AR_MAX_OUTSTANDING	8	Maximum outstanding AR transactions.
STROBE_EVERY_BEAT	0	Declared for compatibility with STREAM. The completion strobe in this engine is generated on the AR handshake regardless of its value.

The module also accepts the short aliases NC, AW, DW, IW, SCW and CIW (the channel id width, $clog2(NUM_CHANNELS)).

8.2.1 Derived Constants

Table 2.3.2: AXI Read Engine Derived Constants

Constant	Definition	Meaning
BYTES_PER_BEAT	DW / 8	Bytes per beat.
AXSIZE	$clog2(BYTES_PER_BEAT)	Value driven on ARSIZE. Full-width beats.
SD_BEATS	1 << (SCW - 1)	Buffer depth in beats implied by SEG_COUNT_WIDTH.
XFER_MAX	SD_BEATS < 256 ? SD_BEATS - 1 : 254	Largest burst length value the engine will use.

8.3 Port List

8.3.1 Clock, Reset and Configuration

Table 2.3.3: Clock, Reset and Configuration

Signal	Direction	Width	Description
clk	input	1	Clock.
rst_n	input	1	Active-low reset.
cfg_axi_rd_xfer_beats	input	8	Configured burst length (ARLEN value, 0 to 255). Clamped to XFER_MAX.
cfg_channel_reset	input	NC	Per-channel reset, level or pulse. Stops the channel's ARs, drains and discards its in-flight R data, and clears its error flag.

8.3.2 Scheduler Interface

Table 2.3.4: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_valid	input	NC	Channel has a read request.
sched_rd_addr	input	NC x AW	Working source byte address.
sched_rd_beats	input	NC x 32	Beats still to read for the channel.
sched_rd_done_strobe	output	NC	Registered pulse, one cycle after the AR handshake for the channel.
sched_rd_beats_done	output	NC x 32	Beats issued by that AR (ARLEN + 1).
sched_rd_error	output	NC	Sticky read error for the channel, set by any non-OKAY R response.

The address is a byte address and may carry an offset in its low bits. The scheduler advances it to the next beat boundary after each done strobe, so only the first burst of a descriptor can see a non-zero offset.

8.3.3 SRAM Allocation and Fill Interface

Table 2.3.5: SRAM Allocation and Fill Interface

Signal	Direction	Width	Description
`axi_rd_alloc_req`	output	1	Registered pulse, one cycle after the AR handshake, reserving space for the burst.
`axi_rd_alloc_size`	output	8	Beats to reserve.
`axi_rd_alloc_id`	output	IW	Channel that owns the reservation.
`axi_rd_alloc_space_free`	input	NC x SCW	Free space per channel. A channel is granted only if it fits the burst.
`axi_rd_sram_valid`	output	1	Follows `m_axi_rvalid`.
`axi_rd_sram_ready`	input	1	SRAM controller accepts the beat. Drives `m_axi_rready`.
`axi_rd_sram_id`	output	IW	Follows `m_axi_rid`.
`axi_rd_sram_data`	output	DW	Follows `m_axi_rdata`.

8.3.4 AXI4 Read Master Interface

Table 2.3.6: AXI4 Read Master Interface

Signal	Direction	Width	Description
`m_axi_arid`	output	IW	Read address ID (the channel).
`m_axi_araddr`	output	AW	Beat-aligned read address.
`m_axi_arlen`	output	8	Burst length

Signal	Direction	Width	Description
			minus one.
m_axi_arsize	output	3	AXSIZE, full-width beats.
m_axi_arburst	output	2	INCR.
m_axi_arvalid	output	1	Address valid.
m_axi_arready	input	1	Address ready.
m_axi_rid	input	IW	Response ID.
m_axi_rdata	input	DW	Read data.
m_axi_rresp	input	2	Read response.
m_axi_rlast	input	1	Last beat of the burst.
m_axi_rvalid	input	1	Data valid.
m_axi_rready	output	1	Data ready.

8.3.5 Debug

Table 2.3.7: Debug Outputs

Signal	Direction	Width	Description
dbg_rd_all_complete	output	NC	Channel has no outstanding reads.
dbg_r_beats_rcvd	output	32	Total R beats received.
dbg_sram_writes	output	32	Total beats written to the SRAM controller.
dbg_arb_request	output	NC	Arbiter request vector.

8.4 Operation

8.4.1 Burst Length Cap (rapids BUG-009)

cfg_axi_rd_xfer_beats is an ARLEN value, but a burst must fit in the buffer that stages it. The engine clamps the configured value to XFER_MAX, giving the working limit w_xfer_cfg. Without the clamp an ARLEN of 255 wraps the 8-bit ARLEN + 1 and a channel is granted with no space reserved. This is the same fix as in RAPIDS Beats.

8.4.2 Byte-Granular Burst Sizing

The engine sizes each burst from the channel's current byte address and remaining beats. Three steps produce ARLEN:

1. Distance to the next 4 KB boundary, in beats: `w_beats_to_4k = (4096 - {sched_rd_addr[11:AXSIZE], AXSIZE'b0}) >> AXSIZE`. The offset bits are dropped first, so the distance is measured from the aligned beat.
2. Burst cap: `w_cap_beats = min(w_xfer_cfg + 1, w_beats_to_4k)`.
3. Burst length: `w_transfer_size = (sched_rd_beats <= w_cap_beats) ? sched_rd_beats - 1 : w_cap_beats - 1`.

`m_axi_araddr` is `{sched_rd_addr[grant][AW-1:AXSIZE], AXSIZE'b0}`. The offset never reaches the bus. The first beat of the burst carries the bytes before the offset, and the data path discards them.

Because the cap always rounds toward the boundary, a burst can be shorter than the configured length, and a later burst restarts on the boundary. The scheduler needs no knowledge of this. It only sees `beats_done` per burst.

8.4.3 Streaming Pipeline

An arbiter picks a channel that has `sched_rd_valid` set and enough `axi_rd_alloc_space_free` for its burst. When the AR handshakes, the engine pulses `axi_rd_alloc_req` one cycle later with the burst size so the SRAM controller reserves the space, and pulses `sched_rd_done_strobe` for the channel. R beats pass straight through: `axi_rd_sram_valid`, `axi_rd_sram_id` and `axi_rd_sram_data` follow the R channel, and `m_axi_rready` is `axi_rd_sram_ready`.

With `PIPELINE = 1` the next AR for a channel can issue while earlier data is still in flight. With `PIPELINE = 0` the channel waits for the burst to finish. The top and core default to 1.

8.4.4 Figure 2.3.2: Burst Segmentation with a Mid-Beat Start

The example uses 32-byte beats (OFF_W 5) and a configured burst of 16 beats. A descriptor reads 8,200 bytes from address 0xFF8. The scheduler computes offset 0x18 (24) and `beats_total = (24 + 8200 + 31) >> 5 = 257`.

Table 2.3.8: Burst Segmentation Example

Burst	Channel address	ARADDR	Beats to 4 KB	ARLEN	Beats left after
1	0x0FF8	0x0FE0	1	0	256
2	0x1000	0x1000	128	15	240
3	0x1200	0x1200	112	15	224
…				15	continues

The first burst is one beat because the aligned start is 32 bytes below the 4 KB boundary. From then on each burst is the configured 16 beats until the remainder runs out. The scheduler advances its address to the next beat boundary after each done strobe, which is why burst 2 starts at 0x1000, not 0x0FF8 plus a beat.

8.4.5 Done Strobe

The engine pulses `sched_rd_done_strobe[ch]` once per AR handshake, with `sched_rd_beats_done[ch]` equal to `ARLEN + 1`. The scheduler subtracts this from its remaining count and advances the address. The strobe therefore reports beats issued, not beats returned.

8.5 Error Handling

Table 2.3.9: Error Handling

Error	Detection	Response
AXI SLVERR	`m_axi_rresp == 2'b10` on an accepted beat	Set sticky `sched_rd_error` for the channel
AXI DECERR	`m_axi_rresp == 2'b11` on an accepted beat	Set sticky `sched_rd_error` for the channel
Any other non-zero response	`m_axi_rresp != 2'b00` on an accepted beat	Same: the test is not-OKAY, and the channel is taken from `RID`

The scheduler turns the sticky error into its ERROR state and a MonBus event. The engine does not abort the burst: the remaining R beats still pass through, so the interconnect is not left mid-burst.

8.5.1 Channel Reset

`cfg_channel_reset[ch]` clears the sticky `sched_rd_error[ch]`, so a channel reset recovers a read error without `aresetn`. The reset also has to settle AR bursts that are already on the bus, because an AXI master cannot recall them:

- A channel in reset, or flushing, issues no new AR and does not enter the allocation pipeline.
- `r_rd_flush[ch]` is set by the reset and holds until the channel has no burst outstanding. While it holds, each R beat whose `RID` names the channel is accepted (`m_axi_rready` is forced high) and dropped: `axi_rd_sram_valid` stays low, so nothing reaches the SRAM, and its `RRESP` does not set the error flag.

- The completion strobe and pending allocation of the channel are cleared, so the scheduler does not see a stale done pulse after the reset.
- Other channels are not affected: every term is indexed by channel, and R beats of other channels pass as usual.

The channel's SRAM partition is cleared by the SRAM controller wrapper (see Sink Data Path and Source Data Path AXIS). The discard is what guarantees that no beat of the reset channel reaches the partition after it has been cleared.

8.6 Differences from RAPIDS Beats

Table 2.3.10: Read Engine Delta

Item	RAPIDS Beats	RAPIDS
ARADDR	Channel address as given (always aligned)	Aligned down to the beat
Burst cap	Configured length and remaining beats	Also capped at the 4 KB boundary
Everything else		Unchanged

The beat-count chapter for the beats engine is AXI Read Engine (RAPIDS Beats). The scheduler that feeds this engine is described in Scheduler.

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9 AXI Write Engine

Module: `axi_write_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-09-30

9.1 Overview

The AXI write engine drains data from the SRAM controller and issues AXI4 write bursts to system memory. It is a streaming pipeline with no state machine. It also tracks the write responses, which is how the scheduler learns that data has been committed.

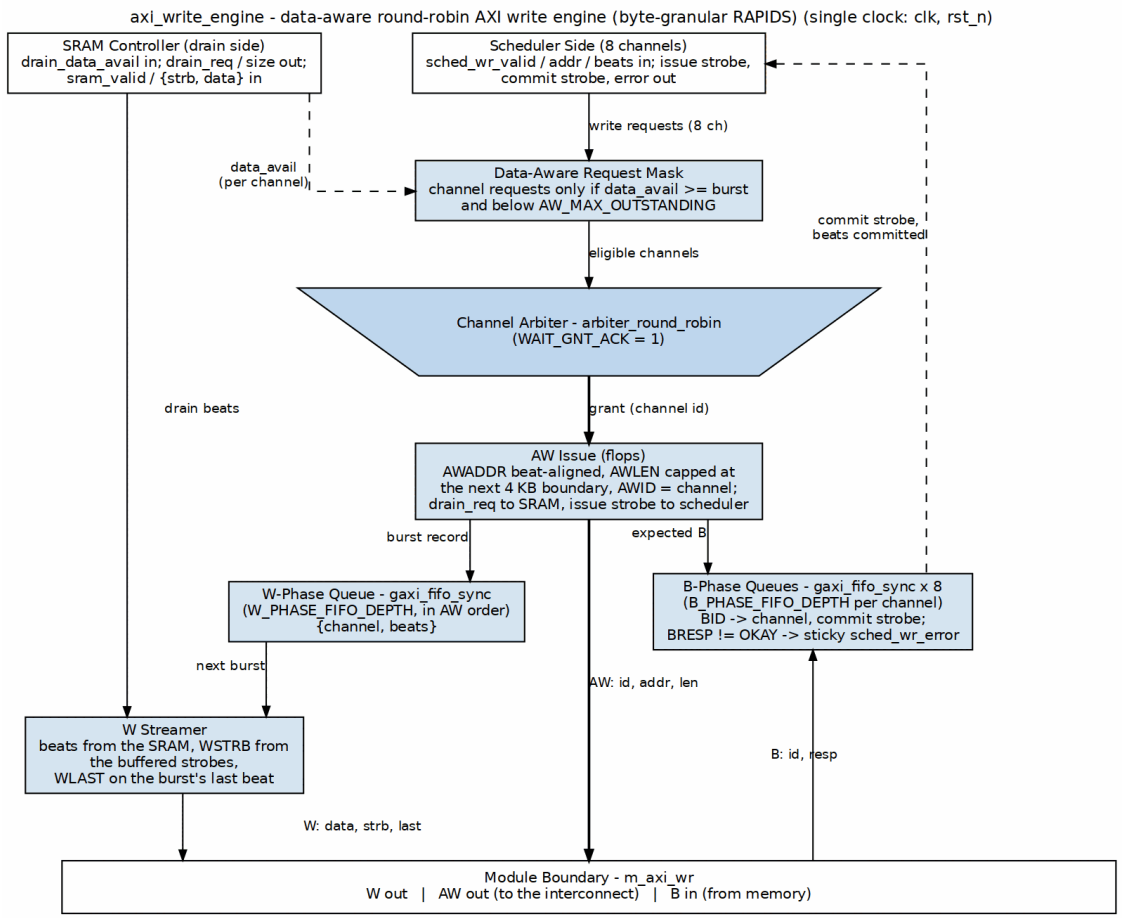
Byte-granular RAPIDS changes the engine in three places. Write strobes now come from the SRAM instead of being all ones, so a partial first or last beat writes only the bytes that belong to the packet. Burst lengths are capped at the 4 KB boundary. `AWADDR` is beat aligned.

9.1.1 Key Features

- **Streaming pipeline:** no FSM, round-robin arbitration across channels.
- **Real write strobes:** `m_axi_wstrb` is the strobe word stored beside each data beat (`axi_wr_sram_strb`). Lanes outside the packet are not written.
- **Aligned AW address:** `AWADDR` is the channel address rounded down to a beat boundary.
- **4 KB burst cap:** the same cap as the read engine, applied to both the burst length and the final-burst test.
- **Configurable burst length:** `cfg_axi_wr_xfer_beats`, clamped to what the buffer can hold (rapids BUG-009).
- **Two completion strobes:** done on the AW handshake, commit on the B response.
- **Sticky error:** a non-OKAY B response sets `sched_wr_error`.

9.1.2 Block Diagram

9.1.3 Figure 2.4.1: AXI Write Engine Block Diagram



AXI write engine block diagram

Source: 06_axi_write_engine.dot

9.2 Parameters

Table 2.4.1: AXI Write Engine Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels. Channel id is carried in the AXI ID.
ADDR_WIDTH	64	AXI address width.
DATA_WIDTH	512	AXI data width in bits. The

Parameter	Default	Description
		Genesys 2 build uses 256.
ID_WIDTH	8	AXI ID width.
USER_WIDTH	8	Width of m_axi_wuser, which carries the active channel.
SEG_COUNT_WIDTH	8	Width of the per-channel available-data count from the SRAM controller. Sets the burst clamp.
PIPELINE	1	1 allows further AWs before earlier B responses return. 0 waits for the B.
AW_MAX_OUTSTANDING	8	Maximum outstanding AW transactions per channel.
W_PHASE_FIFO_DEPTH	64	Depth of the W-phase transaction FIFO, kept in order with the AW issue.
B_PHASE_FIFO_DEPTH	16	Depth of the B-phase transaction FIFO used to match responses.

The module also accepts the short aliases NC, AW, DW, IW, UW, SCW and CIW (the channel id width).

9.2.1 Derived Constants

Table 2.4.2: AXI Write Engine Derived Constants

Constant	Definition	Meaning
BYTES_PER_BEAT	DW / 8	Bytes per beat.
AXSIZE	$clog2(BYTES_PER_BEAT)	Value driven on AWSIZE.
XFER_MAX	min(2^(SCW-1) - 1, 254)	Largest burst length value the engine will use.

9.3 Port List

9.3.1 Clock, Reset and Configuration

Table 2.4.3: Clock, Reset and Configuration

Signal	Direction	Width	Description
`clk`	input	1	Clock.
`rst_n`	input	1	Active-low reset.
`cfg_axi_wr_xfer_beats`	input	8	Configured burst length (AWLEN value, 0 to 255). Clamped to `XFER_MAX`.
`cfg_channel_reset`	input	NC	Per-channel reset, level or pulse. Stops new bursts, completes open bursts with null beats, and clears the channel's error flag.

9.3.2 Scheduler Interface

Table 2.4.4: Scheduler Interface

Signal	Direction	Width	Description
`sched_wr_valid`	input	NC	Channel has a write request.
`sched_wr_ready`	output	NC	Pulses when the B response of the last burst of a descriptor arrives.
`sched_wr_addr`	input	NC x AW	Working destination byte address.
`sched_wr_beats`	input	NC x 32	Beats still to write for the channel.
`sched_wr_burst_len`	input	NC x 8	Requested burst length. Present on the port; the engine sizes bursts from `cfg_axi_wr_xfer_beats` and the 4 KB cap.
`sched_wr_don`	output	NC	Pulses on the AW

Signal	Direction	Width	Description
e_strobe			handshake for the channel.
sched_wr_beats_done	output	NC x 32	Beats issued by that AW (AWLEN + 1).
sched_wr_commit_strobe	output	NC	Pulses when the B response for the channel arrives.
sched_wr_commit_beats	output	NC x 32	Beats confirmed by that B response, recovered from the B-phase FIFO.
sched_wr_error	output	NC	Sticky write error for the channel.

The done strobe advances the scheduler's destination address, which lets the next AW issue without waiting for a response. The commit strobe is what the scheduler counts when it decides the descriptor is complete. Both strobes are per channel and report beat counts, not bytes.

9.3.3 SRAM Reservation and Drain Interface

Table 2.4.5: SRAM Drain Interface

Signal	Direction	Width	Description
axi_wr_drain_req	output	NC	Pulses for the granted channel on the AW handshake.
axi_wr_drain_size	output	NC x 8	Beats to reserve (AWLEN + 1).
axi_wr_drain_data_avail	input	NC x SCW	Data available per channel after reservations.
axi_wr_sram_valid	input	NC	Per-channel valid, registered. Used for arbitration.
axi_wr_sram_valid_comb	input	NC	Per-channel valid, combinational. Gates m_axi_wvalid.
axi_wr_sram_	output	1	Drain request. Equal to

Signal	Direction	Width	Description
drain			wvalid && wready.
axi_wr_sram_id	output	CIW	Channel selected for the drain.
axi_wr_sram_data	input	DW	Data from the selected channel.
axi_wr_sram_strb	input	DW/8	Byte strobes stored with that data beat. New in RAPIDS.

9.3.4 AXI4 Write Master Interface

Table 2.4.6: AXI4 Write Master Interface

Signal	Direction	Width	Description
m_axi_awid	output	IW	Write address ID (the channel).
m_axi_awaddr	output	AW	Beat-aligned write address.
m_axi_awlen	output	8	Burst length minus one.
m_axi_awsize	output	3	AXSIZE, full-width beats.
m_axi_awburst	output	2	INCR.
m_axi_awvalid	output	1	Address valid.
m_axi_awready	input	1	Address ready.
m_axi_wdata	output	DW	Write data.
m_axi_wstrb	output	DW/8	Write strobes, equal to axi_wr_sram_strb.
m_axi_wlast	output	1	Last beat of the burst.
m_axi_wuser	output	UW	Channel of the beat being written.
m_axi_wvalid	output	1	Data valid.
m_axi_wready	input	1	Data ready.
m_axi_bid	input	IW	Response ID.
m_axi_bresp	input	2	Write response.

Signal	Direction	Width	Description
m_axi_bvalid	input	1	Response valid.
m_axi_bready	output	1	Response ready.

9.3.5 Debug and Status

Table 2.4.7: Debug and Status Outputs

Signal	Direction	Width	Description
dbg_wr_all_complete	output	NC	Channel has no outstanding writes.
dbg_aw_transactions	output	32	Total AW handshakes.
dbg_w_beats	output	32	Total W beats.
o_active_channel_id	output	CIW	Channel whose burst is driving, or about to drive, the W bus.
o_active_channel_valid	output	1	o_active_channel_id is meaningful.

9.4 Operation

9.4.1 Burst Length Cap (rapids BUG-009)

cfg_axi_wr_xfer_beats is an AWLEN value, but a burst cannot be larger than the SRAM that stages it: the engine waits for the whole burst to be present before issuing the AW, and a burst the buffer cannot hold would wait forever. The engine clamps the configured value to XFER_MAX. Without the clamp an AWLEN of 255 wraps the 8-bit AWLEN + 1 and the engine issues an AW on an empty buffer. This is the same fix as in RAPIDS Beats.

9.4.2 Byte-Granular Burst Sizing

The sizing follows the read engine:

1. w_beats_to_4k is the distance from the aligned channel address to the next 4 KB boundary, in beats.
2. w_cap_beats = min(w_xfer_cfg + 1, w_beats_to_4k).
3. w_transfer_size (the AWLEN) is sched_wr_beats - 1 if sched_wr_beats <= w_cap_beats, otherwise w_cap_beats - 1.

A channel may issue when the buffer holds at least `w_transfer_size + 1` beats (`w_has_data`), or when this is the final burst. The final-burst test is `sched_wr_beats > 0`, `sched_wr_beats <= w_cap_beats`, and the available data covers all remaining beats. The 4 KB cap therefore applies to both tests, so a descriptor that ends just past a boundary still finishes on a short final burst.

`m_axi_awaddr` is `{sched_wr_addr[ch][AW-1:AXSIZE], AXSIZE'b0}`. The offset of a descriptor that starts mid-beat never reaches the bus. The first W beat of that descriptor carries `WSTRB` bits only for the bytes at and above the offset, and the last carries only the bytes up to the end of the packet.

9.4.3 Write Strobes

Each SRAM word holds a data beat and its strobe word side by side (see Sink Data Path). The write engine takes both from the drain port and presents them together, so the strobe always matches the data. In RAPIDS Beats `WSTRB` was tied high.

9.4.4 Streaming Pipeline

Each channel with `sched_wr_valid` and a burst that passes the data test requests arbitration. On grant the AW is issued, and on the AW handshake the engine pulses `axi_wr_drain_req` with the burst size so the drain controller reserves the beats. W beats stream from the drain port that `axi_wr_sram_id` selects. `axi_wr_sram_drain` equals `m_axi_wvalid && m_axi_wready`. `m_axi_wlast` closes each burst, and `m_axi_wuser` carries the channel.

With `PIPELINE = 1` a channel may issue further AWs before earlier B responses return, up to `AW_MAX_OUTSTANDING`. With `PIPELINE = 0` the channel waits for each B. That costs throughput, so the top and core default to 1.

9.4.5 Two Completion Strobes

Table 2.4.8: Completion Strobes

Strobe	Raised when	Scheduler use
`sched_wr_done_strobe`	AW handshake for the channel	Advance the destination address and the issued-beat count.
`sched_wr_commit_strobe`	B response for the channel	Count committed beats. The descriptor is complete when all beats are committed.

`sched_wr_ready` is a third, coarser pulse: it fires when the B response belongs to the last burst of a descriptor.

9.5 Error Handling

Table 2.4.9: Error Handling

Error	Detection	Response
AXI SLVERR	`m_axi_bresp == 2'b10`	Set sticky `sched_wr_error` for the channel
AXI DECERR	`m_axi_bresp == 2'b11`	Set sticky `sched_wr_error` for the channel
Any other non-zero response	`m_axi_bresp != 2'b00`	Same: the test is not-OKAY, and the channel is taken from `BID`

The sink data path ORs `sched_wr_error` with its own packet-length error before passing it to the scheduler.

9.5.1 Channel Reset

`cfg_channel_reset[ch]` clears the sticky `sched_wr_error[ch]`, so a channel reset recovers a B-response error without `aresetn`. A burst whose AW has been accepted must still receive all of its W beats, so the engine settles in-flight work rather than dropping it:

- `w_kill[ch]` is the reset level ORed with `r_wr_flush[ch]`. The flush bit is set by the reset and holds until the channel has no open burst and no B response due.
- No new AW is arbitrated for a channel with `w_kill` set. `AWADDR` comes from `sched_wr_addr`, which the scheduler does not clear, so the value on the bus stays stable across the reset.
- An open burst is completed with null W beats: `WSTRB` is zero, `WDATA` repeats the last captured value and the SRAM drain is not popped, so the null beats modify no byte at the destination. A beat that was presented on W and not accepted when the reset hit is the one exception: it is replayed unchanged, with its own strobes, from a captured copy, so the W channel does not change its payload while `WVALID` is high. At most one beat per reset is replayed this way.
- B responses of a channel with `w_kill` set are consumed. The commit strobe, `sched_wr_ready`, the done strobe and the error flag are suppressed for them.
- Other channels are not affected.

The reset therefore leaves no partial state in the engine: the B-phase FIFO drains through the flush, and the channel restarts with clean counters when the scheduler next raises `sched_wr_valid`.

9.6 Differences from RAPIDS Beats

Table 2.4.10: Write Engine Delta

Item	RAPIDS Beats	RAPIDS
`WSTRB`	All ones	`axi_wr_sram_strb`
SRAM drain port	Data only	Data plus strobe word
`AWADDR`	Channel address as given (always aligned)	Aligned down to the beat
Burst cap and final-burst test	Configured length and remaining beats	Also capped at the 4 KB boundary
Everything else		Unchanged

The beats chapter is AXI Write Engine (RAPIDS Beats). The read-side counterpart is AXI Read Engine.

Last Updated: 2026-09-30

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10 Beats Alloc Control Specification

Module: `alloc_ctrl_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

> **No longer in the SRAM datapath (2026-09-26).** RAPIDS' SRAM path now uses STREAM's `stream_alloc_ctrl` / `stream_drain_ctrl` / `stream_latency_bridge`, reached through the `snk_/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and still verified – it keeps its own test, its own filelist and its `rapids_all.f` entry – it simply is not instantiated by the SRAM controllers any more. Treat this page as the FUB's own specification, not as a description of the current SRAM path.

10.1 Overview

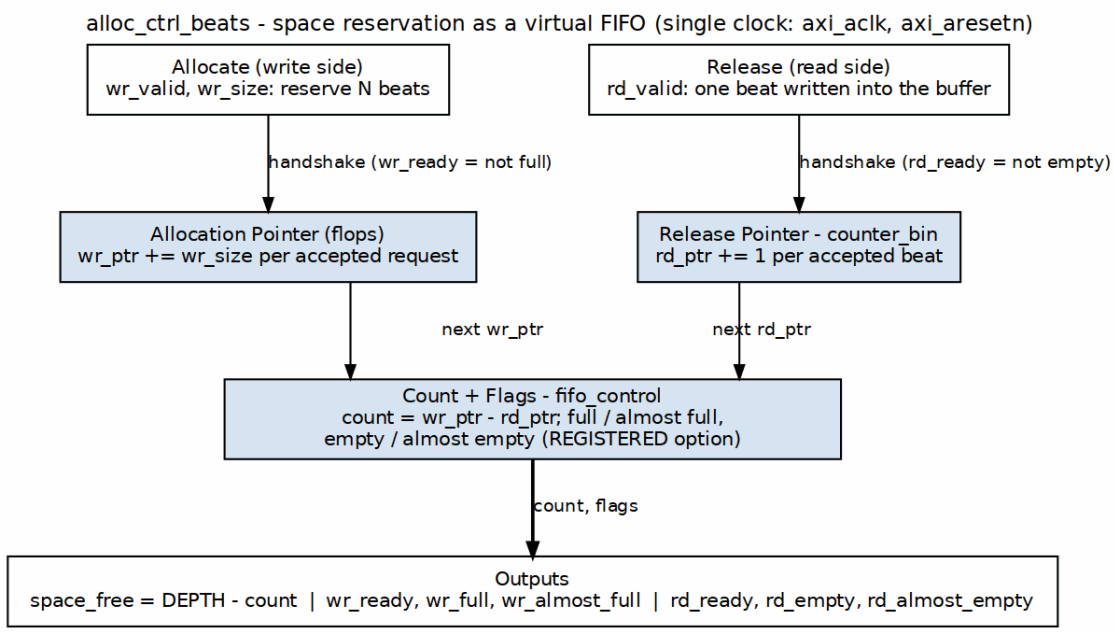
The Beats Alloc Control is a "virtual FIFO" that tracks space allocation without storing actual data. It uses FIFO pointer logic to manage pre-allocation for AXI engines, preventing race conditions between allocation and actual data arrival.

10.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Variable-Size Allocation:** Supports multi-beat allocation requests
- **Single-Beat Release:** Data writes release one beat at a time
- **Status Flags:** Full, almost full, empty, almost empty
- **Space Tracking:** Reports available space for flow control

10.1.2 Block Diagram

10.1.3 Figure 2.5.1: Beats Alloc Control Block Diagram



Alloc control block diagram

Source: 05_alloc_ctrl.dot

10.2 Concept: Virtual FIFO

Unlike a traditional FIFO that stores data, alloc_ctrl_beats only tracks **space reservations**:

	Traditional FIFO	Virtual FIFO (alloc_ctrl_beats)
Write side	`wr_data` into storage	`wr_size` advances a pointer
Read side	`rd_data` out of storage	one beat advances a pointer
Output	data	`space_free`

10.2.1 Use Case: Pre-Allocation for AXI Reads

```
1. AXI engine requests N beats of space
   - alloc_ctrl.wr_valid = 1
   - alloc_ctrl.wr_size = N
```

```
   - wr_ptr advances by N

2. As AXI read data arrives (1 beat at a time)
   - alloc_ctrl.rd_valid = 1 (each beat)
   - rd_ptr advances by 1

3. Space becomes available again
   - space_free reflects available capacity
```

10.3 Parameters

```
parameter int DEPTH = 512;                    // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;           // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;           // Almost empty margin
parameter int REGISTERED = 1;                 // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);             // Address width
```

: Table 2.5.1: Beats Alloc Control Parameters

10.4 Port List

10.4.1 Clock and Reset

Table 2.5.2: Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low reset

10.4.2 Write Interface (Allocation Requests)

Table 2.5.3: Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	Allocate space
wr_size	input	8	Number of beats to allocate

Signal	Direction	Width	Description
wr_ready	output	1	Space available for allocation

10.4.3 Read Interface (Data Written)

Table 2.5.4: Read Interface

Signal	Direction	Width	Description
rd_valid	input	1	One beat of data written
rd_ready	output	1	Not full (can accept)

10.4.4 Status Outputs

Table 2.5.5: Status Outputs

Signal	Direction	Width	Description
space_free	output	AW+1	Available space in beats
wr_full	output	1	No space available
wr_almost_full	output	1	Near full
rd_empty	output	1	No allocations pending
rd_almost_empty	output	1	Near empty

10.5 Operation

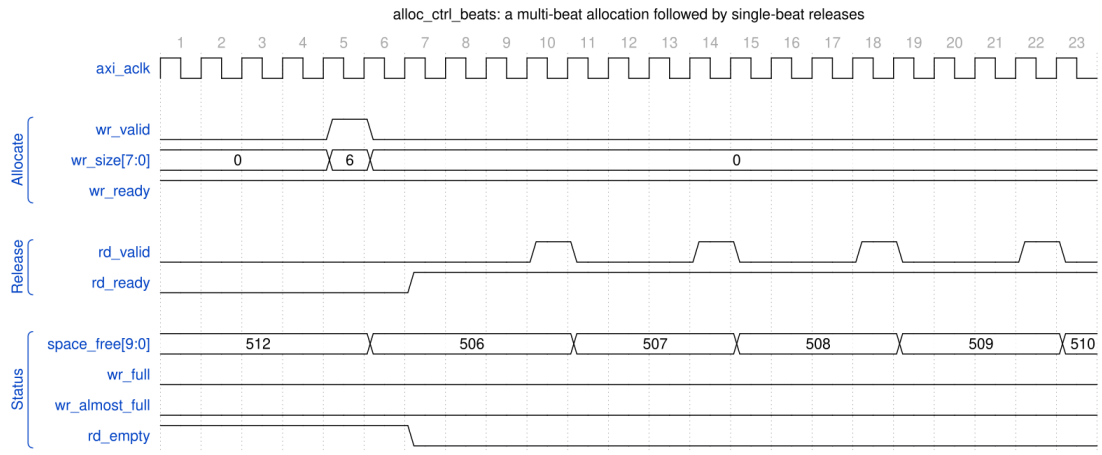
10.5.1 Pointer Arithmetic

```
space_free = DEPTH - (wr_ptr - rd_ptr)

Full condition:    wr_ptr - rd_ptr >= DEPTH
Empty condition:   wr_ptr == rd_ptr
```

10.5.2 Timing Diagram

10.5.3 Figure 2.5.2: Allocation and Release Timing



Allocation control - allocate then release

Source: alloc_ctrl_beats_alloc_release.json, captured from `dv/tests/fub_beats/test_alloc_ctrl_beats.py` (`basic_alloc_drain`, depth 512, seed 7) with `WAVES=1`.

Reading it: one `wr_valid` handshake with `wr_size = 6` reserves six beats and `space_free` drops 512 to 506 the next cycle; `rd_empty` falls because reserved space is now outstanding. Each later `rd_valid` handshake releases one beat and `space_free` climbs back one at a time (507, 508, ...). `wr_full` and `wr_almost_full` stay low: the virtual FIFO is nowhere near its depth.

10.6 Integration Example

```
// Source path: Pre-allocate SRAM space before AXI read
alloc_ctrl_beats #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_WR_MARGIN(8)
) u_src_alloc (
    .axi_aclk       (clk),
    .axi_aresetn    (rst_n),

    // Scheduler allocates space before issuing AXI read
    .wr_valid       (sched_rd_valid),
    .wr_size        (sched_rd_beats[7:0]),
    .wr_ready       (alloc_ready),

    // AXI read data arrival releases allocation
```

```
    .rd_valid       (m_axi_rvalid && m_axi_rready),
    .rd_ready       (),  // Not used for single-beat

    // Status
    .space_free     (sram_space_available),
    .wr_full        (sram_full),
    .wr_almost_full (sram_almost_full)
);
```

Last Updated: 2025-01-10

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11 Beats Drain Control Specification

Module: `drain_ctrl_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

> **No longer in the SRAM datapath (2026-09-26).** RAPIDS' SRAM path now uses STREAM's `stream_alloc_ctrl` / `stream_drain_ctrl` / `stream_latency_bridge`, reached through the `snk_/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and still verified – it keeps its own test, its own filelist and its `rapids_all.f` entry – it simply is not instantiated by the SRAM controllers any more. Treat this page as the FUB's own specification, not as a description of the current SRAM path.

11.1 Overview

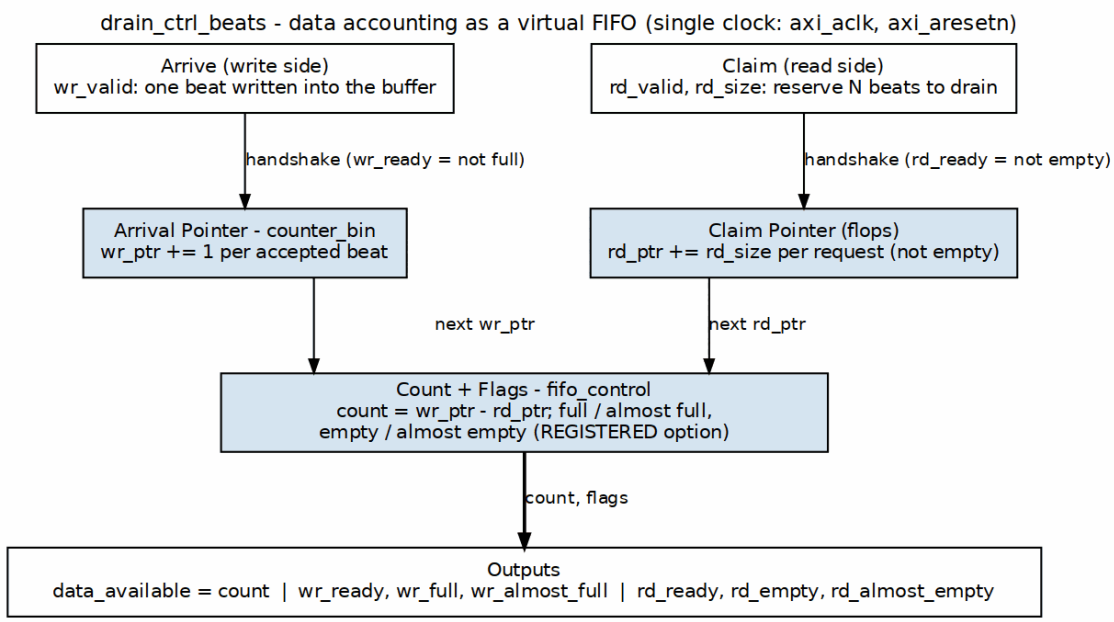
The Beats Drain Control is a "virtual FIFO" that tracks data availability without storing actual data. It complements alloc_ctrl_beats by tracking when data has been written to SRAM and is available for draining.

11.1.1 Key Features

- **Virtual FIFO:** Pointer tracking without data storage
- **Single-Beat Write:** Data arrivals tracked one beat at a time
- **Variable-Size Drain:** Supports multi-beat drain requests
- **Status Flags:** Full, almost full, empty, almost empty
- **Data Availability:** Reports beats available for drain

11.1.2 Block Diagram

11.1.3 Figure 2.6.1: Beats Drain Control Block Diagram



Drain control block diagram

Source: 06_drain_ctrl.dot

11.2 Concept: Data Availability Tracking

The drain_ctrl tracks when data has been written to SRAM:

```
alloc_ctrl:  Tracks SPACE reservations (allocation -> actual write)
drain_ctrl:  Tracks DATA availability (data write -> drain request)
```

11.2.1 Complementary Roles

```
Source Path:
  1. Scheduler allocates space (alloc_ctrl.wr)
  2. AXI reads data into SRAM
  3. Data written to SRAM (drain_ctrl.wr)
  4. Drain consumer reads (drain_ctrl.rd)

Sink Path:
  1. Fill allocates space (alloc_ctrl.wr)
  2. Fill writes to SRAM (alloc_ctrl.rd)
  3. Data available in SRAM (drain_ctrl.wr)
  4. AXI write drains SRAM (drain_ctrl.rd)
```

11.3 Parameters

```
parameter int DEPTH = 512;                    // Virtual FIFO depth
parameter int ALMOST_WR_MARGIN = 1;           // Almost full margin
parameter int ALMOST_RD_MARGIN = 1;           // Almost empty margin
parameter int REGISTERED = 1;                 // Registered outputs

// Derived
parameter int AW = $clog2(DEPTH);             // Address width
```

: Table 2.6.1: Beats Drain Control Parameters

11.4 Port List

11.4.1 Clock and Reset

Table 2.6.2: Clock and Reset

Signal	Direction	Width	Description
axi_aclk	input	1	System clock
axi_aresetn	input	1	Active-low reset

11.4.2 Write Interface (Data Written to SRAM)

Table 2.6.3: Write Interface

Signal	Direction	Width	Description
wr_valid	input	1	One beat written to SRAM
wr_ready	output	1	Can track more data

11.4.3 Read Interface (Drain Requests)

Table 2.6.4: Read Interface

Signal	Direction	Width	Description
rd_valid	input	1	Drain request

Signal	Direction	Width	Description
rd_size	input	8	Number of beats to drain
rd_ready	output	1	Data available for drain

11.4.4 Status Outputs

Table 2.6.5: Status Outputs

Signal	Direction	Width	Description
data_available	output	AW+1	Beats available to drain
wr_full	output	1	Tracking full
wr_almost_full	output	1	Near full
rd_empty	output	1	No data to drain
rd_almost_empty	output	1	Near empty

11.5 Operation

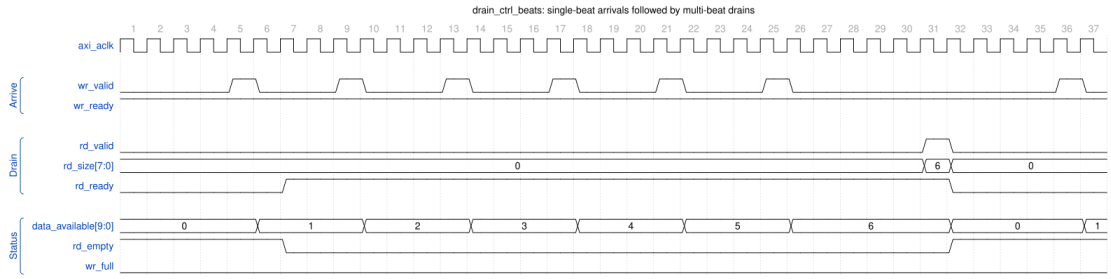
11.5.1 Pointer Arithmetic

```
data_avail = wr_ptr - rd_ptr

Full condition:    wr_ptr - rd_ptr >= DEPTH
Empty condition:   wr_ptr == rd_ptr
```

11.5.2 Timing Diagram

11.5.3 Figure 2.6.2: Data Arrival and Drain Timing



Drain control - single-beat arrivals then a multi-beat drain

Source: drain_ctrl_beats_arrive_drain.json, captured from `dv/tests/fub_beats/test_drain_ctrl_beats.py` (`basic_write_drain`, depth 512, seed 7) with `WAVES=1`.

Reading it: six `wr_valid` handshakes, one beat each, raise `data_available` 1, 2, … 6 and drop `rd_empty` on the first arrival. A single `rd_valid` handshake with `rd_size = 6` then reserves all six at once: `data_available` returns to 0 and `rd_empty` rises the next cycle. The counter tracks beats, not bytes, and never touches the data itself.

11.6 Integration Example

```
// Sink path: Track data availability for AXI write engine
drain_ctrl_beats #(
    .DEPTH(SRAM_DEPTH),
    .ALMOST_RD_MARGIN(8)
) u_snk_drain (
    .axi_aclk       (clk),
    .axi_aresetn    (rst_n),

    // Fill writes data to SRAM
    .wr_valid       (fill_valid && fill_ready),
    .wr_ready       (),  // Single-beat, always ready

    // AXI write engine drains data
    .rd_valid       (axi_wr_drain_req),
    .rd_size        (axi_wr_burst_len),
    .rd_ready       (data_ready_for_drain),

    // Status
    .data_available (beats_to_write),
```

```
    .rd_empty       (no_data_to_write)
);
```

11.7 Relationship: alloc_ctrl vs drain_ctrl

Table 2.6.6: alloc_ctrl vs drain_ctrl Comparison

Aspect	alloc_ctrl_beats	drain_ctrl_beats
Tracks	Space reservations	Data availability
Write port	Multi-beat allocation	Single-beat arrival
Read port	Single-beat release	Multi-beat drain
Output	`space_free`	`data_available`
Use case	Pre-allocate before fill	Know when to drain

Last Updated: 2025-01-10

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12 Latency Bridge (beats) Specification

Module: `latency_bridge_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub_beats/` **Status:** Implemented, tested, not in the SRAM path **Last Updated:** 2026-09-27

> **Not in the SRAM datapath since 2026-09-26.** RAPIDS' SRAM path is STREAM's `sram_controller`, which carries its own `stream_latency_bridge` (the same design), reached through the `snk_/src_sram_controller_beats` naming wrappers (`bdf4e0dff`). This FUB is still built and verified – it keeps its test, its filelist and its `rapids_all.f` entry – but nothing in `rapids_core_beats` instantiates it. This page is the FUB's own specification.

12.1 Overview

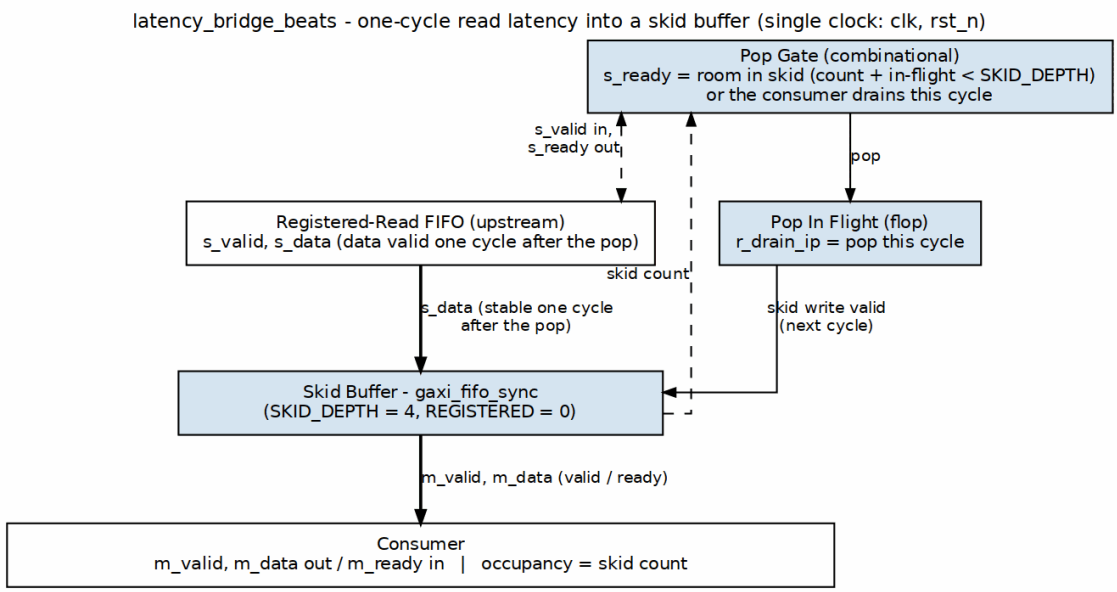
`latency_bridge_beats` turns a **registered-read FIFO** – one whose data appears the cycle *after* the read handshake – into a plain **valid/ready stream** the consumer can stall at will. It is a data-width bridge: it moves `DATA_WIDTH` bits per beat and carries nothing else. There is no beat count, no channel ID and no programmable delay in it; an earlier version of this page described such an interface, and it never existed in this module.

12.1.1 Key Features

- **One-cycle glue, one skid buffer:** a single flop (`r_drain_ip`) remembers that a FIFO read was accepted, so the data that lands the next cycle is pushed into a `gaxi_fifo_sync` skid of depth `SKID_DEPTH`
- **Full throughput when the consumer is ready:** back-to-back reads are accepted as long as the skid keeps draining
- **Backpressure absorbed in the skid:** `s_ready` drops only when a skid write is actually stalled, so the FIFO is never read into a slot that does not exist
- **Occupancy reported:** `occupancy` counts the beats the bridge holds (the one in flight plus the skid contents), which is what an occupancy-based data-available counter needs
- **Debug taps:** `dbg_r_pending` and `dbg_r_out_valid` expose the in-flight flop and the skid output valid, for catching stuck data from a test

12.1.2 Block Diagram

12.1.3 Figure 2.7.1: Latency Bridge Block Diagram



Latency bridge block diagram

Source: 07_latency_bridge.dot

12.2 Concept: bridging a registered read

A FIFO with `REGISTERED=1` read (the SRAM style this repo uses so BRAM infers cleanly) answers a read handshake with data **one cycle later**. A consumer speaking valid/ready expects data *with* valid, and may deassert ready at any time. Without a bridge the FIFO has already popped a beat the consumer did not take.

```
cycle 0   s_valid && s_ready          FIFO read accepted
r_drain_ip <= 1
cycle 1   data arrives on s_data      skid_wr_valid = r_drain_ip, data
pushed
cycle N   m_valid && m_ready          consumer drains the skid at its
own pace
```

The bridge never guesses about downstream readiness: `s_ready` is derived from the skid's room, counting only a write that is *stalled* this cycle as pending (a write that completes this cycle frees its slot at the same edge). That is why full throughput holds while writes complete immediately, and why the FIFO is never over-read when they do not.

12.3 Parameters

```
parameter int DATA_WIDTH = 64;    // Beat width
parameter int SKID_DEPTH = 4;     // Skid buffer depth (2-4
recommended)
parameter int DW = DATA_WIDTH;    // Short alias
```

: Table 2.7.1: Latency Bridge Parameters

`occupancy` is 3 bits wide and counts 0..5 at the default depth (one beat in flight plus four in the skid); a larger `SKID_DEPTH` needs the port widened.

12.4 Port List

12.4.1 Clock and Reset

Table 2.7.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

12.4.2 Upstream Interface (from the registered FIFO)

Table 2.7.3: Upstream Interface

Signal	Direction	Width	Description
`s_valid`	input	1	FIFO has a beat to read (not empty)
`s_ready`	output	1	Bridge accepts a read this cycle; the data lands NEXT cycle
`s_data`	input	DW	Read data, valid one cycle after the `s_valid && s_ready` handshake

12.4.3 Downstream Interface (to the consumer)

Table 2.7.4: Downstream Interface

Signal	Direction	Width	Description
`m_valid`	output	1	Beat available

Signal	Direction	Width	Description
m_ready	input	1	Consumer takes the beat
m_data	output	DW	Beat data, valid with m_valid

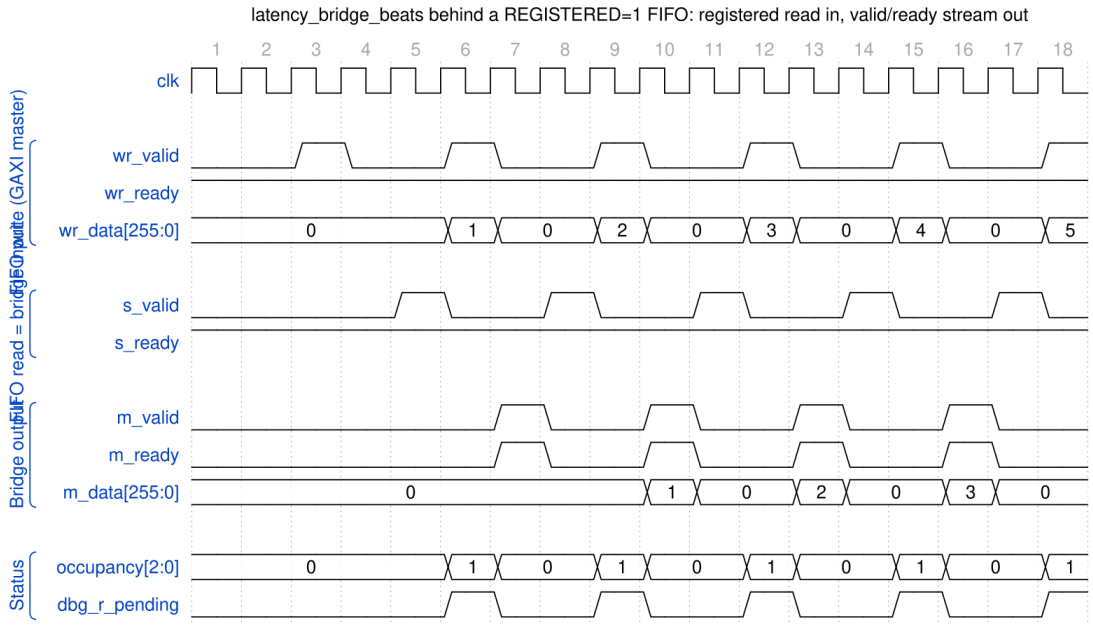
12.4.4 Status and Debug

Table 2.7.5: Status and Debug

Signal	Direction	Width	Description
occupancy	output	3	Beats held by the bridge: the in-flight read plus the skid contents (0..5)
dbg_r_pending	output	1	r_drain_ip: a read was accepted and its data is due next cycle
dbg_r_out_valid	output	1	Skid output valid (mirrors m_valid)

12.5 Operation

12.5.1 Figure 2.7.2: Latency Bridge Timing (consumer always ready)



Latency bridge behind a registered-read FIFO

Source: latency_bridge_beats_streaming.json, captured from `dv/tests/fub_beats/test_latency_bridge_beats.py` (`streaming`, 256-bit, seed 7) with `WAVES=1`. The test drives a `REGISTERED = 1` `gaxi_fifo_sync` in front of the bridge (`dv/tb/latency_bridge_beats_tb_top.sv`), because that is the only way to produce the bridge's upstream contract: a valid/ready master presents data *with* valid, a registered FIFO presents it one cycle *after* the read handshake.

Reading it: each `wr_valid` beat lands in the FIFO and `s_valid` (FIFO not empty) rises a cycle later; the bridge reads it at once (`s_ready` stays high) and `dbg_r_pending` marks the cycle the data is in flight. Two cycles after the write the beat is on `m_valid`/`m_data` with its payload intact (1, 2, 3, …), and `occupancy` shows exactly one beat inside the bridge per transfer. With a consumer that is always ready the bridge adds the FIFO's read latency and no more.

An earlier revision of this test drove `s_*` directly from the BFM and never compared `m_data`; every beat came out 0 and the test passed (rapids TASK-003). (`dv/tests/fub_beats/test_latency_bridge_beats.py`).

12.6 Integration Context

The bridge sits between a `gaxi_fifo_sync` with `REGISTERED=1` and whatever consumes the stream. The same arrangement, with STREAM's twin `stream_latency_bridge`, is what `sram_controller_unit` uses inside the shared SRAM controller today; `occupancy` there is what the drain accounting must allow for (stream BUG-011).

```
latency_bridge_beats #(
    .DATA_WIDTH (512),
    .SKID_DEPTH (4)
) u_bridge (
    .clk             (clk),
    .rst_n           (rst_n),
    // registered-read FIFO side
    .s_valid         (fifo_rd_valid),
    .s_ready         (fifo_rd_ready),
    .s_data          (fifo_rd_data),
    // consumer side
    .m_valid         (drain_valid),
    .m_ready         (drain_read),
    .m_data          (drain_data),
    // status / debug
    .occupancy       (bridge_occupancy),
    .dbg_r_pending   (dbg_bridge_pending),
    .dbg_r_out_valid (dbg_bridge_out_valid)
);
```

12.7 Design Considerations

Table 2.7.6: Skid Depth Selection

SKID_DEPTH	Behaviour
2	Minimum: one stall of the consumer costs a bubble on the next read
4	Default: absorbs a short consumer stall with no upstream bubble
8	Deeper stall absorption; `occupancy` must be widened past 3 bits

The skid does not need to cover the FIFO's read latency (that is the one `r_drain_ip` flop's job); it needs to cover how long the *consumer* stalls without the bridge stopping upstream reads.

Last Updated: 2026-09-27

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13 Control-Read Engine Specification

Module: `ctrlrd_engine.sv` **Location:** `projects/components/dma-ip/rapids/rtl/fub/`
Status: Implemented **Last Updated:** 2026-07-12

13.1 Overview

The Control-Read Engine implements the `CTRL_READ` descriptor opcode (`DESC_OP_CTRL_READ = 2'b01`) as a **consumer gate**. When the scheduler decodes a `CTRL_READ` descriptor, it hands the engine a poll address, an expected value, and a compare mask. The engine issues an AXI4 read to the poll address, applies the mask to the returned data, and compares `(read_data & mask)` against the expected value. If they match, the gate is satisfied and the descriptor chain proceeds; if not, the engine retries. Retries are bounded so a gate that never matches cannot hang the channel.

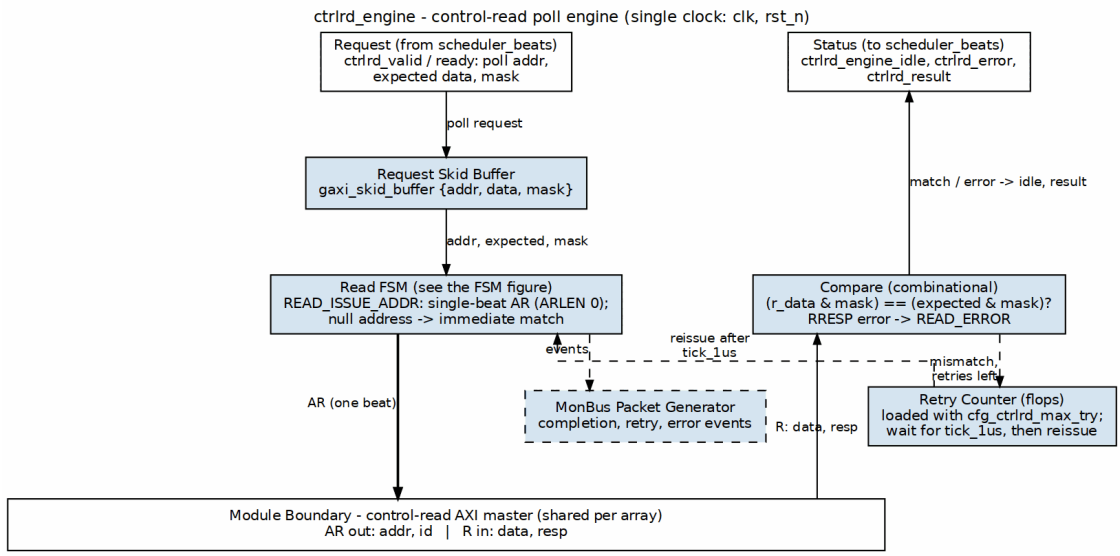
One Control-Read Engine is instantiated per channel in `scheduler_group_beats.sv` (`u_ctrlrd_engine`), configured for 32-bit AXI reads.

13.1.1 Key Features

- **Consumer Gate Semantics:** Polls memory until `(read_data & mask) == expected`
- **Masked Comparison:** Only the masked bits of the polled word participate in the match
- **Bounded Retry:** `cfg_ctrlrd_max_try` (0-511) caps the number of poll attempts
- **1 microsecond Retry Pacing:** Retries wait for `tick_1us` from the scheduler group
- **Error on Exhaustion:** Asserts `ctrlrd_error` if the gate never matches within budget
- **Null Address Fast-Path:** A poll address of zero completes immediately as a match
- **Channel Reset Support:** `cfg_channel_reset` returns the FSM to idle and drains any read still on the fabric
- **MonBus Integration:** Completion, retry, and error events

13.1.2 Block Diagram

13.1.3 Figure 2.8.1: Control-Read Engine Block Diagram



Control-read engine block diagram

Source: 08_ctrlrd_engine.dot

13.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 32;                 // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Channel index
width (derived)
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int AXI_DATA_WIDTH = 64;               // AXI read data
width (>= 32)
parameter int AXI_ID_WIDTH = 8;                  // AXI ID width (>=
CHAN_WIDTH)

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID  = 16'h0030; // Control-Read
Engine Agent ID
parameter logic [7:0]  MON_UNIT_ID   = 8'h02;    // Unit identifier
parameter logic [8:0]  MON_CHANNEL_ID = 9'h000;  // Base channel ID
```

: Table 2.8.1: Control-Read Engine Parameters

Parameter validation enforces `AXI_ID_WIDTH >= CHAN_WIDTH` and `AXI_DATA_WIDTH >= 32` (the engine consumes only the lower 32 bits of the read data). The per-channel instance in `scheduler_group_beats.sv` overrides `AXI_DATA_WIDTH` to 32.

13.3 Port List

13.3.1 Clock and Reset

Table 2.8.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

13.3.2 Scheduler Interface

Table 2.8.3: Scheduler Interface

Signal	Direction	Width	Description
`ctrlrd_valid`	input	1	Gate request valid from scheduler
`ctrlrd_ready`	output	1	Engine ready to accept a request
`ctrlrd_pkt_addr`	input	AW	Poll address (descriptor `poll_addr[63:0]`)
`ctrlrd_pkt_data`	input	32	Expected value (descriptor `expected[95:64]`)
`ctrlrd_pkt_mask`	input	32	Compare mask (descriptor `mask[127:96]`)
`ctrlrd_error`	output	1	Gate failed (retries exhausted or AXI error)
`ctrlrd_result`	output	32	Last read value (lower 32 bits)
`ctrlrd_engin`	output	1	Engine idle (post-issue

Signal	Direction	Width	Description
e_idle			completion indicator)

13.3.3 Configuration Interface

Table 2.8.4: Configuration Interface

Signal	Direction	Width	Description
cfg_ctrlrd_max_try	input	9	Poll retry budget, 0-511 (from CTRL_CONFIG.CTRLRD_MAX_TRY)
cfg_channel_reset	input	1	Per-channel reset request
tick_1us	input	1	1 microsecond tick that paces retries

13.3.4 AXI4 Read Master Interface

Table 2.8.5: AXI4 Read Master Interface

Signal	Direction	Width	Description
ar_valid	output	1	Read address valid
ar_ready	input	1	Read address ready
ar_addr	output	AW	Read address (poll address)
ar_len	output	8	Burst length (8'h00, single beat)
ar_size	output	3	Burst size (3'b010, 4 bytes)
ar_burst	output	2	Burst type (2'b01, INCR)
ar_id	output	ID	Transaction ID (channel-derived)
ar_lock	output	1	Lock (1'b0, normal access)
ar_cache	output	4	Cache (4'b0010, non-cacheable bufferable)
ar_prot	output	3	Protection (3'b000)

Signal	Direction	Width	Description
ar_qos	output	4	Quality of service (4'h0)
ar_region	output	4	Region (4'h0)
r_valid	input	1	Read data valid (shared channel)
r_ready	output	1	Read data ready
r_data	input	ADW	Read data (lower 32 bits used)
r_id	input	ID	Read response ID (matched to our request)
r_resp	input	2	Read response
r_last	input	1	Read last beat

13.3.5 MonBus Interface

Table 2.8.6: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	64	Side-band monitor timestamp
mon_valid	output	1	Monitor packet valid
mon_ready	input	1	Monitor consumer ready
mon_packet	output	128	Monitor packet (monitor_packet_t)
mon_timestamp	output	64	Monitor packet timestamp

13.4 FSM States

13.4.1 Figure 2.8.2: Control-Read Engine FSM

States and transitions of the `ctrlrd_engine` poll FSM.

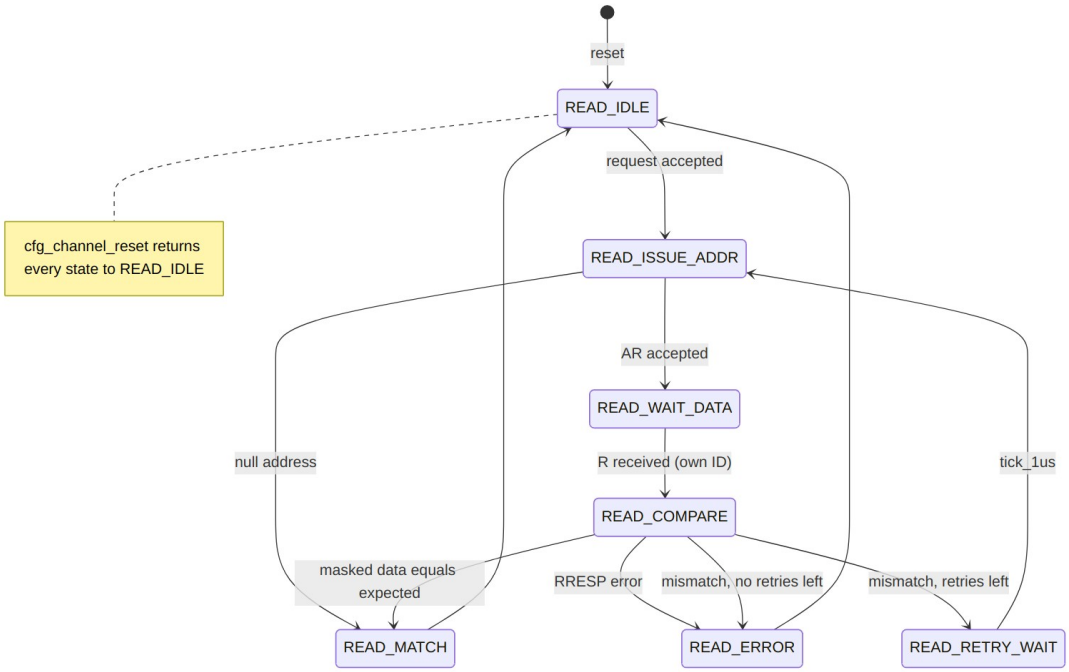


Diagram 2

State summary:

- **READ_IDLE:** Waits for a request from the scheduler (via a two-deep skid buffer). Latches poll address, expected value, mask, and the AXI ID; initializes the retry counter from `cfg_ctrlrd_max_try`.
- **READ_ISSUE_ADDR:** Drives `ar_valid`. A null address (`64'h0`) shortcuts to `READ_MATCH`; otherwise, once `ar_ready` handshakes, advances to `READ_WAIT_DATA`.
- **READ_WAIT_DATA:** Accepts the read response whose `r_id` matches the channel ID, capturing the lower 32 bits and the response code.
- **READ_COMPARE:** On an AXI error response, goes to `READ_ERROR`. On a masked match, goes to `READ_MATCH`. Otherwise, if retries remain, decrements the counter and goes to `READ_RETRY_WAIT`; if the budget is exhausted, goes to `READ_ERROR`.
- **READ_RETRY_WAIT:** Waits for `tick_1us`, then reissues the read.

- **READ_MATCH:** Gate satisfied. Clears the error flag and returns to READ_IDLE.
- **READ_ERROR:** Sets ctrlrd_error and returns to READ_IDLE.

13.5 Operation

13.5.1 Masked Compare

The engine gates on a masked equality check:

```
w_masked_expected = ctrlrd_pkt_data & ctrlrd_pkt_mask
w_masked_actual   = r_data[31:0]    & ctrlrd_pkt_mask
w_data_match      = (w_masked_expected == w_masked_actual)
```

Only the bits set in mask participate. A mask of 32'hFFFFFFFF compares the full word; a mask of a single bit turns the gate into a flag-poll.

13.5.2 Retry Budget and Pacing

The retry counter is loaded from cfg_ctrlrd_max_try when the request is accepted. On each non-matching, non-error compare with retries remaining, the counter decrements and the engine parks in READ_RETRY_WAIT until the next tick_1us, which paces successive polls at roughly 1 microsecond intervals. When the counter reaches zero without a match, the engine transitions to READ_ERROR and asserts ctrlrd_error, so a never-matching gate cannot stall the channel indefinitely.

cfg_ctrlrd_max_try is driven by the CTRL_CONFIG register at offset 0x240, field CTRLRD_MAX_TRY[8:0] (0-511, reset 16).

13.5.3 Null Address Fast-Path

If the latched poll address is 64'h0, the engine treats the gate as already satisfied: READ_ISSUE_ADDR transitions directly to READ_MATCH without issuing an AXI read. This lets software emit an unconditional (no-op) gate.

13.5.4 AXI Read Transaction

Polls are single-beat, 4-byte INCR reads (ar_len = 0, ar_size = 3'b010, ar_burst = 2'b01). The ar_id is derived from CHANNEL_ID, and only responses whose r_id matches are consumed on the shared read data channel. ctrlrd_result exposes the last captured 32-bit value; in the per-channel instantiation the scheduler gates on match/error rather than the raw value, so ctrlrd_result is left unconnected there.

13.5.5 Channel Reset

`cfg_channel_reset` is registered into `r_channel_reset_active`. While asserted, the engine refuses new requests (`ctrlrd_ready` is forced low), the retry and error state is cleared, and the FSM is driven back to `READ_IDLE`.

What the reset cannot do is cancel a read already on the fabric, because AXI has no cancel. The engine therefore drains it (rapids TASK-014, 2026-09-27):

- an AR that was raised and not yet accepted stays raised until the slave takes it (`r_drain_ar`), then
- the R beat that the accepted AR is owed is taken and discarded (`r_drain_pending`, which alone lifts `r_ready` outside `READ_WAIT_DATA`).

Until the beat is gone `ctrlrd_engine_idle` stays low and the request path stays closed, so the next read on this ID cannot be answered by the stale beat. Before this the reset dropped the FSM to idle with `r_ready` low, the beat sat on the bus, and a re-kicked channel took it as its own answer.

`ctrlrd_engine_idle` asserts only in `READ_IDLE` with no pending request, no active channel reset and nothing left to drain.

13.6 MonBus Reporting

The engine emits monitor packets on notable events:

Table 2.8.7: MonBus Events

Event	Packet Type	Code	Payload
Gate satisfied (non-null)	`PktTypeCompletion`	`CORE_COMPL_CTRLRD_COMPLETED`	Poll address
Retry attempt	`PktTypePerf`	`CORE_PERF_CTRLRD_RETRY`	Remaining retry count
Retries exhausted	`PktTypeError`	`CORE_ERR_CTRLRD_MAX_RETRIES`	Poll address
AXI error response	`PktTypeError`	`AXI_ERR_RESP_SLVERR` / `AXI_ERR_RESP_DECERR`	Response code

All packets carry MON_AGENT_ID, MON_UNIT_ID, and MON_CHANNEL_ID, and are timestamped from i_mon_time.

13.7 Descriptor Field Mapping

The CTRL_READ opcode reinterprets the shared 256-bit descriptor (the descriptor engine's existing field extraction is reused):

Table 2.8.8: CTRL_READ Descriptor Field Mapping

Descriptor Field	Bits	Engine Port
poll_addr	[63:0]	ctrlrd_pkt_addr
expected	[95:64]	ctrlrd_pkt_data
mask	[127:96]	ctrlrd_pkt_mask
max_try	[143:128]	Per-descriptor budget (0 = use cfg_ctrlrd_max_try default)

Last Updated: 2026-07-12

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14 Control-Write Engine Specification

Module: ctrlwr_engine.sv **Location:** projects/components/dma-ip/rapids/rtl/fub/
Status: Implemented **Last Updated:** 2025-01-10

14.1 Overview

The Control-Write Engine executes a CTRL_WRITE control descriptor (DESC_OP_CTRL_WRITE = 2'b10). A CTRL_WRITE is a producer doorbell: the engine performs a single-beat AXI4 write of the descriptor's 32-bit wr_data value to its 64-bit wr_addr target, then completes so the descriptor chain continues. One instance is dedicated per channel.

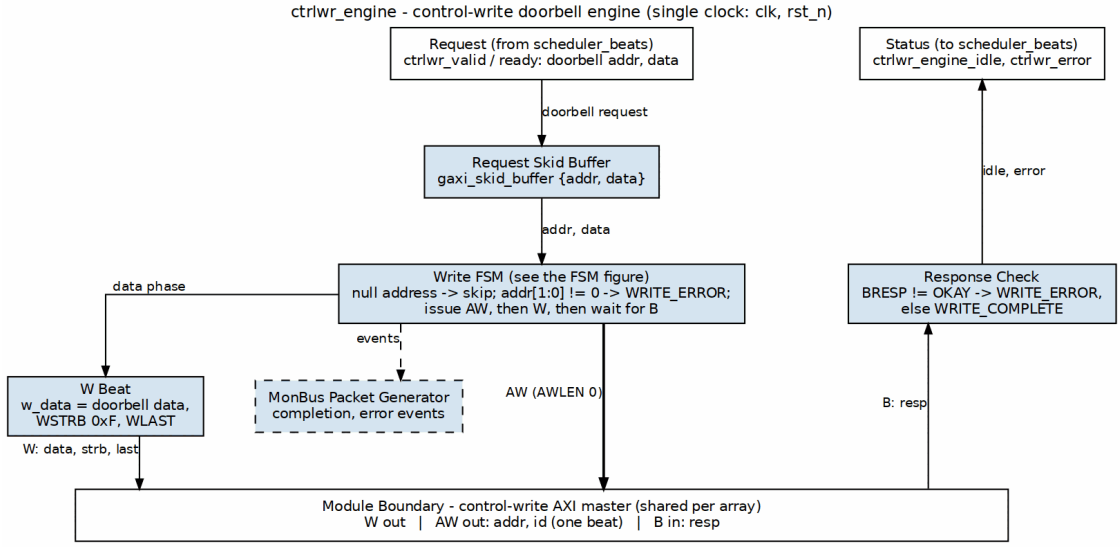
The write target address must be 4-byte aligned because the transfer is a 32-bit (aw_size = 3'b010) single-beat write. A null address (wr_addr == 0) is treated as a no-op and the request completes without issuing an AXI transaction.

14.1.1 Key Features

- **Single-Beat AXI4 Write:** One 32-bit doorbell write per request (aw_len = 0, w_last = 1)
- **Producer Doorbell Semantics:** Write value to target, then release the chain (no poll, no retry)
- **Address Alignment Check:** Requires 4-byte-aligned wr_addr; misalignment raises an error
- **Null-Address Skip:** wr_addr == 0 completes as a no-op
- **Channel Reset Support:** cfg_channel_reset returns the FSM to idle and drains any write still on the fabric
- **MonBus Integration:** Completion and error event reporting

14.1.2 Block Diagram

14.1.3 Figure 2.9.1: Control-Write Engine Block Diagram



Control-write engine block diagram

Source: 09_ctrlwr_engine.dot

14.2 Parameters

```
parameter int CHANNEL_ID = 0;                    // Channel identifier
parameter int NUM_CHANNELS = 32;                 // Total channels
parameter int CHAN_WIDTH = $clog2(NUM_CHANNELS); // Derived channel
index width
parameter int ADDR_WIDTH = 64;                   // Address bus width
parameter int AXI_ID_WIDTH = 8;                  // AXI transaction ID
width

// Monitor Bus Parameters
parameter logic [15:0] MON_AGENT_ID  = 16'h0020; // Ctrlwr Engine
Agent ID
parameter logic [7:0]  MON_UNIT_ID   = 8'h01;    // Unit identifier
parameter logic [8:0]  MON_CHANNEL_ID = 9'h000;  // Base channel ID
```

: Table 2.9.1: Control-Write Engine Parameters

Note: `AXI_ID_WIDTH` must be `>= CHAN_WIDTH` (checked by an elaboration-time `$fatal`). When instantiated in `scheduler_group_beats.sv` as `u_ctrlwr_engine`, `MON_AGENT_ID` is overridden to `16'h0021` (`CTRLWR_MON_AGENT_ID`) to distinguish it from the Control-Read Engine's `16'h0020`.

14.3 Port List

14.3.1 Clock and Reset

Table 2.9.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low asynchronous reset

14.3.2 Configuration Interface

Table 2.9.3: Configuration Interface

Signal	Direction	Width	Description
`cfg_channel_reset`	input	1	Per-channel reset: blocks new requests, drains transaction, forces idle

14.3.3 Scheduler Interface

Table 2.9.4: Scheduler Interface

Signal	Direction	Width	Description
`ctrlwr_valid`	input	1	Request valid (doorbell write requested)
`ctrlwr_ready`	output	1	Ready to accept request
`ctrlwr_pkt_addr`	input	AW	Doorbell target address (`wr_addr`, 4-byte aligned)
`ctrlwr_pkt_data`	input	32	Doorbell write value (`wr_data`)
`ctrlwr_error`	output	1	Error flag (alignment or AXI response error)
`ctrlwr_engine_idle`	output	1	Engine idle (in `WRITE_IDLE`, no active transaction, not in channel reset)

14.3.4 AXI4 Write Master Interface

Table 2.9.5: AXI4 Write Master Interface

Signal	Direction	Width	Description
`aw_valid`	output	1	Write address valid
`aw_ready`	input	1	Write address ready
`aw_addr`	output	AW	Write address (= `wr_addr`)
`aw_len`	output	8	Burst length (`0` = single beat)
`aw_size`	output	3	Transfer size (`3'b010` = 4 bytes / 32-bit)
`aw_burst`	output	2	Burst type (`2'b01` = INCR)
`aw_id`	output	AXI_ID_WIDTH	Transaction ID (derived from `CHANNEL_ID`)
`aw_lock`	output	1	Lock type (`0` = normal)
`aw_cache`	output	4	Cache type (`4'b0010` = non-cacheable bufferable)
`aw_prot`	output	3	Protection type (`3'b000`)
`aw_qos`	output	4	Quality of service (`0`)
`aw_region`	output	4	Region identifier (`0`)
`w_valid`	output	1	Write data valid
`w_ready`	input	1	Write data ready
`w_data`	output	32	Write data (= `wr_data`)
`w_strb`	output	4	Write strobes (`4'b1111` = all bytes)
`w_last`	output	1	Last beat (`1` = single beat)
`b_valid`	input	1	Write response valid
`b_ready`	output	1	Write response ready
`b_id`	input	AXI_ID_WIDTH	Response ID (matched against expected AXI ID)
`b_resp`	input	2	Write response code

14.3.5 MonBus Interface

Table 2.9.6: MonBus Interface

Signal	Direction	Width	Description
`i_mon_time`	input	64	Monitor timestamp side-band input
`mon_valid`	output	1	Monitor packet valid
`mon_ready`	input	1	Monitor packet ready
`mon_packet`	output	128	Monitor packet (`monitor_packet_t`)
`mon_timestamp`	output	64	Monitor packet timestamp

14.4 FSM States

14.4.1 Figure 2.9.2: Control-Write Engine FSM

States and transitions of the `ctrlwr_engine` doorbell FSM.

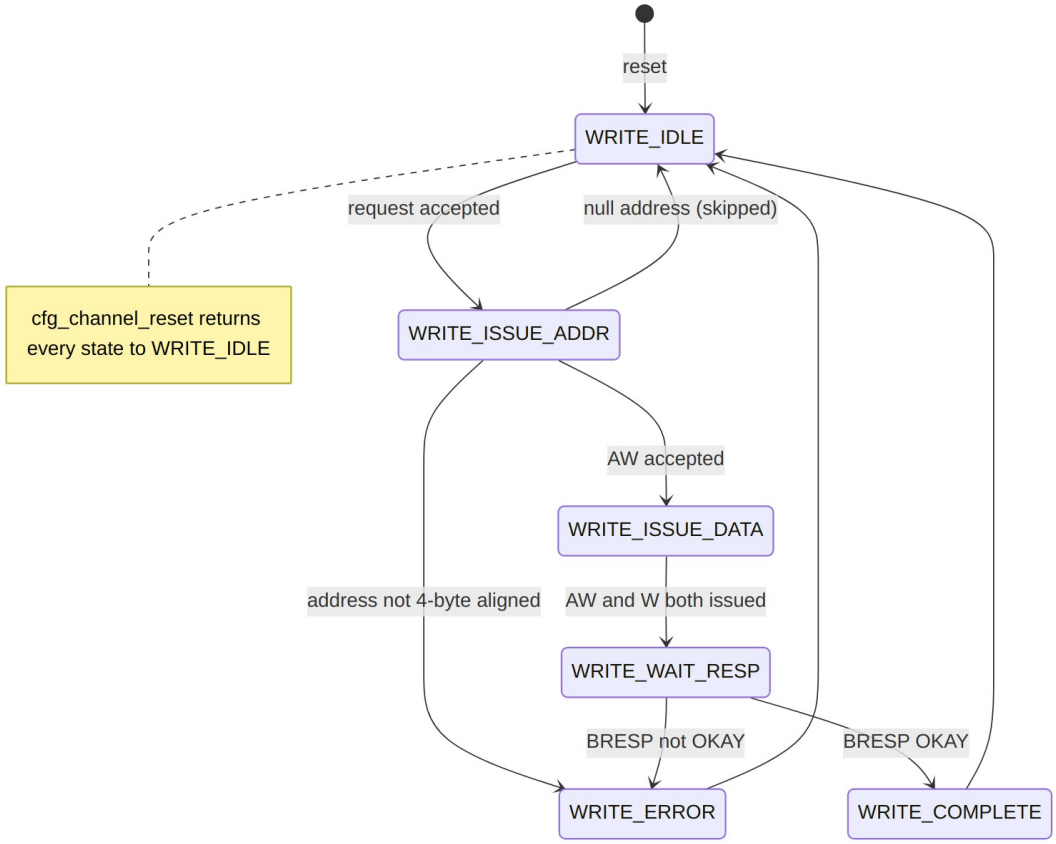


Diagram 3

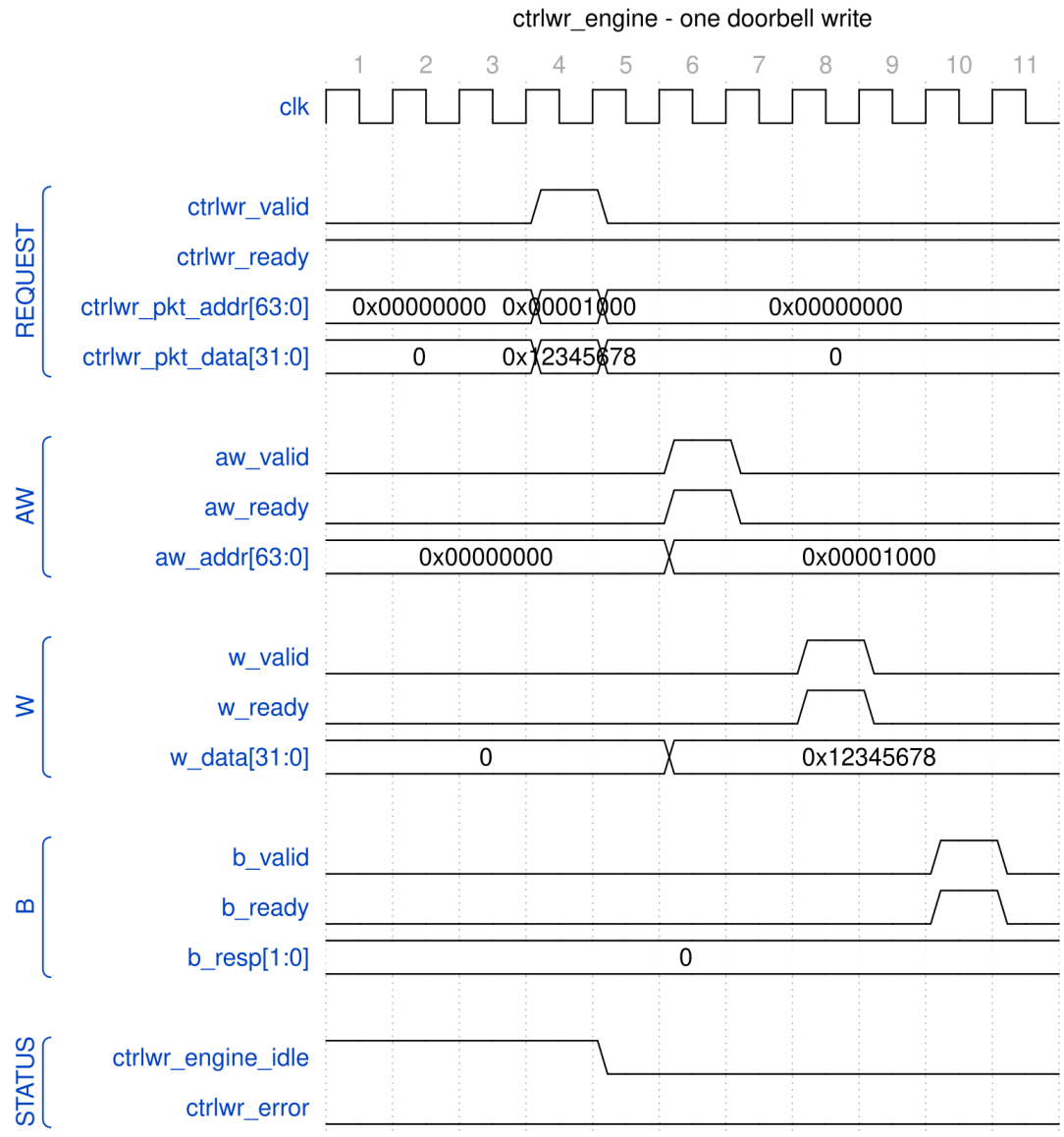
State summary:

- **WRITE_IDLE:** Wait for a request from the request skid buffer; latch `wr_addr`/`wr_data` and the expected AXI ID.
- **WRITE_ISSUE_ADDR:** Drive `aw_valid` until `aw_ready`. A misaligned address routes to `WRITE_ERROR`; a null address returns to `WRITE_IDLE` (no-op skip).
- **WRITE_ISSUE_DATA:** Drive `w_valid` until `w_ready`; advance once both address and data phases have been issued.
- **WRITE_WAIT_RESP:** Accept the matching `b`-channel response; a non-OKAY `b_resp` routes to `WRITE_ERROR`, otherwise to `WRITE_COMPLETE`.
- **WRITE_COMPLETE:** Emit a completion MonBus packet (unless null address) and return to `WRITE_IDLE`.
- **WRITE_ERROR:** Latch `ctrlwr_error`, emit an error MonBus packet, and return to `WRITE_IDLE`. The error flag persists until reset or `cfg_channel_reset`.

14.5 Operation

14.5.1 Doorbell Write Sequence

14.5.2 Figure 2.9.3: Control-Write Doorbell Timing



ctrlwr_engine - one doorbell write

Source: ctrlwr_engine_doorbell.json, captured from `dv/tests/fub/test_ctrlwr_engine.py` (`basic_write`, channel 0, 64-bit address, `REG_LEVEL=GATE`) with `WAVES=1`; the AXI side is the framework write slave.

Reading it: the request handshakes in one cycle (`ctrlwr_ready` is held high by the skid buffer) with the doorbell address 0x1000 and data 0x12345678, and the engine leaves idle the next cycle. AW goes out two cycles after the request, W two cycles after AW, and the slave answers B (OKAY) two cycles after W; `ctrlwr_engine_idle` returns the cycle after the B handshake, eight cycles after the request. Nothing is pipelined across doorbells: the engine takes one request at a time.

14.5.3 Request Handshake and Skid Buffer

Incoming requests are captured through a depth-2 `gaxi_skid_buffer`. The engine accepts a request (`ctrlwr_ready`) only when not in an active channel reset. The FSM pops the buffered request in `WRITE_IDLE`, latching `{wr_addr, wr_data}` and computing the expected AXI ID from `CHANNEL_ID`:

```
aw_id = {{(AXI_ID_WIDTH-CHAN_WIDTH){1'b0}}, CHANNEL_ID[CHAN_WIDTH-
1:0]};
```

The b-channel response is matched against this expected ID (`b_id == r_expected_axi_id`) so a shared write master can be demultiplexed per channel.

14.5.4 Address Validation

```
Null (skip)  : wr_addr == 64'h0
Align error  : wr_addr[1:0] != 2'b00  (and not null)
```

A null address completes immediately as a no-op with no AXI activity and no completion packet. A misaligned address never issues on AXI; it transitions to `WRITE_ERROR`, asserts `ctrlwr_error`, and emits an error MonBus packet.

14.5.5 Error Reporting

`ctrlwr_error` is a registered, sticky flag. It is set on an alignment error or on a non-OKAY AXI write response (`b_resp != 2'b00`), and it persists until `rst_n` or `cfg_channel_reset`. The MonBus error packet carries either the offending address (alignment error) or the captured `b_resp` code (AXI response error).

14.5.6 Channel Reset

Asserting `cfg_channel_reset` registers `r_channel_reset_active`, which blocks new requests (`ctrlwr_ready` low), clears the error flag and the phase-issued state, and drives the FSM back to `WRITE_IDLE`.

A write already on the fabric cannot be cancelled, so the engine drains it (rapids TASK-014, 2026-09-27): a raised AW stays up until accepted (`r_drain_aw`), an accepted AW is followed by its W beat with the latched data (`r_drain_w`), and the B response is taken and discarded (`r_drain_b`, which alone lifts `b_ready` outside `WRITE_WAIT_RESP`). The write that was already issued

therefore lands. Until B is taken `ctrlwr_engine_idle` stays low and the request path stays closed. Before this the reset abandoned whatever phase was in flight and left a dangling response on the channel's ID.

`ctrlwr_engine_idle` asserts only in `WRITE_IDLE` with no pending request, no active channel reset and nothing left to drain.

14.6 Contrast with the Control-Read Engine

The Control-Write Engine is the producer counterpart to the Control-Read Engine (Section 2.8). Both are per-channel control-descriptor executors, but they implement opposite half of the producer/consumer handshake:

Aspect	Control-Read Engine (2.8)	Control-Write Engine (2.9)
Opcode	`DESC_OP_CTRL_READ` (`2'b01`)	`DESC_OP_CTRL_WRITE` (`2'b10`)
Role	Consumer gate	Producer doorbell
AXI direction	Read (poll)	Single-beat write
Semantics	Poll until `(rd & mask) == expected`	Write `wr_data` to `wr_addr`, then continue
Retry budget	`max_try` retry budget (config default)	None (single unconditional write)
Chain effect	Holds off the chain until the gate passes	Releases the chain after the write completes
Descriptor fields	`poll_addr` / `expected` / `mask` / `max_try`	`wr_addr` / `wr_data`

A `CTRL_WRITE` therefore has no maximum-try configuration: it is one unconditional 32-bit write, whereas `CTRL_READ` polls with a bounded retry budget before either passing the gate or timing out.

Last Updated: 2026-07-12

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15 Scheduler Group

Module: `scheduler_group.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

15.1 Overview

The scheduler group wraps one channel: a descriptor engine, the scheduler, the two control engines (control read and control write) and a MonBus arbiter that merges their monitor packets. It is the unit the scheduler group array instantiates once per channel.

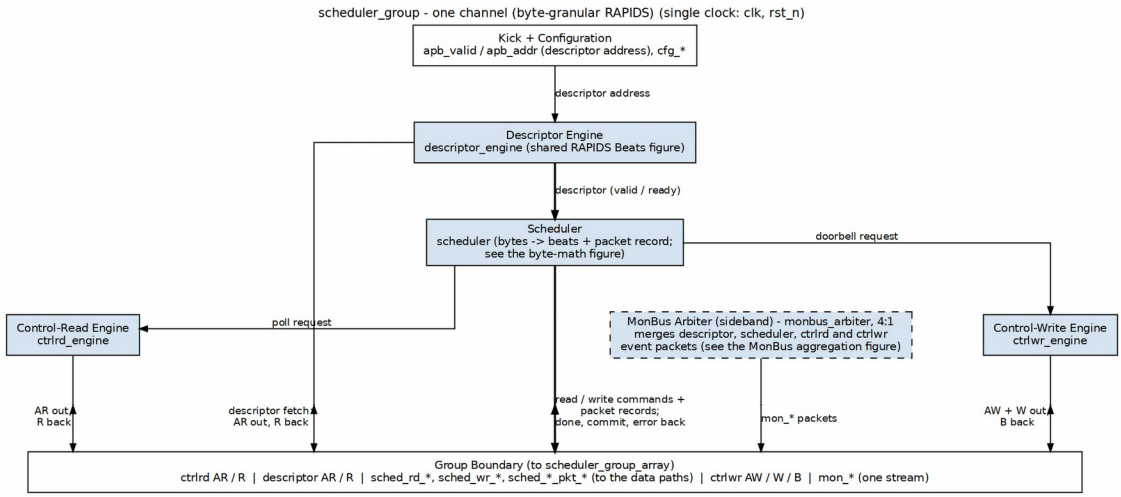
The structure is identical to the RAPIDS Beats group. The one change is the packet-record port set: the scheduler now publishes a per-direction record of the packet's byte length and start offset, and the group passes it through unchanged.

15.1.1 Key Features

- **One channel, four engines:** descriptor engine, scheduler, control read engine, control write engine.
- **Packet-record pass-through:** `sched_rd_pkt_*` and `sched_wr_pkt_*` connect the scheduler straight to the group boundary.
- **MonBus aggregation:** a four-client arbiter merges the four engines' packets.
- **Completion gating:** `cfg_sched_compl_enable` drops CORE Completion packets from the descriptor engine and the scheduler at the arbiter input and acknowledges the emitters, so nothing stalls (rapids ISSUE-005). Errors are never gated.
- **Status aggregation:** idle, FSM state and sticky error are exported.

15.1.2 Block Diagram

15.1.3 Figure 3.1.1: Scheduler Group Block Diagram



Scheduler group block diagram

Source: 07_scheduler_group.dot

15.2 Parameters

Table 3.1.1: Scheduler Group Parameters

Parameter	Default	Description
CHANNEL_ID	0	Channel identifier.
NUM_CHANNELS	8	Total channels.
CHAN_WIDTH	$clog2(NUM_CHANNELS)	Channel id width.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits. Sets the beat size seen by the scheduler.
AXI_ID_WIDTH	8	AXI ID width for the descriptor and control masters.
DESC_MON_AGENT_ID	16	MonBus agent id of the descriptor engine.

Parameter	Default	Description
SCHED_MON_AGENT_ID	48	MonBus agent id of the scheduler.
CTRLRD_MON_AGENT_ID	32	MonBus agent id of the control read engine.
CTRLWR_MON_AGENT_ID	33	MonBus agent id of the control write engine.
MON_UNIT_ID	1	MonBus unit id.
MON_CHANNEL_ID	0	Base MonBus channel id.
EN_READ	1	Enable the read direction of the scheduler.
EN_WRITE	1	Enable the write direction of the scheduler.
USE_ROW_COL_MAJOR_ADDRESSING	1	Descriptor engine addressing mode.
GEN_MON	1	Enable monitor packet generation.
OFF_W	$clog2(DATA_WIDTH/8)	Width of the packet-record byte offset. New in RAPIDS.

15.3 Port List

15.3.1 Clock, Reset and Kick-Off

Table 3.1.2: Clock, Reset and Kick-Off

Signal	Direction	Width	Description
clk	input	1	System clock.
rst_n	input	1	Active-low reset.
apb_valid	input	1	Channel kick-off.
apb_ready	output	1	Descriptor engine ready for kick-off.

Signal	Direction	Width	Description
apb_addr	input	AW	First descriptor address.

15.3.2 Configuration Interface

Table 3.1.3: Configuration Interface

Signal	Direction	Width	Description
cfg_channel_enable	input	1	Enable this channel.
cfg_channel_reset	input	1	Per-channel soft reset.
cfg_sched_timeout_cycles	input	32	Write-progress timeout window in cycles.
cfg_sched_timeout_limit	input	8	Consecutive timeout windows before a fatal error. 0 means never.
cfg_sched_timeout_enable	input	1	Enable timeout detection.
cfg_sched_err_enable	input	1	Enable error reporting.
cfg_sched_compl_enable	input	1	1 passes CORE Completion packets to the monitor bus; 0 drops them at the arbiter inputs.
cfg_sched_perf_enable	input	1	Enable performance monitoring.
cfg_ctrlrd_max_try	input	9	Control-read poll retry budget, 0 to 511.
tick_1us	input	1	1 us tick for control-read retry spacing.
cfg_desceng_prefetch	input	1	Enable descriptor prefetch.
cfg_desceng_fifo_thresh	input	4	Prefetch threshold.
cfg_desceng_addr0_base	input	AW	Valid descriptor address

Signal	Direction	Width	Description
			range 0 base.
cfg_desceng_addr0_limit	input	AW	Valid descriptor address range 0 limit.
cfg_desceng_addr1_base	input	AW	Valid descriptor address range 1 base.
cfg_desceng_addr1_limit	input	AW	Valid descriptor address range 1 limit.

15.3.3 Status

Table 3.1.4: Status Interface

Signal	Direction	Width	Description
descriptor_engine_idle	output	1	Descriptor engine idle.
scheduler_idle	output	1	Scheduler idle.
scheduler_state	output	7	Scheduler FSM state, one-hot.
sched_error	output	1	Scheduler error, sticky.

15.3.4 Descriptor AXI Master Interface

Table 3.1.5: Descriptor AXI Master Interface

Signal	Direction	Width	Description
desc_ar_valid / desc_ar_ready	output / input	1	AR handshake.
desc_ar_addr	output	AW	Descriptor fetch address.
desc_ar_len	output	8	Burst length minus one.
desc_ar_size	output	3	Burst size.
desc_ar_burst	output	2	Burst type.
desc_ar_id	output	IW	Transaction ID.
desc_ar_lock,	output	1, 4, 3, 4, 4	Attribute fields.

Signal	Direction	Width	Description
_cache, _prot, _qos, _region			
desc_r_valid / desc_r_ready	input / output	1	R handshake.
desc_r_data	input	256	Descriptor payload, fixed 256 bits.
desc_r_resp	input	2	Read response.
desc_r_last	input	1	Last beat.
desc_r_id	input	IW	Response ID.

15.3.5 Control Read and Control Write AXI Interfaces

The control read engine fetches a 32-bit semaphore word: ctrlrd_ar_* carries the AR channel (address, length, size, burst, id and the same attribute fields as above) and ctrlrd_r_valid, ctrlrd_r_ready, ctrlrd_r_data[31:0], ctrlrd_r_id, ctrlrd_r_resp and ctrlrd_r_last the R channel. The control write engine writes a 32-bit word: ctrlwr_aw_* carries the AW channel, ctrlwr_w_valid, ctrlwr_w_ready, ctrlwr_w_data[31:0], ctrlwr_w_strb[3:0] and ctrlwr_w_last the W channel, and ctrlwr_b_valid, ctrlwr_b_ready, ctrlwr_b_id and ctrlwr_b_resp the B channel. These are unchanged from RAPIDS Beats. See Control Read Engine and Control Write Engine.

15.3.6 Scheduler Data Interface

Table 3.1.6: Scheduler Data Interface

Signal	Direction	Width	Description
sched_rd_valid	output	1	Read request.
sched_rd_addr	output	AW	Working source byte address.
sched_rd_beats	output	32	Beats still to read.
sched_wr_valid	output	1	Write request.
sched_wr_ready	input	1	Write engine reports the last burst of the descriptor committed.
sched_wr_addr	output	AW	Working destination

Signal	Direction	Width	Description
			byte address.
sched_wr_beats	output	32	Beats still to write.
sched_rd_done_strobe	input	1	Read burst issued.
sched_rd_beats_done	input	32	Beats in that burst.
sched_wr_done_strobe	input	1	Write AW issued.
sched_wr_beats_done	input	32	Beats in that AW.
sched_wr_commit_strobe	input	1	Write B response received.
sched_wr_commit_beats	input	32	Beats committed by that B response.
sched_rd_error	input	1	Sticky read error from the read engine.
sched_wr_error	input	1	Sticky write error, including a packet-length error from the sink path.

15.3.7 Packet Record Interface

Table 3.1.7: Packet Record Interface

Signal	Direction	Width	Description
sched_rd_pkt_valid	output	1	One-cycle pulse: a read-side packet record is available.
sched_rd_pkt_ready	input	1	Consumer has room in its record queue.
sched_rd_pkt_bytes	output	32	Packet length in bytes.
sched_rd_pkt_offset	output	OFF_W	Byte offset of the packet start within its first memory beat.
sched_wr_pkt_valid	output	1	One-cycle pulse: a write-side packet record is

Signal	Direction	Width	Description
			available.
`sched_wr_pkt_ready`	input	1	Consumer has room in its record queue.
`sched_wr_pkt_bytes`	output	32	Packet length in bytes.
`sched_wr_pkt_offset`	output	OFF_W	Byte offset of the packet start within its first memory beat.

The group wires these signals directly to the scheduler. The read-side record goes to the source data path and the write-side record to the sink data path. The scheduler holds a descriptor in its fetch state while a required record queue is full, using the ready inputs. See Packet Record Interface.

15.3.8 MonBus Interface

Table 3.1.8: MonBus Interface

Signal	Direction	Width	Description
`i_mon_time`	input	`monbus_timestamp_t`	Shared monitor timebase.
`mon_valid`	output	1	Monitor packet valid.
`mon_ready`	input	1	Consumer ready.
`mon_packet`	output	`monitor_packet_t`	Monitor packet.
`mon_timestamp`	output	`monbus_timestamp_t`	Packet timestamp.

15.4 Internal Architecture

15.4.1 Descriptor Path

The descriptor engine fetches 256-bit descriptors from the address range the configuration allows and hands each one to the scheduler with its type, end-of-stream, end-of-list and error flags. The scheduler reports `scheduler_idle` back to the descriptor engine as `channel_idle`. No byte or beat arithmetic happens in the descriptor engine.

15.4.2 Control Engines

When a descriptor asks the scheduler to wait on or update a semaphore, the scheduler talks to the control engines through valid and ready handshakes with an address, data and mask. Both engines share the group's `AXI_ID_WIDTH` and each raises its own MonBus packets.

15.4.3 MonBus Aggregation

A four-client arbiter merges the descriptor engine, scheduler, control read and control write packets. `cfg_sched_compl_enable` gates only the CORE Completion packets of the descriptor engine and the scheduler: with it low, those packets are consumed at the arbiter input and never reach `mon_packet`. Error packets are never gated. `GEN_MON` masks every arbiter input when clear.

15.5 Differences from RAPIDS Beats

Table 3.1.9: Scheduler Group Delta

Item	RAPIDS Beats	RAPIDS
Parameter `OFF_W`	absent	present, `$clog2(DATA_WIDTH/8)`
`sched_rd_pkt_*`, `sched_wr_pkt_*` ports	absent	present, wired to the scheduler
Everything else		Unchanged

The beats chapter is Beats Scheduler Group. The scheduler is described in Scheduler.

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16 Scheduler Group Array

Module: `scheduler_group_array.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

16.1 Overview

The scheduler group array instantiates one scheduler group per channel and shares three AXI masters among them: the descriptor fetch master, the control-read master and the control-write master. It also owns the descriptor-AXI monitor and merges every channel's MonBus output with the monitor's into one stream.

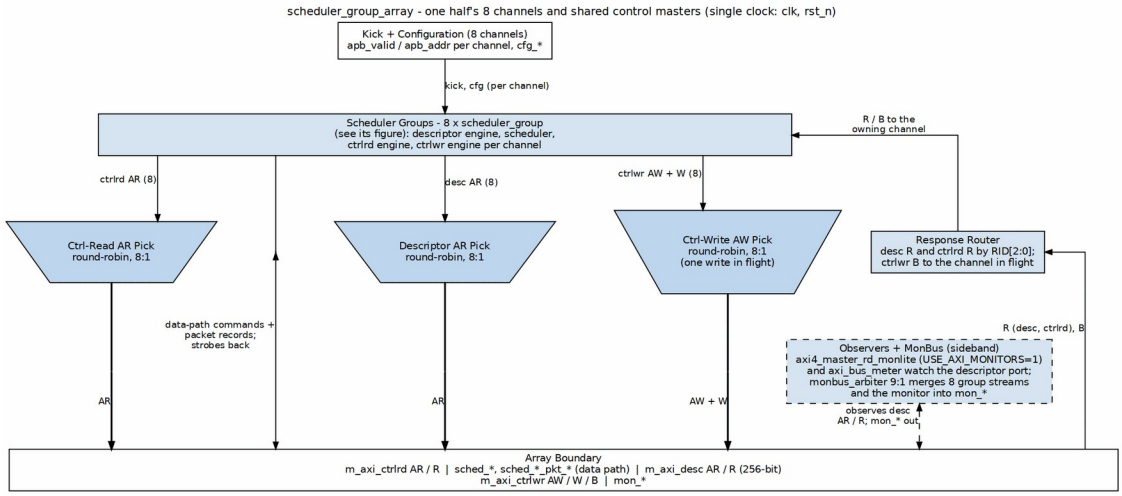
Compared with the RAPIDS Beats array, the sharing, arbitration and monitoring are unchanged. The array adds one parameter, `OFF_W`, and the per-channel packet-record ports, which it connects to each group by channel index.

16.1.1 Key Features

- **One group per channel:** `NUM_CHANNELS` scheduler groups.
- **Shared masters:** three round-robin arbiters with grant acknowledgment (`u_desc_ar_arbiter`, `u_ctrlrd_ar_arbiter`, `u_ctrlwr_aw_arbiter`).
- **Descriptor-AXI monitor:** a lite AXI read monitor on the shared descriptor master, plus an always-on bus meter for the perf window.
- **Unified MonBus:** `NUM_CHANNELS + 1` sources through one round-robin arbiter.
- **Direction gating:** `EN_READ` and `EN_WRITE` let the source array be read-only and the sink array write-only.
- **Per-channel packet records:** vector ports for the byte length and offset of each channel's packet.

16.1.2 Block Diagram

16.1.3 Figure 3.2.1: Scheduler Group Array Block Diagram



Scheduler group array block diagram

Source: 08_scheduler_group_array.dot

16.2 Parameters

Table 3.2.1: Scheduler Group Array Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels.
CHAN_WIDTH	$clog2(NUM_CHANNELS)	Channel id width.
ADDR_WIDTH	64	Address width.
DATA_WIDTH	512	Data width in bits.
AXI_ID_WIDTH	8	AXI ID width.
DESC_MON_BASE_AGENT_ID	16	Base MonBus agent id of the descriptor engines.
SCHED_MON_BASE_AGENT_ID	48	Base MonBus agent id of the schedulers.
DESC_AXI_MON_AGENT_ID	8	MonBus agent id of the descriptor AXI monitor.
MON_UNIT_ID	1	MonBus unit id.

Parameter	Default	Description
MON_MAX_TRANSACTIONS	16	Transaction table depth of the descriptor monitor.
EN_READ	1	Enable the read direction in every scheduler. Clear for a sink array.
EN_WRITE	1	Enable the write direction in every scheduler. Clear for a source array.
USE_AXI_MONITORS	1	0 omits the descriptor-AXI monitor hardware and ties its MonBus source to zero.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out the extended addressing path.
GEN_MON	1	0 omits the per-channel completion and error emitters.
OFF_W	$clog2(DATA_WIDTH/8)	Width of the packet-record byte offset. New in RAPIDS.

16.3 Port List

Vector ports are NUM_CHANNELS wide (NC), indexed by channel.

16.3.1 Clock, Reset and Kick-Off

Table 3.2.2: Clock, Reset and Kick-Off

Signal	Direction	Width	Description
clk	input	1	System clock.
rst_n	input	1	Active-low reset.
apb_valid	input	NC	Per-channel kick-off.
apb_ready	output	NC	Per-channel

Signal	Direction	Width	Description
			ready.
apb_addr	input	NC x AW	First descriptor address per channel.
cfg_channel_enable	input	NC	Per-channel enable.
cfg_channel_reset	input	NC	Per-channel soft reset.

16.3.2 Global Scheduler and Descriptor Engine Configuration

These inputs are applied to every channel.

Table 3.2.3: Global Configuration

Signal	Direction	Width	Description
cfg_sched_enable	input	1	Master scheduler enable.
cfg_sched_timeout_cycles	input	32	Timeout window in cycles.
cfg_sched_timeout_limit	input	8	Consecutive timeouts before a fatal error. 0 means never.
cfg_sched_timeout_enable	input	1	Enable timeout detection.
cfg_sched_err_enable	input	1	Enable error reporting.
cfg_sched_compl_enable	input	1	Enable CORE Completion packets.
cfg_sched_perf_enable	input	1	Enable performance monitoring.
cfg_desceng_enable	input	1	Master descriptor engine enable.
cfg_desceng_prefetch	input	1	Enable descriptor prefetch.
cfg_desceng_fifo_thresh	input	4	Prefetch threshold.

Signal	Direction	Width	Description
cfg_desceng_addr0_base, _limit	input	AW	Valid descriptor address range 0.
cfg_desceng_addr1_base, _limit	input	AW	Valid descriptor address range 1.
cfg_ctrlrd_max_try	input	9	Control-read poll retry budget.
tick_1us	input	1	1 us tick for control-read retry spacing.

16.3.3 Descriptor Monitor Configuration and Status

Table 3.2.4: Descriptor Monitor Configuration and Status

Signal	Direction	Width	Description
cfg_desc_mon_enable	input	1	Enable the descriptor AXI monitor.
cfg_desc_mon_err_enable	input	1	Enable error packets.
cfg_desc_mon_perf_enable	input	1	Enable performance packets.
cfg_desc_mon_perf_run	input	1	Perf window control.
cfg_desc_mon_timeout_enable	input	1	Enable timeout packets.
cfg_desc_mon_timeout_cycles	input	32	Timeout threshold.
cfg_desc_mon_latency_thresh	input	32	Latency threshold.
cfg_desc_mon_pkt_mask	input	16	Packet type mask.
cfg_desc_mon_err_select	input	4	Error select.
cfg_desc_mon_err_mask, _timeout_mas	input	8 each	Per-class event masks.

Signal	Direction	Width	Description
k, _compl_mask, _thresh_mask, _perf_mask, _addr_mask, _debug_mask			
cfg_sts_desc_mon_busy	output	1	Monitor busy.
cfg_sts_desc_mon_active_txns	output	8	Active transactions.
cfg_sts_desc_mon_error_count	output	16	Error count.
cfg_sts_desc_mon_txn_count	output	32	Transaction count.
cfg_sts_desc_mon_conflict_error	output	1	Conflict error.

16.3.4 Descriptor-AXI Perf Window

The perf counters come from an always-on bus meter on the descriptor R channel, not from the monitor. `cfg_desc_mon_perf_run` opens the window: counters clear on its rising edge, count while it is high and hold while it is low. `USE_AXI_MONITORS` does not gate the meter.

Table 3.2.5: Descriptor-AXI Perf Window

Signal	Direction	Width	Description
sts_desc_mon_win_active	output	1	Window open.
sts_desc_mon_win_cycles	output	32	Cycles in the window.
sts_desc_mon_prod_cycles	output	32	R valid and ready.
sts_desc_mon_bp_cycles	output	32	R valid, not ready.
sts_desc_mon_starv_cycles	output	32	R ready, not valid.

Signal	Direction	Width	Description
sts_desc_mon_idle_cycles	output	32	Neither valid nor ready.
sts_desc_mon_beat_count	output	32	R handshakes.
sts_desc_mon_byte_count	output	64	Bytes moved on R (32 per descriptor beat).
sts_desc_mon_burst_count	output	32	AR handshakes.

16.3.5 Status

Table 3.2.6: Status

Signal	Direction	Width	Description
descriptor_engine_idle	output	NC	Per-channel descriptor engine idle.
scheduler_idle	output	NC	Per-channel scheduler idle.
scheduler_state	output	NC x 7	Per-channel FSM state, one-hot.
sched_error	output	NC	Per-channel scheduler error, sticky.

16.3.6 Shared AXI Masters

The three shared masters carry the same signals as the per-group ports in Scheduler Group, prefixed `desc_axi_`, `ctrlrd_axi_` and `ctrlwr_axi_`. The descriptor master returns 256-bit data. The control read master returns 32-bit data. The control write master is 32 bits wide with a 4-bit strobe. Each is a single interface: the arbiter selects one channel's request onto it and steers the response back by ID.

16.3.7 Per-Channel Scheduler Data Interface

Table 3.2.7: Per-Channel Scheduler Data Interface

Signal	Direction	Width	Description
sched_rd_valid	output	NC	Read request per channel.
sched_rd_addr	output	NC x AW	Working source byte address.
sched_rd_bea	output	NC x 32	Beats still to read.

Signal	Direction	Width	Description
ts			
sched_wr_valid	output	NC	Write request per channel.
sched_wr_ready	input	NC	Last burst of the descriptor committed.
sched_wr_addr	output	NC x AW	Working destination byte address.
sched_wr_beats	output	NC x 32	Beats still to write.
sched_rd_done_strobe	input	NC	Read burst issued.
sched_rd_beats_done	input	NC x 32	Beats in that burst.
sched_wr_done_strobe	input	NC	Write AW issued.
sched_wr_beats_done	input	NC x 32	Beats in that AW.
sched_wr_commit_strobe	input	NC	Write B response received.
sched_wr_commit_beats	input	NC x 32	Beats committed.
sched_rd_error	input	NC	Sticky read error.
sched_wr_error	input	NC	Sticky write error.

16.3.8 Per-Channel Packet Record Interface

Table 3.2.8: Per-Channel Packet Record Interface

Signal	Direction	Width	Description
sched_rd_pkt_valid	output	NC	Read-side record pulse per channel.
sched_rd_pkt_ready	input	NC	Read-side consumer has room.
sched_rd_pkt_bytes	output	NC x 32	Packet length in bytes.
sched_rd_pkt_offset	output	NC x OFF_W	Start offset in the first memory beat.

Signal	Direction	Width	Description
sched_wr_pkt_valid	output	NC	Write-side record pulse per channel.
sched_wr_pkt_ready	input	NC	Write-side consumer has room.
sched_wr_pkt_bytes	output	NC x 32	Packet length in bytes.
sched_wr_pkt_offset	output	NC x OFF_W	Start offset in the first memory beat.

Element [ch] of each vector connects to the matching port of scheduler_group[ch]. The array adds no logic between them. A source array uses only the read set and a sink array only the write set.

16.3.9 MonBus Interface

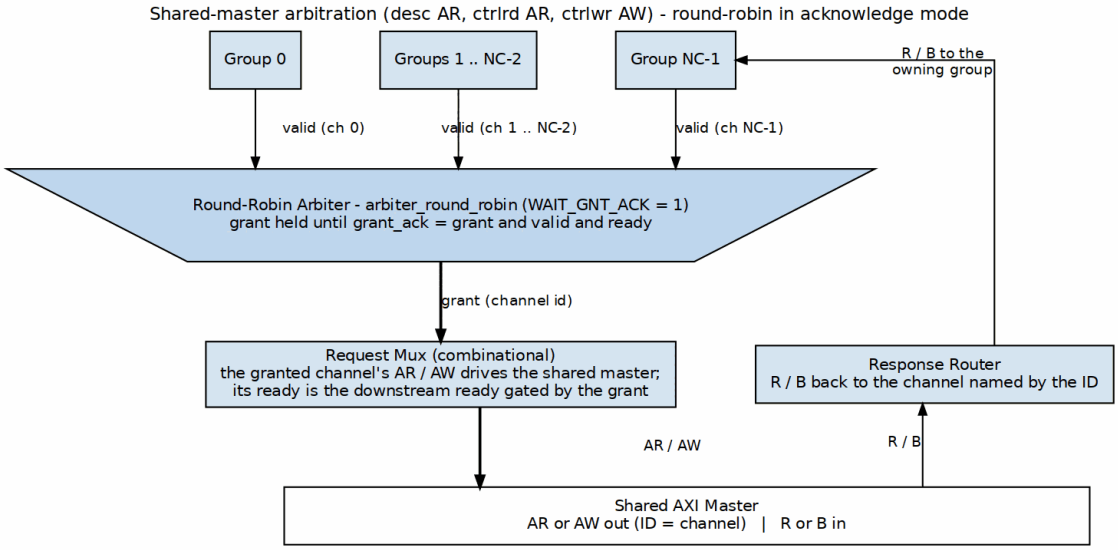
Table 3.2.9: MonBus Interface

Signal	Direction	Width	Description
i_mon_time	input	monbus_timestamp_t	Shared monitor timebase.
mon_valid	output	1	Packet valid.
mon_ready	input	1	Consumer ready.
mon_packet	output	monitor_packet_t	Packet.
mon_timestamp	output	monbus_timestamp_t	Timestamp.

16.4 Shared Master Arbitration

Each shared master has a round-robin arbiter over NUM_CHANNELS requesters in acknowledge mode. The grant is held until the granted channel's request is accepted (grant_ack), so a granted channel always sees its own valid and ready. The AR or AW ready seen by a channel is the downstream ready gated by that channel's grant.

16.4.1 Figure 3.2.2: Shared Master Arbitration



Shared master arbitration

Source: 11_shared_master_arbitration.dot

16.5 MonBus Aggregation

Table 3.2.10: MonBus Source Assignment

Source	Origin	Content
0 to NC-1	`scheduler_group[ch]`	Descriptor engine, scheduler, control read and control write packets of the channel
NC	Descriptor AXI monitor	Descriptor fetch monitor packets

One round-robin arbiter with acknowledge mode merges the `NUM_CHANNELS + 1` sources into `mon_*`. Each source carries its own timestamp alongside the packet.

16.6 Differences from RAPIDS Beats

Table 3.2.11: Array Delta

Item	RAPIDS Beats	RAPIDS
Parameter `OFF_W`	absent	present
`sched_rd_pkt_*`, `sched_wr_pkt_*` vectors	absent	present, wired by channel
Everything else		Unchanged

The beats chapter is Beats Scheduler Group Array.

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17 Sink Data Path

Module: `snk_data_path.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

17.1 Overview

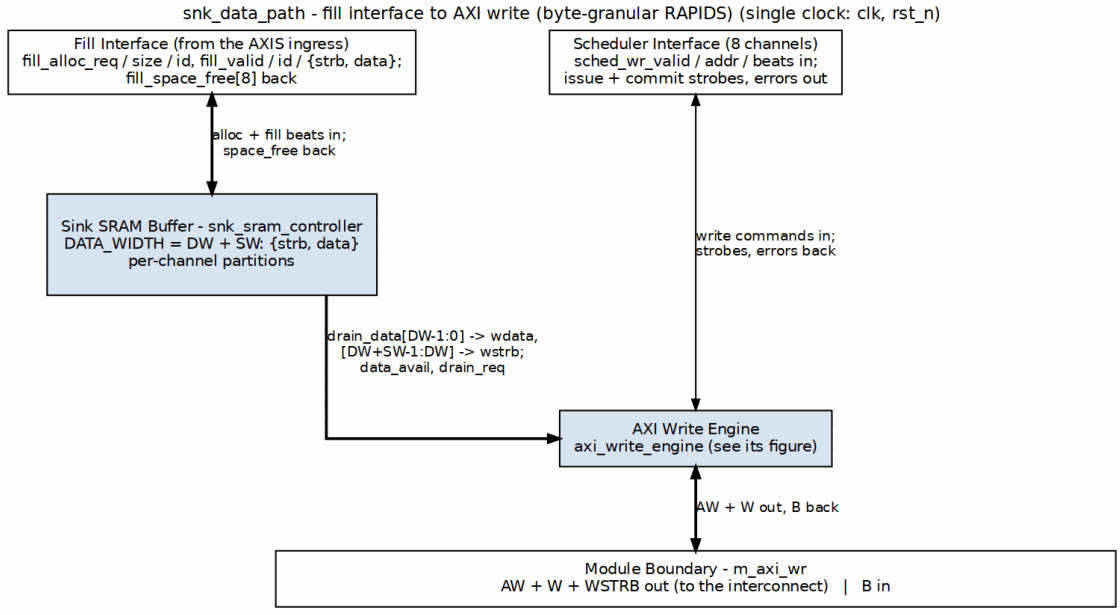
The sink data path joins the per-channel SRAM controller to the AXI write engine. Data enters through the fill interface, waits in the channel's SRAM partition and leaves as AXI4 write bursts.

In RAPIDS every SRAM word carries the byte enables of its data. The fill interface widens from `DW` to `DW + SW` bits, where `SW = DW/8`, and the write engine drives `m_axi_wstrb` from the stored enables. Everything else, including the allocation protocol, the scheduler interface and the burst rules, is as in RAPIDS Beats.

17.1.1 Key Features

- **Per-channel SRAM partitions:** flow control by allocation and space-free counts.
- **Byte enables stored with the data:** the SRAM is `DW + SW` bits wide.
- **AXI write engine:** 4 KB-safe, beat-aligned bursts. See AXI Write Engine.
- **Scheduler integration:** beat counts in, issue and commit strobes out.

17.1.2 Figure 3.3.1: Sink Data Path Block Diagram



Sink data path block diagram

Source: 09_snk_data_path.dot

17.2 Parameters

Table 3.3.1: Sink Data Path Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Number of channels (alias NC).
ADDR_WIDTH	64	Address width (alias AW).
DATA_WIDTH	512	Data width in bits (alias DW). The board uses 256.
AXI_ID_WIDTH	8	AXI ID width (alias IW).
SRAM_DEPTH	512	Words per channel partition (alias SD).
SEG_COUNT_WIDTH	$clog2(SRAM_DEPTH) + 1	Space counter width (alias SCW).
PIPELINE	1	SRAM read pipeline setting.
AW_MAX_OUTSTANDING	8	AW commands the write engine may have in flight.
W_PHASE_FIFO_DEPTH	64	Depth of the write engine W-phase queue.
B_PHASE_FIFO_DEPTH	16	Depth of the write engine B-phase queue.
CIW	$clog2(NC)	Channel id width.
SW	DW / 8	Byte enables per beat. New in RAPIDS.

17.3 Interfaces

17.3.1 Fill Interface

Table 3.3.2: Fill Interface

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Space allocation request.
`fill_alloc_size`	input	8	Beats to allocate.
`fill_alloc_id`	input	CIW	Channel to allocate in.
`fill_space_free`	output	NC x SCW	Free space per channel.
`fill_valid`	input	1	Fill beat valid.
`fill_ready`	output	1	Ready for a fill beat.
`fill_id`	input	CIW	Channel of the fill beat.
`fill_data`	input	DW + SW	`{byte enables, data}`. Bits `[DW-1:0]` are data. Bits `[DW+SW-1:DW]` are enables.

17.3.2 Configuration

Table 3.3.3: Configuration

Signal	Direction	Width	Description
`cfg_axi_wr_xfer_beats`	input	8	Write burst size cap in beats, minus one. Renamed from `cfg_axi_wr_xfer`.
`cfg_channel_reset`	input	NC	Per-channel reset, level or pulse. Forwarded to the write engine and the SRAM controller wrapper.

17.3.3 Scheduler Interface

Table 3.3.4: Scheduler Interface

Signal	Direction	Width	Description
sched_wr_valid	input	NC	Channel requests writes.
sched_wr_ready	output	NC	Last burst of a descriptor committed.
sched_wr_addr	input	NC x AW	Working destination byte address.
sched_wr_beats	input	NC x 32	Beats still to write.
sched_wr_burst_len	input	NC x 8	Requested burst length. Not used for sizing.
sched_wr_done_strobe	output	NC	AW issued.
sched_wr_beats_done	output	NC x 32	Beats in that AW.
sched_wr_commit_strobe	output	NC	B response received.
sched_wr_commit_beats	output	NC x 32	Beats committed by that B.
sched_wr_error	output	NC	Sticky write error (bad B response).

17.3.4 AXI4 Write Master

The write master is `m_axi_aw*`, `m_axi_w*` and `m_axi_b*` with `IW`-bit IDs and `DW`-bit data. See AXI4 Interface Specification. `m_axi_wstrb` is `DW/8` bits and carries the stored byte enables.

17.4 Byte Enable Path

1. The upstream stage presents `{strb, data}` on `fill_data` with `fill_valid`.
2. The SRAM controller stores the whole `DW + SW` word. It is instantiated with `DATA_WIDTH = DW + SW`.
3. The drain side returns the same word as `drain_data`.
4. `drain_data[DW-1:0]` goes to `axi_wr_sram_data` and `drain_data[DW+SW-1:DW]` goes to `axi_wr_sram_strb` on the write engine.

5. The write engine drives `m_axi_wdata` and `m_axi_wstrb` from those two.

The SRAM depth counts words, so capacity in beats is unchanged. Only the width grows, by one bit per data byte.

17.4.1 Channel Reset

`cfg_channel_reset[ch]` reaches two blocks. The AXI Write Engine stops issuing for the channel, completes an open burst with null beats and clears its error flag. The SRAM controller wrapper (`snk_sram_controller`) holds one single-channel `sram_controller` per channel. Each instance's reset is a registered flop, `r_ch_rst_n[ch]`, which goes low one cycle after `cfg_channel_reset[ch]` rises, so the partition's pointers and counters return to empty. The shared STREAM `sram_controller` has no channel reset of its own and is not modified; giving each channel its own instance and reset is what makes the reset per channel. A reset of one channel leaves the instances of the other channels untouched.

17.5 Differences from RAPIDS Beats

Table 3.3.5: Sink Data Path Delta

Item	RAPIDS Beats	RAPIDS
`fill_data` width	`DW`	`DW + SW`
SRAM word width	`DW`	`DW + SW`
`m_axi_wstrb`	all ones	stored per-beat enables
Config port name	`cfg_axi_wr_xfer`	`cfg_axi_wr_xfer_beats`
Scheduler port names	`sched_wr`, `sched_wr_done`, `sched_wr_commit`	`sched_wr_beats`, `sched_wr_beats_done`, `sched_wr_commit_beats`
Fill allocation, drain, error export		Unchanged

The beats chapter is Sink Data Path. Its timing diagrams apply to the AW, W and B sequencing unchanged.

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18 Sink Data Path with AXIS (Ingress)

Module: `snk_data_path_axis.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

18.1 Overview

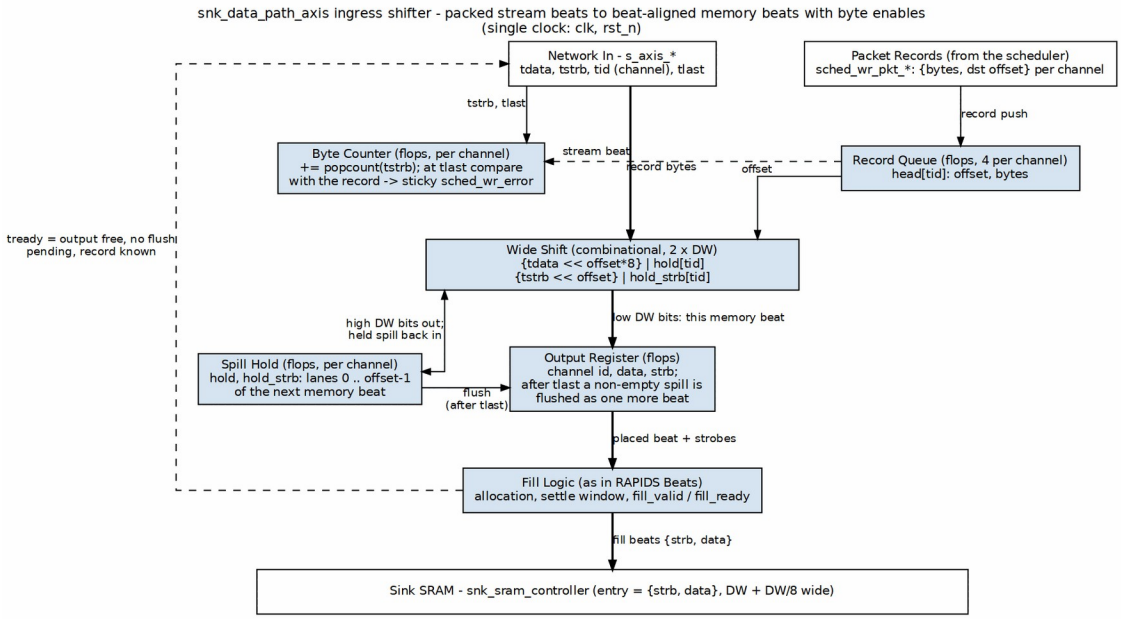
This module puts an AXI-Stream slave in front of the Sink Data Path and converts a packed byte stream into beat-aligned memory beats with byte enables. It is the block that makes the sink side byte-granular.

A packet arrives with its bytes packed from lane 0. Its destination in memory starts at a byte offset inside the first memory beat. The scheduler supplies that offset in a packet record. The module shifts each stream beat up by the offset, holds the bytes that spill past the beat boundary, and merges them into the next memory beat. After the last stream beat, a non-empty spill becomes one more, partial, memory beat.

18.1.1 Key Features

- **Barrel shift by the destination offset:** stream lane `l` lands in memory lane `offset + l`.
- **Byte enables follow the data:** the shifted strobe is stored beside the data and becomes WSTRB.
- **Per-channel state:** spill hold, received-byte count and length-error flag per channel, so packets of different channels interleave by beat.
- **Packet record queue:** four records per channel, fed by the scheduler.
- **Descriptor gating:** the stream is accepted only when the channel has a packet record.
- **Length check:** a packet whose byte count differs from the descriptor sets a sticky channel error.
- **Allocation logic unchanged from RAPIDS Beats**, including the settle wait.

18.1.2 Figure 3.4.1: Ingress Shifter



Ingress shifter

Source: 01_ingress_shifter.dot

18.2 Parameters

Parameters of Sink Data Path apply unchanged. The additions are:

Table 3.4.1: Additional Parameters

Parameter	Default	Description
`AXIS_ID_WIDTH`	8	`tid` width. The low `CIW` bits select the channel.
`AXIS_DEST_WIDTH`	4	`tdest` width. Not used to select a channel.
`AXIS_USER_WIDTH`	1	`tuser` width. Not used.
`SW`	`DW / 8`	Byte lanes per beat.
`OFF_W`	`$clog2(SW)`	Width of the packet-record offset.

18.3 Interfaces

18.3.1 AXI-Stream Slave

Table 3.4.2: AXI-Stream Slave

Signal	Direction	Width	Description
`s_axis_tdata`	input	DW	Packed data.
`s_axis_tstrb`	input	SW	Byte strobes. Contiguous from lane 0. Partial only on the `tlast` beat.
`s_axis_tlast`	input	1	Last beat of the packet.
`s_axis_tid`	input	AXIS_ID_WIDTH	Channel select.
`s_axis_tdest`	input	AXIS_DEST_WIDTH	Unused.
`s_axis_tuser`	input	AXIS_USER_WIDTH	Unused.
`s_axis_tvalid`	input	1	Beat valid.
`s_axis_tready`	output	1	Beat accepted.

`s_axis_tready` is high when the output register is free, no flush beat is pending and the addressed channel has a packet record queued. A channel's stream therefore stalls until its scheduler has fetched the descriptor. `s_axis_tready` is a single signal qualified by the TID of the beat on the bus, so a beat for a channel that has no record blocks the stream, and any channel's beats behind it, until that record arrives. Channels on one stream are not isolated from each other; a system that needs isolation gives each channel its own stream or guarantees the descriptor is queued before the packet is sent. `s_axis_tready` is also low for a beat whose TID names a channel in reset, and high without a record for a channel that is discarding the tail of a cut packet (see Channel Reset).

18.3.2 Scheduler Interface

The write request and completion ports are those of Sink Data Path. The packet record ports are new:

Table 3.4.3: Packet Record Ports

Signal	Direction	Width	Description
`sched_wr_pkt`	input	NC	Record pulse from the

Signal	Direction	Width	Description
_valid			scheduler.
sched_wr_pkt_ready	output	NC	Record queue not full.
sched_wr_pkt_bytes	input	NC x 32	Packet length in bytes.
sched_wr_pkt_offset	input	NC x OFF_W	Destination offset in the first memory beat.
cfg_channel_reset	input	NC	Per-channel reset, level or pulse. Clears the channel's queue, hold, byte count and length error, and discards the tail of a cut packet.

`sched_wr_error` is the OR of the write engine's B-response error and the packet length error of the channel. Both clear on `cfg_channel_reset` (see Channel Reset).

18.3.3 Other Ports

The AXI4 write master is that of Sink Data Path. `cfg_alloc_size` (8 bits) is the allocation segment per request, with 0 treated as 1. The debug outputs are `dbg_sram_bridge_pending`, `dbg_sram_bridge_out_valid`, `dbg_axis_beats_received` and `dbg_axis_packets_received`. `o_active_channel_id` and `o_active_channel_valid` carry the write engine's active channel to the per-channel bus meter.

18.4 Ingress Operation

18.4.1 Packet Records

Each channel has a four-entry queue of `{offset, bytes}`. A record is pushed on `sched_wr_pkt_valid` when the queue is not full and popped when the packet completes. The head record supplies the offset for every beat of the packet and the expected length at `tlast`.

18.4.2 Placement

For an accepted stream beat, with `off` the head record's offset:

- The wide word is `(tdata_m << off*8) | hold_data`, of width `2*DW`, where `tdata_m` is `tdata` with every lane whose `tstrb` bit is clear forced to zero.
- The wide strobe is `(tstrb << off) | hold_strb`, of width `2*SW`.

- The low half goes to the output register as the memory beat and its enables.
- The high half becomes the new hold for the channel.

The strobe mask matters on a partial last beat: the non-strobed lanes carry whatever the sender left there, and unmasked they would shift into the spill hold. A packet that ends without a spill never flushes the hold, so that junk would be ORed into the first memory beat of the channel's next packet. Masking by `tstrb` keeps the hold to real payload bytes.

The first memory beat has its low `off` strobe bits clear, so the bytes below the start address are not written.

18.4.3 Flush and Completion

On `tlast` the received-byte total is compared with the head record. If the wide strobe's high half is non-zero, a flush is scheduled and `s_axis_tready` drops until the hold has been emitted as the last memory beat. The packet then completes and its record pops. If the high half is empty the packet completes on the `tlast` beat itself.

18.4.4 Length Check

`r_pkt_rx_bytes` counts the popcount of `tstrb` over the packet. At `tlast`, a total that differs from the record's byte count sets the channel's sticky error. Nothing is padded or dropped. The error is visible on `sched_wr_error` and the scheduler treats it as fatal. It clears on `cfg_channel_reset`.

18.4.5 Channel Reset

`cfg_channel_reset[ch]` is a register level in the system (`SNK.CHANNEL_RESET.CH_RST` ORed with `GLOBAL_CTRL.GLOBAL_RST`), so a pulse and a held level must both work. The ingress registers it once into `r_rst_d1` and uses `w_rst[ch] = cfg_channel_reset[ch] | r_rst_d1[ch]`, which stretches a one-cycle pulse to two cycles and lets every clear see the reset on a cycle where it is asserted. The stretch is what settles an output-register beat and an allocation that were launched on the cycle of the reset. The reset is written for one channel; every term is indexed by channel, so the other channels are not affected.

Table 3.4.4: Channel Reset Actions

State	Action on `w_rst[ch]`
Packet record queue	Write and read pointers cleared. A record arriving on the reset cycle is not pushed.
Spill hold, `r_pkt_rx_bytes`	Cleared.
`r_pkt_len_error`	Cleared. The channel's `sched_wr_error` contribution drops.
`r_pending_alloc`,	Cleared, so no allocation request of the old packet is issued.

State	Action on `w_rst[ch]`
`r_alloc_settle`	
Output register (`r_out_valid`), flush beat (`r_flush_valid`)	Dropped when the beat belongs to the channel, so no beat of the old packet enters the SRAM after the reset.
`s_axis_tready`	Low for a beat whose TID names a channel in reset.
`r_discard[ch]`	Set when a packet was part-way through the stream (received bytes non-zero and no flush of its own pending).

With `r_discard[ch]` set the ingress accepts the channel's beats without a record and drops them: `s_axis_tready` does not wait for a packet record, and no state is updated. `r_discard[ch]` clears on the dropped beat that carries `tlast`. The next beat is then an ordinary first beat of a new packet. The sender is not asked to do anything special: it finishes the packet it was sending and the tail is discarded. A packet whose beats had all been accepted (the flush beat aside) is not discarded.

Head-of-line blocking is unchanged. A beat for a channel with no record, and not discarding, still holds `s_axis_tready` low for every channel. After a reset the next packet for the channel waits for its descriptor as before.

The channel's write engine and SRAM partition are cleared by the blocks that own them; see AXI Write Engine and Sink Data Path. All of them settle within a few cycles of the reset, and the scheduler's own reset returns it to CH_IDLE, so a channel reset recovers the channel without `aresetn`.

18.4.6 Allocation

The output register holds one memory beat for the fill interface. Allocation follows the beats design:

- A channel requests `cfg_alloc_size` beats, or the remainder when less is free.
- The next allocation of a channel waits three cycles after its previous one so `fill_space_free` is current.
- A beat leaves the output register only when the fill interface accepts it.

18.5 Differences from RAPIDS Beats

Table 3.4.5: Ingress Delta

Item	RAPIDS Beats	RAPIDS
Stream beat to memory beat	one to one, full lanes	shifted by the offset, spill held
`s_axis_tstrb`	full only	partial on the `tlast` beat
Byte enables into SRAM	none	`{strb, data}` word
Packet record queue	absent	four per channel
Stream gating	buffers before the descriptor	waits for the packet record
Length mismatch	not checked	sticky channel error, cleared by channel reset
Channel reset	not applicable	per-channel clear, cut packet tail discarded

The beats chapter is Sink Data Path with AXIS.

Last Updated: 2026-09-30

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19 Sink SRAM Controller Specification

Module: snk_sram_controller_beats.sv **Location:** projects/components/dma-ip/rapids/rtl/macro_beats/ **Status:** Implemented **Last Updated:** 2025-01-10

19.1 Overview

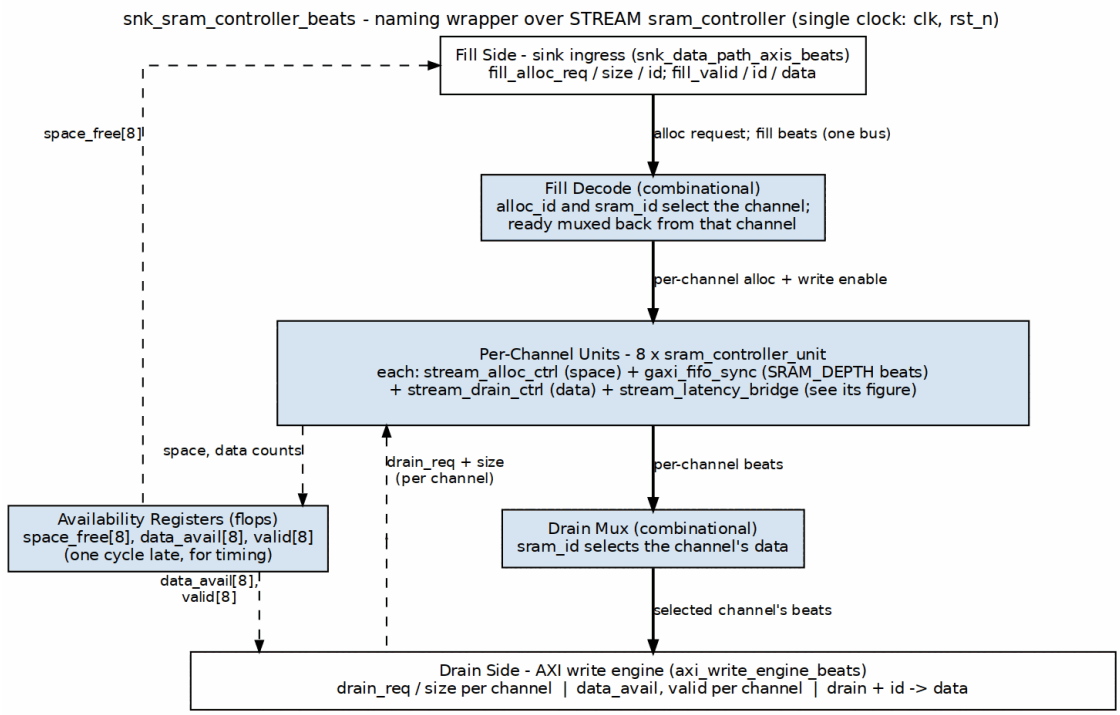
The Sink SRAM Controller is a naming wrapper, with no logic of its own, around STREAM's sram_controller – the one per-channel SRAM implementation both DMAs share since bdf4e0dff. It maps the RAPIDS fill_*/drain_* names onto STREAM's ports; the eight per-channel units (allocation counter, FIFO, drain counter, latency bridge) live inside STREAM's block. It manages 8 channel units, providing per-channel buffering with channel arbitration for the shared AXI write engine.

19.1.1 Key Features

- **One shared implementation:** instantiates STREAM's sram_controller once; it holds 8 sram_controller_units, each a gaxi_fifo_sync FIFO with stream_alloc_ctrl, stream_drain_ctrl and stream_latency_bridge
- **Channel Arbitration:** none inside the controller – the AXI write engine picks a channel and presents it on drain_id
- **Flow Control Integration:** per-channel stream_alloc_ctrl / stream_drain_ctrl virtual FIFOs; the drain counter's virtual depth is 2 x SRAM_DEPTH (stream BUG-011)
- **Space Tracking:** Reports available space per channel

19.1.2 Block Diagram

19.1.3 Figure 3.5.1: Sink SRAM Controller Block Diagram



Sink SRAM controller block diagram

Source: 15_snk_sram_controller.dot

19.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;                    // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
```

: Table 3.5.1: Sink SRAM Controller Parameters

19.3 Port List

19.3.1 Clock and Reset

Table 3.5.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

19.3.2 Fill Interface (ID-Selected)

Table 3.5.3: Fill Interface (ID-Selected)

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Allocation request (single, ID-selected)
`fill_alloc_size`	input	8	Beats to allocate
`fill_alloc_id`	input	CIW	Transaction ID selects the channel
`fill_space_free`	output	NC x SCW	Available space per channel
`fill_valid`	input	1	Fill data valid
`fill_ready`	output	1	Ready for fill data
`fill_id`	input	CIW	Transaction ID selects the channel
`fill_data`	input	DW	Fill data

19.3.3 Arbitrated Drain Interface (to AXI Write Engine)

Table 3.5.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
`drain_data_avail`	output	NC x SCW	Data available per channel
`drain_req`	input	NC	Drain reservation request per channel
`drain_size`	input	NC x 8	Beats to reserve

Signal	Direction	Width	Description
`drain_valid`	output	NC	Drain valid, registered (for arbitration)
`drain_valid_comb`	output	NC	Drain valid, combinational (gates `m_axi_wvalid`)
`drain_read`	input	1	Consumer read strobe
`drain_id`	input	CIW	Channel ID select for drain
`drain_data`	output	DW	Drain data (muxed from selected channel)

19.3.4 Debug Interface

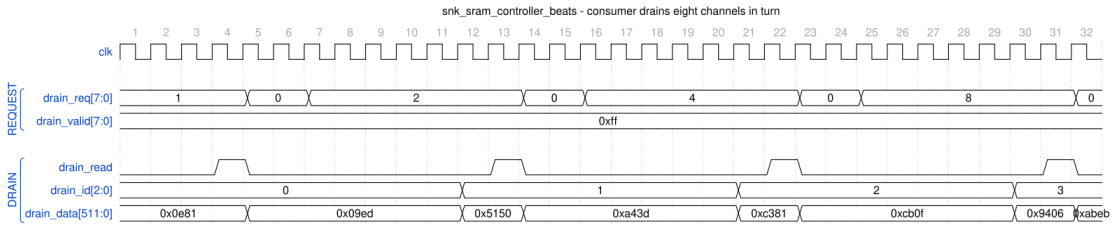
Table 3.5.5: Debug Interface

Signal	Direction	Width	Description
`dbg_bridge_pending`	output	NC	Bridge has a pending beat per channel
`dbg_bridge_out_valid`	output	NC	Bridge output valid per channel

19.4 Drain Selection

There is no arbiter inside the controller. Every channel unit raises its own `drain_valid[ch]` / `drain_size[ch]` when it has data; the AXI write engine picks a channel, drives it on `drain_id`, and the controller steers that channel's beats to the single `drain_*` port. Switching channels costs nothing beyond the consumer's own decision, and a channel keeps presenting `drain_valid` until it is empty.

19.4.1 Figure 3.5.2: Drain Selection by the Consumer



snk_sram_controller_beats - consumer drains eight channels in turn

Source: snk_sram_controller_drain_select.json, captured from `dv/tests/macro_beats/test_snk_sram_controller_beats.py` (`multi_channel`, 8 channels, 512-bit data, 512-entry SRAM, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: all eight channels have data (`drain_valid` = 0xff) and the consumer walks them in turn. It raises one bit of `drain_req` at a time; the controller answers with `drain_id` naming that channel and `drain_data` presenting its head word on the same cycle, and the consumer's `drain_read` pulse pops it. Each channel's turn lasts nine cycles here because the test releases the request and re-arms the next one; the controller itself needs no gap between channels.

19.5 Integration with AXI Write Engine

```
snk_sram_controller_beats #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_snk_sram_ctrl (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Per-channel fill interface
    .fill_alloc_req         (fill_alloc_req),
    .fill_alloc_size        (fill_alloc_size),
    .fill_space_free        (fill_space_free),
    .fill_valid             (fill_valid),
    .fill_ready             (fill_ready),
    .fill_data              (fill_data),

    // Arbitrated drain interface
    .drain_req              (sram_drain_req),
    .drain_id               (sram_drain_id),
    .drain_size             (sram_drain_beats),

    // Status
    .drain_data_avail       (channel_data_avail),
    .dbg_bridge_pending     (channel_empty),
    .dbg_bridge_out_valid   (channel_full)
);
```

Last Updated: 2025-01-10

20 Source Data Path Specification

Module: `src_data_path_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

20.1 Overview

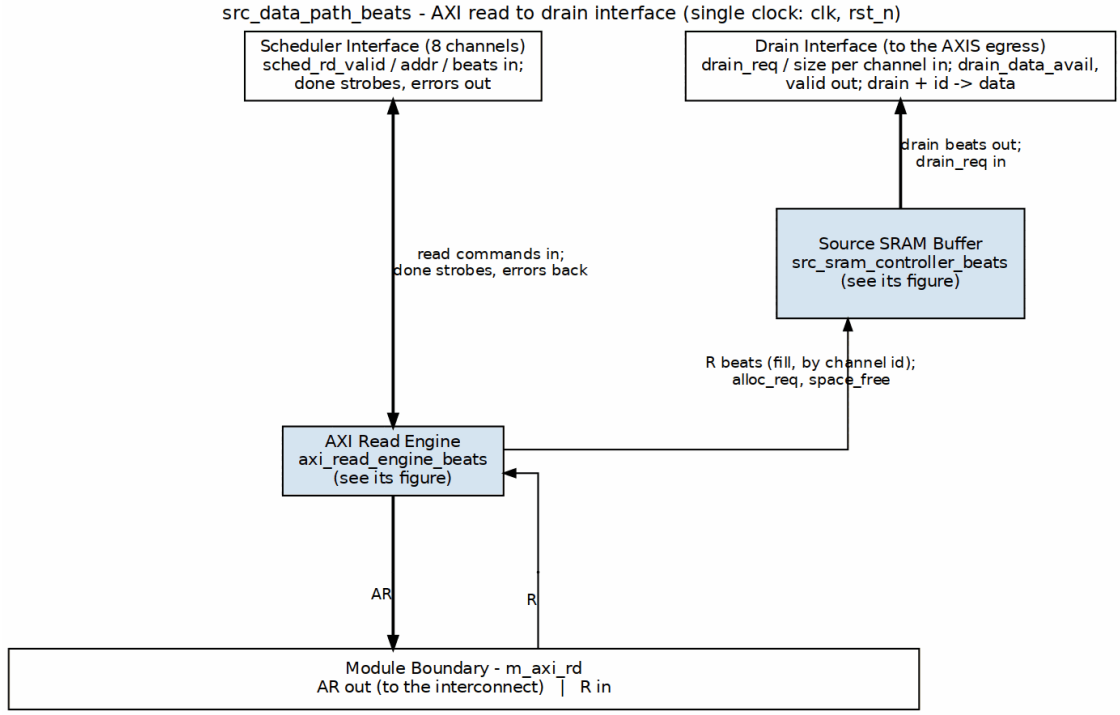
The Source Data Path integrates the AXI read engine and SRAM controller for memory-to-network transfers. Data is read from system memory via AXI4, buffered in SRAM, and delivered through the drain interface to the network.

20.1.1 Key Features

- **AXI Read Engine:** Streaming burst reads from system memory
- **8-Channel SRAM Controller:** Per-channel buffering with flow control
- **Beat-Level Tracking:** Precise flow control at beat granularity
- **Scheduler Integration:** Receives transfer requests from scheduler

20.1.2 Block Diagram

20.1.3 Figure 3.6.1: Source Data Path Block Diagram



Source data path block diagram

Source: 16_source_data_path.dot

20.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int ADDR_WIDTH = 64;
parameter int DATA_WIDTH = 512;
parameter int AXI_ID_WIDTH = 8;
parameter int SRAM_DEPTH = 512;
parameter int AR_MAX_OUTSTANDING = 8;
parameter int R_PHASE_FIFO_DEPTH = 64;
```

: Table 3.6.1: Source Data Path Parameters

20.3 Data Flow

20.3.1 Figure 3.6.2: Source Path Data Flow Sequence

```
1. Scheduler Read Request
   - sched_rd_valid = 1
   - sched_rd_addr = source address
   - sched_rd_beats = N
   - sched_rd_id = channel
        |
        v
2. Space Check (stream_alloc_ctrl inside STREAM's sram_controller)
   - Check src_drain_space_free[channel] >= N
   - Allocate N beats of space
        |
        v
3. AXI Read Transaction
   - axi_read_engine issues AR
   - m_axi_araddr = sched_rd_addr
   - m_axi_arlen = calculated from beats
        |
        v
4. R Channel Data
   - Data arrives on m_axi_rdata
   - Written to channel's SRAM partition
        |
        v
5. SRAM Write Complete
   - stream_drain_ctrl notified via the unit's latency bridge
   - Data available for drain
        |
        v
6. Drain Interface
   - src_drain_valid = 1
   - Data delivered to network master
        |
        v
7. Completion
   - sched_rd_done_strobe = 1
   - Scheduler notified of completion
```

20.4 Port List

20.4.1 Clock and Reset

Table 3.6.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

20.4.2 Configuration

Table 3.6.3: Configuration

Signal	Direction	Width	Description
`cfg_axi_rd_xfer_beats`	input	8	AXI read transfer size in beats (all channels)

20.4.3 Scheduler Interface

Table 3.6.4: Scheduler Interface

Signal	Direction	Width	Description
`sched_rd_valid`	input	NC	Channel requests read
`sched_rd_addr`	input	NC x AW	Source addresses
`sched_rd_beats`	input	NC x 32	Beats remaining to read
`sched_rd_done_strobe`	output	NC	Burst completed (pulsed 1 cycle)
`sched_rd_beats_done`	output	NC x 32	Beats completed in burst
`sched_rd_error`	output	NC	Sticky error flag per channel

20.4.4 AXI Read Master Interface

Table 3.6.5: AXI Read Master Interface

Signal	Direction	Width	Description
`m_axi_arvalid`	output	1	AR channel valid

Signal	Direction	Width	Description
m_axi_arready	input	1	AR channel ready
m_axi_araddr	output	AW	Read address
m_axi_arlen	output	8	Burst length
m_axi_arsize	output	3	Burst size
m_axi_arburst	output	2	Burst type
m_axi_arid	output	ID_W	Transaction ID
m_axi_rvalid	input	1	R channel valid
m_axi_rready	output	1	R channel ready
m_axi_rdata	input	DW	Read data
m_axi_rresp	input	2	Read response
m_axi_rid	input	ID_W	Response ID
m_axi_rlast	input	1	Last beat

20.4.5 Drain Interface (Output to Network)

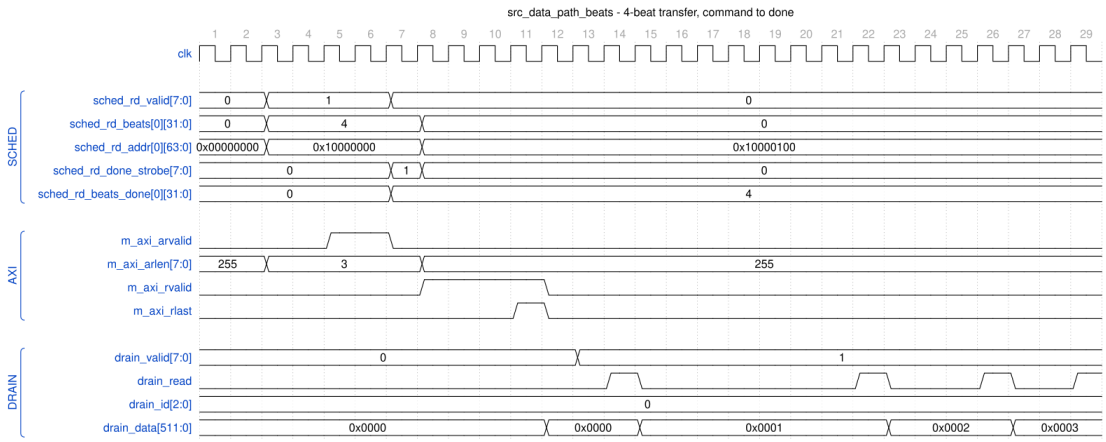
Table 3.6.6: Drain Interface

Signal	Direction	Width	Description
drain_data_avail	output	NC x SCW	Data available per channel
drain_req	input	NC	Drain reservation request per channel
drain_size	input	NC x 8	Beats to reserve
drain_valid	output	NC	Drain data valid per channel
drain_valid_comb	output	NC	Combinational per-channel valid (beat gate)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from

Signal	Direction	Width	Description
			selected channel)

20.5 Timing Diagram

20.5.1 Figure 3.6.3: Source Path Transfer Timing



src_data_path_beats - 4-beat transfer, command to done

Source: src_data_path_transfer.json, captured inside `rapids_core_beats.u_src.u_source_data_path`, captured from `dv/tests/top_beats/test_rapids_core_beats.py` (source path, channel 0, 4 beats, 512-bit data, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: the scheduler holds `sched_rd_valid[0]` with 4 beats at 0x10000000. The read engine issues the AR (`arlen = 3`) two cycles later; the four R beats land from cycle 8, `rlast` on the fourth, and the cycle after it the engine pulses `sched_rd_done_strobe[0]` with `beats_done = 4` – the scheduler advances its address to 0x10000100 on that strobe. The data reaches the source SRAM two cycles after the last R beat (`drain_valid[0]` rises) and the AXIS side drains it one word per `drain_read` pulse at the pace the egress consumer allows.

20.6 Integration Example

```
src_data_path_beats #(
    .NUM_CHANNELS(8),
    .ADDR_WIDTH(64),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_source_path (
    .clk                    (clk),
```

```
    .rst_n                  (rst_n),

    // Scheduler interface
    .sched_rd_valid         (sched_rd_valid),
    .sched_rd_addr          (sched_rd_addr),
    .sched_rd_beats         (sched_rd_beats),
    .sched_rd_done_strobe   (sched_rd_done_strobe),
    .sched_rd_beats_done    (sched_rd_beats_done),
    .sched_rd_error         (sched_rd_error),

    // AXI read master
    .m_axi_arvalid          (src_axi_arvalid),
    .m_axi_arready          (src_axi_arready),
    .m_axi_araddr           (src_axi_araddr),
    .m_axi_arlen            (src_axi_arlen),
    .m_axi_rvalid           (src_axi_rvalid),
    .m_axi_rready           (src_axi_rready),
    .m_axi_rdata            (src_axi_rdata),
    .m_axi_rlast            (src_axi_rlast),

    // Drain interface
    .drain_valid            (src_drain_valid),
    .drain_data             (src_drain_data),
    .drain_id               (src_drain_id),
    .drain_data_avail       (src_data_avail)
);
```

Last Updated: 2025-01-10

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21 Source Data Path with AXIS (Egress)

Module: `src_data_path_axis.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

21.1 Overview

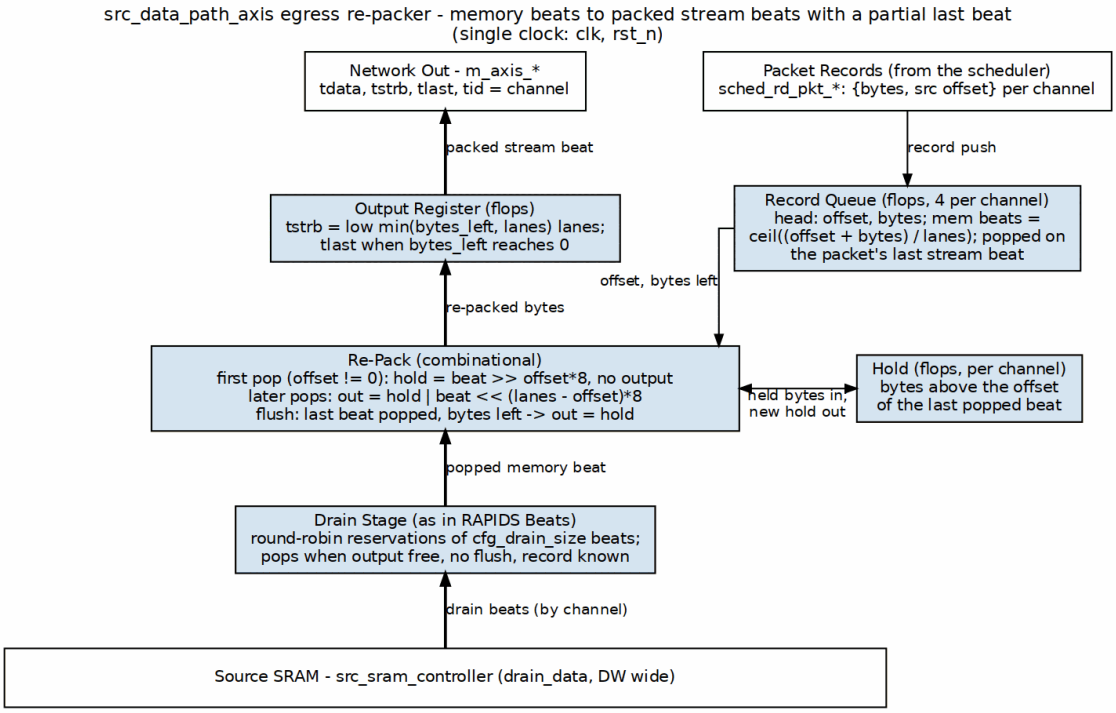
This module puts an AXI-Stream master behind the source data path. It pops beat-aligned memory beats from the SRAM and re-packs them into stream beats that start at lane 0, so a packet leaves as exactly the bytes its descriptor named.

A descriptor's source address may start in the middle of a beat. The scheduler supplies that offset and the byte length in a packet record. The module shifts each memory beat down by the offset, holds the bytes that belong to the next stream beat, and marks the last stream beat with a contiguous strobe and `tlast`. One descriptor produces one AXI-Stream packet.

21.1.1 Key Features

- **Barrel shift by the source offset:** memory lane `offset + l` goes out on stream lane `l`.
- **Partial last beat:** `tstrb` is the low `min(bytes_left, SW)` lanes and `tlast` marks the final beat.
- **Packets follow descriptors,** not drain reservations.
- **Two-stage reservation and drain,** unchanged from RAPIDS Beats. The drain stage advances on memory-beat pops, not on stream handshakes.
- **Per-channel state:** hold, bytes-left and beats-left per channel, so reservations of different channels interleave.
- **Packet record queue:** four records per channel, fed by the scheduler.

21.1.2 Figure 3.7.1: Egress Shifter



Egress shifter

Source: 02_egress_shifter.dot

21.2 Parameters

Table 3.7.1: Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels (alias NC).
ADDR_WIDTH	64	Address width (alias AW).
DATA_WIDTH	512	Data width in bits (alias DW). The board uses 256.
AXI_ID_WIDTH	8	AXI ID width (alias IW).
SRAM_DEPTH	512	Words per channel partition.
SEG_COUNT_WIDTH	$clog2(SRAM_DEPTH) + 1	Space counter width.
PIPELINE	1	SRAM read pipeline setting.

Parameter	Default	Description
AR_MAX_OUTSTANDING	8	ARs the read engine may have in flight.
AXIS_ID_WIDTH	8	tid width.
AXIS_DEST_WIDTH	4	tdest width.
AXIS_USER_WIDTH	1	tuser width.
SW	DW / 8	Byte lanes per beat.
OFF_W	$clog2(SW)	Width of the packet-record offset.

21.3 Interfaces

21.3.1 Configuration

Table 3.7.2: Configuration

Signal	Direction	Width	Description
cfg_axi_rd_xfer_beats	input	8	Read burst size cap in beats, minus one.
cfg_drain_size	input	8	Beats per drain reservation. 0 is treated as 1.
cfg_channel_reset	input	NC	Per-channel reset, level or pulse. Clears the channel's queue, hold, egress counters, drain reservations and error flag.

21.3.2 Scheduler Interface

Table 3.7.3: Scheduler Interface

Signal	Direction	Width	Description
sched_rd_valid	input	NC	Channel requests reads.
sched_rd_addr	input	NC x AW	Working source byte address.

Signal	Direction	Width	Description
sched_rd_beats	input	NC x 32	Beats still to read.
sched_rd_pkt_valid	input	NC	Record pulse from the scheduler.
sched_rd_pkt_ready	output	NC	Record queue not full.
sched_rd_pkt_bytes	input	NC x 32	Packet length in bytes.
sched_rd_pkt_offset	input	NC x OFF_W	Source offset in the first memory beat.
sched_rd_done_strobe	output	NC	AR issued.
sched_rd_beats_done	output	NC x 32	Beats in that AR.
sched_rd_error	output	NC	Sticky read error.

21.3.3 AXI-Stream Master

Table 3.7.4: AXI-Stream Master

Signal	Direction	Width	Description
m_axis_tdata	output	DW	Packed data.
m_axis_tstrb	output	SW	Contiguous from lane 0. Partial only on the `tlast` beat.
m_axis_tlast	output	1	Last beat of the packet.
m_axis_tid	output	AXIS_ID_WIDTH	Channel.
m_axis_tdest	output	AXIS_DEST_WIDTH	Channel.
m_axis_tuser	output	AXIS_USER_WIDTH	Zero.
m_axis_tvalid	output	1	Beat valid.
m_axis_tready	input	1	Consumer ready.

21.3.4 Other Ports

The AXI4 read master is that of the AXI Read Engine, with IW-bit IDs and DW-bit data. The debug outputs are dbg_rd_all_complete, dbg_r_beats_rcvd, dbg_sram_writes, dbg_arb_request, dbg_sram_bridge_pending, dbg_sram_bridge_out_valid, dbg_axis_beats_sent and dbg_axis_packets_sent.

21.4 Egress Operation

21.4.1 Reservation and Drain

Unchanged from RAPIDS Beats. A round-robin decision per cycle selects a channel whose available data, less reservations still in flight, reaches cfg_drain_size beats. It may also select one whose fill is finished and has any data left. Each decision is a one-cycle drain_req of up to cfg_drain_size beats. Up to four reservations queue ahead of the drain stage, which loads the next entry on the last beat of the current one.

21.4.2 Pop

A memory beat is popped when all of these hold:

- The drain stage is active on channel c and has a beat.
- The output register is free or being taken.
- No flush is pending.
- The channel's packet record is known.

Each pop counts against the reservation block, so the drain stage advances on pops.

21.4.3 Shift and Hold

With off the head record's offset:

Table 3.7.5: Egress Shift Rules

Situation	Action
off = 0	Every pop emits a stream beat.
off > 0, first pop	The pop only primes the hold with beat >> off*8. It emits only when the packet fits in that single memory beat.
off > 0, later pops	Emit hold | (beat << (SW - off)*8). The new hold is beat

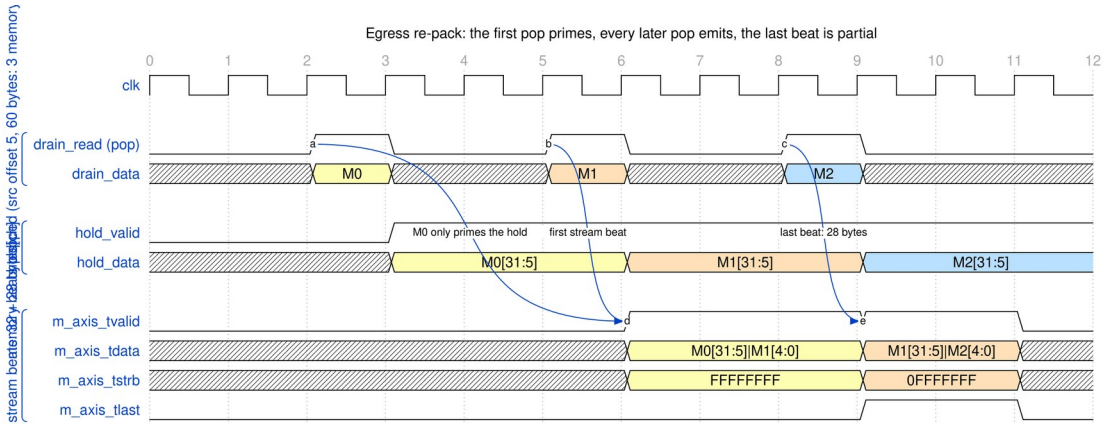
Situation	Action
	`>> off*8`.
Last memory beat popped, bytes left	The flush emits the hold alone.

The memory beat count of a packet is `(off + bytes + SW - 1) >> OFF_W`, computed in 33 bits.

21.4.4 Strobe and Last

Each emitted beat carries `n = min(bytes_left, SW)` bytes. `tstrb` is the low `n` lanes. `tlast` is set on the beat where `bytes_left` equals `n`. That beat also retires the packet record and clears the channel's hold, started flag and counters.

21.4.5 Waveform 3.7: Egress Re-pack



Egress re-pack

Source: 01_egress_repack.json

Offset 5, 60 bytes and a 256-bit beat give three memory beats. The first pop, M0, only primes the hold. The second pop emits the first stream beat with 32 bytes. The third pop emits the last beat with 28 bytes, `tstrb` `0FFFFFFF` and `tlast`.

21.5 Channel Reset

`cfg_channel_reset[ch]` is a register level, so a pulse and a held level must both work. As in the sink ingress the data path forms `w_rst[ch] = cfg_channel_reset[ch] | r_rst_d1[ch]`, a two-cycle stretch of a pulse. The read engine clears its error flag and settles the reads in flight (see AXI Read Engine); the data path clears everything that would otherwise let the old descriptor keep streaming:

Table 3.7.6: Channel Reset Actions

State	Action on `w_rst[ch]`
Packet record queue	Pointers cleared. A record arriving on the reset cycle is not pushed.
Egress state	`r_eg_started`, `r_hold_valid`, `r_hold_data`, `r_eg_bytes_left` and `r_eg_mem_left` cleared.
Drain reservation	Grants for the channel are masked. `r_drain_tminus1[ch]` is cleared.
Reservation queue	Entries that belong to the channel are zeroed. An entry of size 0 loads the drain stage idle, so the drain stage skips it. A reservation being drained for the channel is dropped.
Output register	A beat already in `r_out` completes, so `m_axis` stays stable.

The channel's SRAM partition is cleared by `src_sram_controller`, which holds one single-channel `sram_controller` per channel with a registered per-channel reset `r_ch_rst_n[ch]`. The shared STREAM controller has no channel reset of its own and is not modified. Reads of the old descriptor that are already on the bus are discarded by the read engine, so no stale data reaches the cleared partition.

A packet cut by the reset is left unterminated. If the reset lands after the first beat of a packet has gone out and before its `tlast` beat, the data path does not send a `tlast` and does not send a null terminator. The downstream receiver sees the packet stop. A packet that had not started, or whose `tlast` beat was already in `r_out` when the reset hit, is unaffected. A receiver that needs to resynchronize after a channel reset treats the reset as the end of the packet.

Other channels are unaffected: every term above is indexed by channel, the drain and reservation stages skip only the channel in reset, and their beats keep flowing through the shared `m_axis`.

21.6 Differences from RAPIDS Beats

Table 3.7.7: Egress Delta

Item	RAPIDS Beats	RAPIDS
Stream beat from memory beat	one to one, full lanes	shifted by the offset, hold merged
`m_axis_tstrb`	all ones	contiguous, partial on the last beat
`m_axis_tlast`	on the drain block boundary	on the last byte of

Item	RAPIDS Beats	RAPIDS
		the descriptor
Drain advance	stream handshakes	memory-beat pops
Packet framing	one per drain block	one per descriptor
Packet record queue	absent	four per channel
Channel reset	not applicable	per-channel clear of every stage, cut packet left unterminated
Reservation stage, allocation of drain		Unchanged

The beats chapter is Source Data Path with AXIS.

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22 Source SRAM Controller Specification

Module: `src_sram_controller_beats.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro_beats/` **Status:** Implemented **Last Updated:** 2025-01-10

22.1 Overview

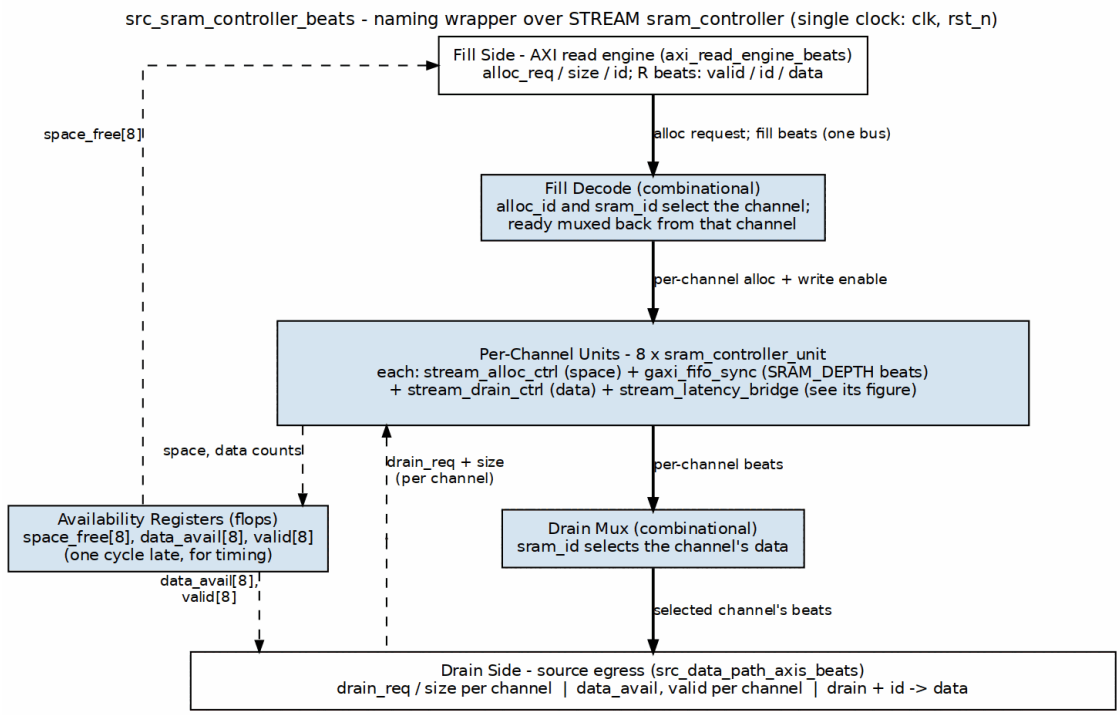
The Source SRAM Controller is a naming wrapper, with no logic of its own, around STREAM's `sram_controller` – the one per-channel SRAM implementation both DMAs share since `bdf4e0dff`. It maps the RAPIDS `fill_*`/`drain_*` names onto STREAM's ports; the eight per-channel units (allocation counter, FIFO, drain counter, latency bridge) live inside STREAM's block. It manages 8 channel units for the source data path, providing per-channel buffering with channel arbitration for data delivery to the network.

22.1.1 Key Features

- **One shared implementation:** instantiates STREAM's `sram_controller` once; it holds 8 `sram_controller_units`, each a `gaxi_fifo_sync` FIFO with `stream_alloc_ctrl`, `stream_drain_ctrl` and `stream_latency_bridge`
- **Channel Arbitration:** none inside the controller – the AXIS egress arbiter picks a channel and presents it on `drain_id`
- **Flow Control Integration:** per-channel `stream_alloc_ctrl` / `stream_drain_ctrl` virtual FIFOs; the drain counter's virtual depth is 2 x SRAM_DEPTH (stream BUG-011)
- **Data Availability Tracking:** Reports available data per channel

22.1.2 Block Diagram

22.1.3 Figure 3.8.1: Source SRAM Controller Block Diagram



Source SRAM controller block diagram

Source: 18_src_sram_controller.dot

22.2 Parameters

```
parameter int NUM_CHANNELS = 8;
parameter int DATA_WIDTH = 512;
parameter int SRAM_DEPTH = 512;                    // Per-channel depth
parameter int ADDR_WIDTH = $clog2(SRAM_DEPTH);
```

: Table 3.8.1: Source SRAM Controller Parameters

22.3 Port List

22.3.1 Clock and Reset

Table 3.8.2: Clock and Reset

Signal	Direction	Width	Description
`clk`	input	1	System clock
`rst_n`	input	1	Active-low reset

22.3.2 Fill Interface (ID-Selected, from AXI Read Engine)

Table 3.8.3: Fill Interface (ID-Selected)

Signal	Direction	Width	Description
`fill_alloc_req`	input	1	Allocation request (single, ID-selected)
`fill_alloc_size`	input	8	Beats to allocate
`fill_alloc_id`	input	CIW	Transaction ID selects the channel
`fill_space_free`	output	NC x SCW	Available space per channel
`fill_valid`	input	1	Fill data valid
`fill_ready`	output	1	Ready for fill data
`fill_id`	input	CIW	Transaction ID selects the channel
`fill_data`	input	DW	Fill data

22.3.3 Arbitrated Drain Interface (to Network)

Table 3.8.4: Arbitrated Drain Interface

Signal	Direction	Width	Description
`drain_data_avail`	output	NC x SCW	Data available per channel
`drain_req`	input	NC	Drain reservation request per channel
`drain_size`	input	NC x 8	Beats to reserve

Signal	Direction	Width	Description
drain_valid	output	NC	Drain data valid per channel
drain_valid_comb	output	NC	Combinational per-channel valid (beat gate)
drain_read	input	1	Consumer read strobe
drain_id	input	CIW	Channel ID select for drain
drain_data	output	DW	Drain data (muxed from selected channel)

22.3.4 Debug Interface

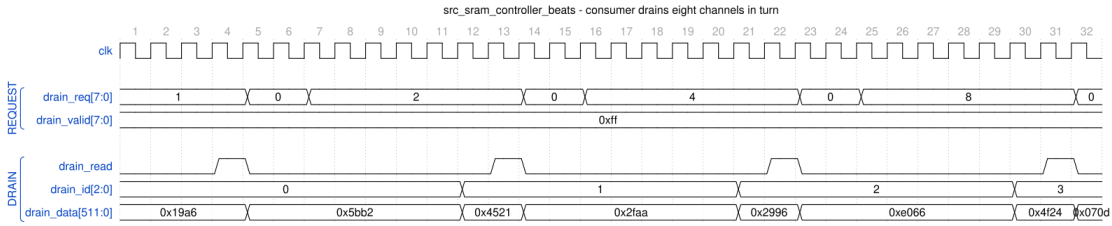
Table 3.8.5: Debug Interface

Signal	Direction	Width	Description
dbg_bridge_pending	output	NC	Bridge has a pending beat per channel
dbg_bridge_out_valid	output	NC	Bridge output valid per channel

22.4 Drain Selection

There is no arbiter inside the controller. Every channel unit raises its own `drain_valid[ch]` / `drain_size[ch]` when it has data; the AXIS egress arbiter picks a channel, drives it on `drain_id`, and the controller steers that channel's beats to the single `drain_*` port. Switching channels costs nothing beyond the consumer's own decision, and a channel keeps presenting `drain_valid` until it is empty.

22.4.1 Figure 3.8.2: Drain Selection by the Consumer



src_sram_controller_beats - consumer drains eight channels in turn

Source: src_sram_controller_drain_select.json, captured from `dv/tests/macro_beats/test_src_sram_controller_beats.py` (`multi_channel`, 8 channels, 512-bit data, 512-entry SRAM, `REG_LEVEL=GATE`) with `WAVES=1`.

Reading it: the same selection contract as the sink controller (Figure 3.5.2). With `drain_valid` = 0xff the consumer asks for one channel at a time through `drain_req`; `drain_id` and `drain_data` follow the request on the same cycle and `drain_read` pops the word. The nine-cycle turn per channel is the test's pacing, not the controller's.

22.5 Comparison: Sink vs Source SRAM Controller

Table 3.8.6: Sink vs Source Controller Comparison

Aspect	Sink Controller	Source Controller
Fill Source	Network ingress	AXI read data
Drain Destination	AXI write engine	Network egress
Fill Interface	Per-channel from network	Per-channel from AXI R
Drain Interface	Arbitrated SRAM read	Arbitrated drain to network
Primary Use	Network -> Memory	Memory -> Network

22.6 Integration Example

```
src_sram_controller_beats #(
    .NUM_CHANNELS(8),
    .DATA_WIDTH(512),
    .SRAM_DEPTH(512)
) u_src_sram_ctrl (
    .clk                    (clk),
    .rst_n                  (rst_n),

    // Per-channel fill interface (from AXI read engine)
    .fill_alloc_req         (fill_alloc_req),
    .fill_alloc_size        (fill_alloc_size),
    .fill_space_free        (fill_space_free),
    .fill_valid             (fill_valid),
    .fill_ready             (fill_ready),
    .fill_data              (fill_data),

    // Arbitrated drain interface (to network)
```

```
    .drain_valid            (src_drain_valid),
    .drain_data             (src_drain_data),
    .drain_id               (src_drain_id),

    // Status
    .drain_data_avail       (channel_data_avail),
    .dbg_bridge_pending     (channel_empty),
    .dbg_bridge_out_valid   (channel_full)
);
```

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23 RAPIDS Core, Sink Half and Source Half

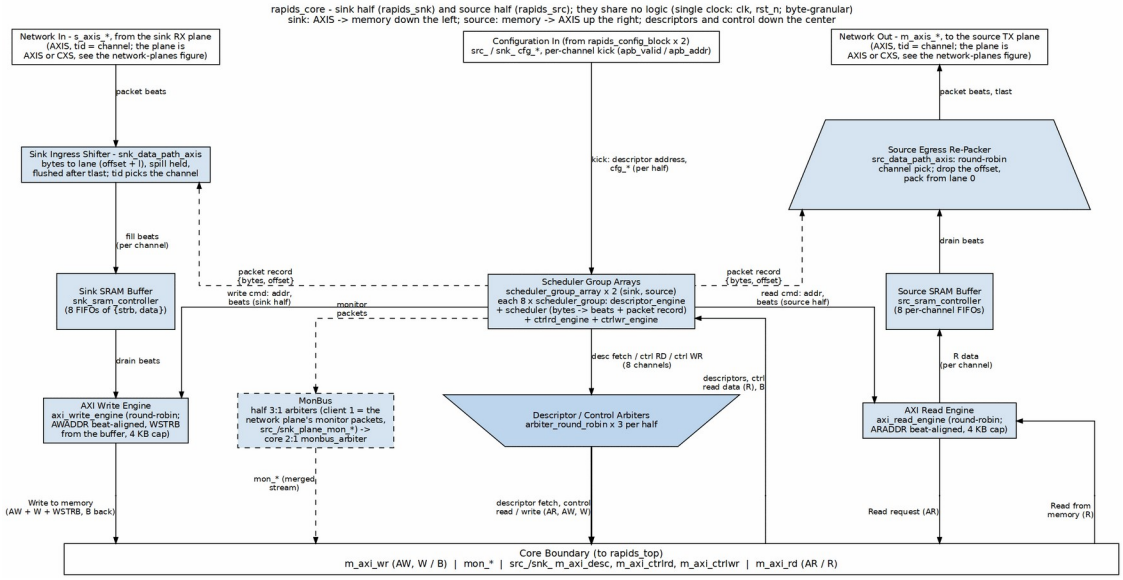
Module: `rapids_core.sv`, `rapids_snk.sv`, `rapids_src.sv` **Location:** `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

23.1 Overview

`rapids_core` is a thin structural wrapper over two independent halves. `rapids_src` moves memory to the network. `rapids_snk` moves the network to memory. The halves share no logic and no signals. The core only presents one boundary and merges the two MonBus streams.

The structure, the port naming convention and the register and monitor plumbing are those of RAPIDS Beats. This chapter records what the byte-granular RTL contains and how it differs. The full port tables are in RAPIDS Core Beats, with the port renames below applied.

23.1.1 Figure 3.9.1: Core Block Diagram



Core block diagram

Source: 04_rapids_core.dot

23.2 Byte-Granular Content

23.2.1 Packet Record Nets

Each half declares the packet record nets of its direction and connects them between its scheduler group array and its AXIS data path:

Table 3.9.1: Packet Record Nets per Half

Half	Nets	Producer	Consumer
rapids_src	sched_rd_pkt_valid, _ready, _bytes, _offset	array read set	src_data_path_axis
rapids_snk	sched_wr_pkt_valid, _ready, _bytes, _offset	array write set	snk_data_path_axis

The offset nets are $clog2(DW/8) bits wide. The unused direction of each array is tied off: valid, bytes and offset are left open and ready is held at all ones so nothing stalls. The record nets are internal, so the core port list has no packet record ports.

23.2.2 Port Renames

The core ports that carry the beat-count unit are renamed. Everything else keeps its Beats name.

Table 3.9.2: Core Port Renames

RAPIDS Beats	RAPIDS
cfg_axi_rd_xfer	cfg_axi_rd_xfer_beats
cfg_axi_wr_xfer	cfg_axi_wr_xfer_beats
src_dbg_r_rcvd	src_dbg_r_beats_rcvd
src_dbg_axis_sent	src_dbg_axis_beats_sent
snk_dbg_axis_received	snk_dbg_axis_beats_received

23.2.3 Data Ports

Table 3.9.3: Data Ports

Interface	Signals	Notes
Source AXIS	m_axis_*	tstrb is contiguous, partial

Interface	Signals	Notes
		on the `tlast` beat.
Sink AXIS	`s_axis_*`	`tid` selects the channel. Partial `tstrb` only on `tlast`.
Source AXI read	`m_axi_rd_*`	ARADDR is beat-aligned, bursts stop at 4 KB.
Sink AXI write	`m_axi_wr_*`	`wstrb` carries the stored byte enables.

23.3 Parameters

Table 3.9.4: Core Parameters

Parameter	Default	Description
`NUM_CHANNELS`	8	Channels.
`ADDR_WIDTH`	64	Address width.
`DATA_WIDTH`	512	Data width in bits. The board uses 256.
`AXI_ID_WIDTH`	8	AXI ID width.
`SRAM_DEPTH`	512	Words per channel.
`PIPELINE`	1	Write engine pipelining.
`AR_MAX_OUTSTANDING`, `AW_MAX_OUTSTANDING`	8	Outstanding bursts.
`W_PHASE_FIFO_DEPTH`, `B_PHASE_FIFO_DEPTH`	64, 16	Write engine queues.
`AXIS_ID_WIDTH`, `AXIS_DEST_WIDTH`, `AXIS_USER_WIDTH`	8, 4, 1	AXIS side widths.
`DESC_MON_BASE_AGENT_ID`	16	Descriptor engine agents.
`SCHED_MON_BASE_AGENT_ID`	48	Scheduler agents.

Parameter	Default	Description
DESC_AXI_MON_AGENT_ID	8	Descriptor AXI monitor.
MON_UNIT_ID, MON_MAX_TRANSACTIONS	1, 16	Monitor unit id and table depth.
USE_AXI_MONITORS	1	0 omits the monitor hardware.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out extended addressing.
GEN_MON	1	0 omits the completion and error emitters.

The network endpoint and its monitor-lite moved to the top's network planes (rapids TASK-024), so SNK_AXIS_MON_AGENT_ID, SRC_AXIS_MON_AGENT_ID and ACLK_MHZ are top parameters now. The core takes each plane's monbus packets on src_plane_mon_* / snk_plane_mon_*. See docs/rapids_planes.md.

23.3.1 Channel Reset Routing

Each half has its own configuration block, and that block drives the half's per-channel reset from its CHANNEL_RESET.CH_RST field ORed with GLOBAL_CTRL.GLOBAL_RST. The core exposes it as src_cfg_channel_reset and snk_cfg_channel_reset, and each half forwards it twice: to its scheduler group array (scheduler, descriptor engine, control engines) and to its byte data path (src_data_path_axis or snk_data_path_axis, through which it reaches the AXI engine and the SRAM wrapper). A reset therefore reaches every block that holds a sticky flag or an in-flight beat of the channel, and a channel reset recovers every error class without aresetn. Table 3.9.5 lists the blocks reached.

Table 3.9.5: Channel Reset Routing

Half	Blocks reached	Chapter
rapids_snk	scheduler, axi_write_engine, snk_sram_controller, snk_data_path_axis	Sink Data Path AXIS
rapids_src	scheduler, axi_read_engine, src_sram_controller, src_data_path_axis	Source Data Path AXIS

The two halves are independent: a reset of a sink channel does not touch the source channel with the same index, so software resets both directions of a channel by writing both halves' reset fields. A reset of one channel never disturbs another channel of the same half.

23.4 Differences from RAPIDS Beats

Table 3.9.6: Core Delta

Item	RAPIDS Beats	RAPIDS
Half structure, monitor merge		Unchanged
Packet record nets inside each half	absent	present
Data path instances	`*_axis_beats`	byte-granular `*_axis`
Beat-unit port names	no suffix	`_beats` suffix, see Table 3.9.2

The Beats chapters for the halves are folded into the core chapter there.

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24 RAPIDS Registers Specification

Module: `rapids_regs.sv` (generated) **Source:** `projects/components/dma-ip/rapids/rtl/macro_beats/rapids_regs.rdl`, `rapids_mon_regs.rdl` **Generated:** `projects/components/dma-ip/rapids/regs/generated/rtl/rapids_regs.sv` **Status:** Implemented (PeakRDL-generated)

24.1 Overview

`rapids_regs` is the PeakRDL-generated register block for the RAPIDS Beats top-level. It exposes configuration and status through a single command/response CPU interface and drives the hardware via a `hwif_out` structure that `rapids_config_block` maps onto the core/monitor `cfg_*` signals.

A single addrmap (one APB slave) contains two regfiles:

- **Base config/status** registers at `0x100-0x3FF`.
- A nested **`rapids_mon_regs`** monitor regfile at base `0x1000`, decoded under `hwif_out.MON.*` (all DAXMON / RDMON / WRMON / MON_FIFO and per-monitor performance registers).

Because the monitor regfile lives at `0x1000`, the register-block address decode requires at least **13 bits** (`cpuif_addr` is a 13-bit compare in the generated RTL). The APB address bus feeding the block (`APB_ADDR_WIDTH`) must therefore be >= 13 to reach the monitor regfile.

24.2 Base Register Map (0x100-0x3FF)

Table 3.10.1: Base Register Map

Offset	Name	Access	Description
0x100	GLOBAL_CTRL	RW	`GLOBAL_EN[0]`, `GLOBAL_RST[1]` (level; write 0 to release)
0x104	GLOBAL_STATUS	RO	`SYSTEM_IDLE[0]` (all channels idle)
0x108	VERSION	RO	`MINOR[7:0]=0x5A`, `MAJOR[15:8]`, `NUM_CHANNELS[23:16]=0x08`

Offset	Name	Access	Description
0x120	CHANNEL_ENABLE	RW	CH_EN[7:0] per-channel enable
0x124	CHANNEL_RESET	RW	CH_RST[7:0] per-channel reset (level; write 0 to release)
0x140	CHANNEL_IDLE	RO	CH_IDLE[7:0] – not driven at the top level, reads 0
0x144	DESC_ENGINE_IDLE	RO	Per-channel descriptor-engine idle
0x148	SCHEDULER_IDLE	RO	Per-channel scheduler idle (excludes CH_ERROR)
0x150-0x16C	CH_STATE[0..7].STATE	RO	Per-channel FSM state (stride 0x4)
0x170	SCHED_ERROR	RO	Per-channel sticky scheduler error
0x174	AXI_RD_COMPLETE	RO	Per-channel read-complete flags (SRC half only; reads 0 in SNK)
0x178	AXI_WR_COMPLETE	RO	Per-channel write-complete flags – not driven, reads 0
0x200	SCHED_TIMEOUT_CYCLES	RW	TIMEOUT_CYCLES[31:0] write-progress window (reset 1000)
0x204	SCHED_CONFIG	RW	SCHED_EN, TIMEOUT_EN, ERR_EN, COMPL_EN, PERF_EN
0x208	SCHED_TIMEOUT_LIMIT	RW	LIMIT[7:0] consecutive-timeout escalation limit (reset 4, 0 = never)
0x220	DESCENG_CONFIG	RW	DESCENG_EN[0], PREFETCH_EN[1], FIFO_THRESH[5:2] (reset 8)
0x224	DESCENG_ADDR0_BASE	RW	Descriptor address range 0 base [31:0]
0x228	DESCENG_AD	RW	Descriptor address range 0

Offset	Name	Access	Description
	DR0_LIMIT		limit [31:0]
0x22C	DESCENG_ADDR1_BASE	RW	Descriptor address range 1 base [31:0]
0x230	DESCENG_ADDR1_LIMIT	RW	Descriptor address range 1 limit [31:0]
0x240	CTRL_CONFIG	RW	CTRLRD_MAX_TRY[8:0] control-read poll retry budget (0-511, reset 16)
0x2A0	AXI_XFER_CONFIG	RW	AXI transfer sizing configuration
0x2B0	PERF_CONFIG	RW	Performance profiler configuration
0x2C0	OBS_CTRL	RW	Observation mux control (channel/category select)
0x2C4	OBS_FLAGS	RO	Observation flags
0x2C8	OBS_DATA0	RO	Observation data word 0
0x2CC	OBS_DATA1	RO	Observation data word 1
0x35C	PERF_CH_SEL	RW	Performance channel select
0x378	HIST_SEL	RW	Histogram bucket select
0x37C	HIST_DATA	RO	Histogram bucket data
0x380	HIST_TOTAL	RO	Histogram total count

24.2.1 Key Register Fields

SCHED_TIMEOUT_LIMIT (0x208) – `LIMIT[7:0]` sets the number of consecutive write-progress timeout windows a channel tolerates before the recoverable timeout escalates to a fatal, sticky `CH_ERROR`. `0` = never escalate (pure soft timeout). Total time to escalate is approximately `LIMIT x TIMEOUT_CYCLES`. Reset value is 4.

DESCENG_CONFIG (0x220) – `PREFETCH_EN[1]` and `FIFO_THRESH[5:2]` control the now-functional descriptor prefetch: prefetch off = on-demand chaining (~1 ahead), prefetch on = buffer up to `FIFO_THRESH` descriptors ahead.

CTRL_CONFIG (0x240) – `CTRLRD_MAX_TRY[8:0]` bounds the control-read poll retry budget (0-511, reset 16) fed to every channel's `ctrlrd_engine`. A `CTRL_READ` descriptor polls its gate address once per retry; if the masked value never matches within `CTRLRD_MAX_TRY` attempts the engine raises `ctrlrd_error` so a never-satisfied gate cannot hang the channel. See the Control-

Read / Control-Write Engine specs (Sections 2.8 / 2.9) and the Scheduler control interface (Section 2.1).

24.3 Monitor Register Map (rapids_mon_regs @ 0x1000)

The monitor regfile is instantiated at base `0x1000` under `hwif_out.MON.*`. It holds the MonBus FIFO status, the three AXI-monitor configuration groups (DAXMON = descriptor monitor, RDMON = read monitor, WRMON = write monitor), and per-monitor performance counters.

Table 3.10.2: Monitor Register Map (base 0x1000)

Offset	Name	Access	Description
0x1000	`MON.MON_FIFO_STATUS`	RO	MonBus capture/error FIFO status
0x1004	`MON.MON_FIFO_COUNT`	RO	MonBus FIFO occupancy
0x10C0	`MON.DAXMON_ENABLE`	RW	Descriptor-monitor enables (incl. `COMPRESS_EN`)
0x10C4	`MON.DAXMON_TIMEOUT`	RW	Descriptor-monitor timeout cycles
0x10C8	`MON.DAXMON_LATENCY_THRESH`	RW	Descriptor-monitor latency threshold
0x10CC	`MON.DAXMON_PKT_MASK`	RW	Descriptor-monitor packet mask
0x10D0	`MON.DAXMON_ERR_CFG`	RW	Descriptor-monitor error select/mask
0x10D4-0x10DC	`MON.DAXMON_MASK1/2/3`	RW	Descriptor-monitor category masks
0x10E0-0x10FC	`MON.RDMON_*`	RW	Source-egress AXIS monitor-lite config (same layout as DAXMON; SRC instance, rapids TASK-015)
0x1100-0x111C	`MON.WRMON_*`	RW	Sink-ingress AXIS monitor-lite config (same layout as DAXMON; SNK instance, rapids TASK-015)

Offset	Name	Access	Description
0x1150-0x1178	MON.DAXMON_PERF_*	RO/RW	Descriptor-monitor performance counters
0x1180-0x11A8	MON.RDMON_PERF_*	RO/RW	Read-monitor performance counters
0x11B0-0x11D8	MON.WRMON_PERF_*	RO/RW	Write-monitor performance counters
0x11E0-0x11EC	MON.{RD,WR}MON_PERF_CH_*	RO	Per-channel producer/backpressure/starve/idle
0x11F0	MON.RDMON_PERF_CH_OVERFLOW	RO	Read-monitor per-channel overflow
0x11F4	MON.WRMON_PERF_CH_OVERFLOW	RO	Write-monitor per-channel overflow

Each *_ENABLE register carries MON_EN, ERR_EN, COMPL_EN, TIMEOUT_EN (and COMPRESS_EN on the write monitor); each *_PERF_* group provides window, producer, backpressure, starve, idle, beat, byte (lo/hi), and burst counters.

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25 RAPIDS Config Block Specification

Module: rapids_config_block.sv **Location:** projects/components/dma-ip/rapids/rtl/macro_beats/ **Status:** Implemented

---

25.1 Overview

rapids_config_block is a combinational adapter that maps the register block's hwif_out structure onto the flat cfg_* signals consumed by rapids_core_beats and the AXI monitors. It contains no state; every output is a direct assignment (with a small number of width expansions and global-enable gates).

The monitor registers arrive under hwif_out.MON.*; base configuration and status arrive at the top level of hwif_out.

25.2 Mapping Groups

25.2.1 Base Configuration

Routes scheduler, channel, and descriptor-engine configuration to the core:

- cfg_channel_enable = CHANNEL_ENABLE.CH_EN & {8{GLOBAL_CTRL.GLOBAL_EN}} (per-channel enable gated by the global master enable).
- cfg_sched_timeout_cycles = SCHED_TIMEOUT_CYCLES.TIMEOUT_CYCLES (32-bit).
- cfg_sched_timeout_limit = SCHED_TIMEOUT_LIMIT.LIMIT (8-bit, direct).
- cfg_sched_{timeout,err,compl,perf}_enable = SCHED_CONFIG.{TIMEOUT,ERR,COMPL,PERF}_EN. COMPL_EN is the one the scheduler group acts on for its monitor packets: 0 drops the scheduler's and descriptor engine's CORE Completion packets at the group (emitters still acknowledged); ERR_EN and PERF_EN reach the group but gate nothing today (errors always reported, perf not implemented).
- cfg_desceng_* from DESCENG_CONFIG (enable / prefetch / FIFO threshold).
- Descriptor address ranges (cfg_desceng_addr{0,1}_{base,limit}) are zero-extended from the 32-bit register fields to the 64-bit ADDR_WIDTH.

25.2.2 Monitor Configuration (hwif_out.MON.*)

Three parallel groups map to the descriptor-read monitor and, since rapids TASK-015, the two AXIS data-path monitor-lites (one per half):

Table 3.11.1: Monitor Register-to-Config Mapping

Register group	cfg_* target
MON.DAXMON_*	cfg_desc_mon_* (descriptor monitor)
MON.RDMON_*	cfg_rdeng_mon_* (source-egress AXIS monitor-lite; wired in the SRC half only)
MON.WRMON_*	cfg_wreng_mon_* (sink-ingress AXIS monitor-lite; wired in the SNK half only)

The *_ENABLE bits map onto the AXIS monitor-lite's class enables: MON_EN (gated by GLOBAL_EN), ERR_EN (error, channel-change and TSTRB checks), COMPL_EN (one Completion/STREAM_END per packet), PERF_EN (stream start/pause/resume and Credit/BACKPRESSURE stall reports), TIMEOUT_EN. *_TIMEOUT is MICROSECONDS in its low 16 bits (0 = never), *_LATENCY_THRESH the stall CYCLES that raise a backpressure report, and *_PKT_MASK is a MASK: bit[type] = 1 masks (drops) that packet type at the monitor-lite and again at the monbus group. It resets to all-masked (0xFFFF); the host clears the bits of the classes it wants to see (rapids BUG-008 corrected the description, which used to read "1 = enable"; the polarity and the reset are as intended).

Within each group the enable/timeout/latency/mask fields map 1:1, e.g. cfg_desc_mon_enable = MON.DAXMON_ENABLE.MON_EN & GLOBAL_CTRL.GLOBAL_EN, cfg_desc_mon_timeout_cycles = MON.DAXMON_TIMEOUT.TIMEOUT_CYCLES, and the category masks (timeout/compl/thresh/perf/addr/debug) pass through directly. Key enables are gated by GLOBAL_CTRL.GLOBAL_EN; masks pass through ungated.

25.2.3 Performance and Observation

PERF_CONFIG, PERF_CH_SEL, and OBS_CTRL map to the profiler and channel-observation-mux cfg_* signals.

Last Updated: 2026-07-02

26 RAPIDS Top

Module: `rapids_top.sv`, `rapids_cxs_top.sv` (generated), `rapids_top_base.sv` **Location:** `projects/components/dma-ip/rapids/rtl/top/` **Status:** Implemented **Last Updated:** 2026-10-16

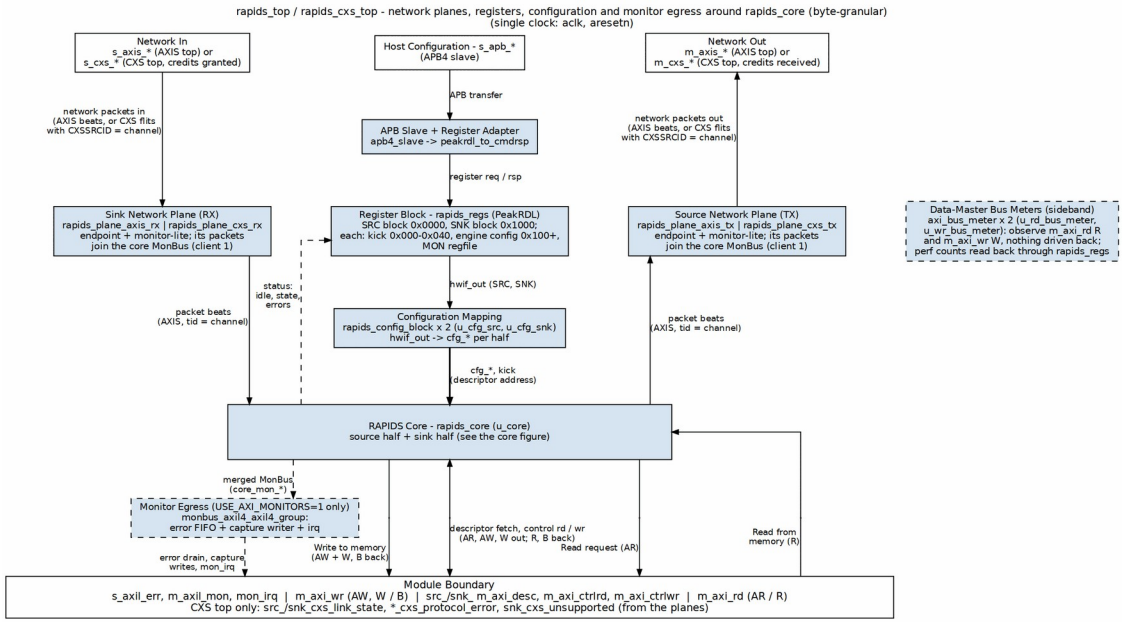
26.1 Overview

`rapids_top` is the synthesizable top level. It wraps RAPIDS Core with an APB4 register interface, two config blocks, the register file, per-channel bus meters and the MonBus AXI-Lite group. It has the same hierarchy as RAPIDS Beats. Only the beat-count unit names differ.

The byte-granular behavior is entirely below the top: the packet record nets are internal to the core halves, so the top port list has no new signals. The full port list is in Top-Level Port List and the top integration description is in RAPIDS Beats Top.

Both tops are generated by `bin/gen_plane_tops.py` from `rapids_top_base`: `rapids_top` adds the AXIS network planes and `rapids_cxs_top` the CXS planes, one CXS link per direction with `CXSSRCID` carrying the channel. The CXS planes carry `tstrb` in `CXSUSER`, so the byte-granular partial last beats and unaligned starts cross the link intact; that caps `DATA_WIDTH` at 1024. The planes are shared with RAPIDS Beats: see Network Planes and the CXS Interface Specification.

26.1.1 Figure 3.12.1: Top Hierarchy



RAPIDS top integration

Source: 10_rapids_top.dot

26.2 APB Address Map

The APB address is 13 bits. Bit 12 selects the half.

Table 3.12.1: APB Address Map

Range	Content
`0x0000` to `0x003F`	Source staged descriptor addresses.
`0x0040`	Source kick-off, one single-pulse bit per channel.
`0x0040` to `0x0FFF`	Source configuration registers.
`0x1000` to `0x103F`	Sink staged descriptor addresses.
`0x1040`	Sink kick-off, one single-pulse bit per channel.
`0x1040` to `0x1FFF`	Sink configuration registers.

The monitor register windows sit at +0x800 of each half's 4 KB space. With USE_MON_REGS clear the windows answer with an APB error, so a host can tell a build without monitors from a zero setting.

26.3 Parameters

Table 3.12.2: Top Parameters

Parameter	Default	Description
NUM_CHANNELS	8	Channels.
DATA_WIDTH	512	Data width in bits. The board uses 256.
ADDR_WIDTH	64	Address width.
AXI_ID_WIDTH	8	AXI ID width.
SRAM_DEPTH	4096	Words per channel at the top.
APB_ADDR_WIDTH, APB_DATA_WIDTH	13, 32	APB widths.
MON_MAX_TRANSACTIONS	16	Monitor table depth.
SNK_AXIS_MON_AGENT_ID, SRC_AXIS_MON_AGENT_ID	9, 10	Network-plane monitor agent ids (AXIS or CXS plane).
ACLK_MHZ	100	Clock in MHz for the network-plane monitor tick.
USE_AXI_MONITORS	1	0 omits the monitors and the MonBus AXI-Lite group.
USE_ROW_COL_MAJOR_ADDRESSING	1	0 compiles out extended addressing.
GEN_MON	1	0 omits the completion and error emitters.
USE_MON_REGS	USE_AXI_MONITORS != 0	Wire the monitor register windows.
AR_MAX_OUTSTANDING, AW_MAX_OUTSTANDING	8	Outstanding bursts.
PIPELINE	1	Write engine pipelining.
AXIS_ID_WIDTH, AXIS_DEST_WIDTH,	8, 4, 1	AXIS side widths.

Parameter	Default	Description
AXIS_USER_WIDTH		

Clock and reset are aclk and aresetn. cam_clear synchronously clears the MonBus group transaction tables.

Each half's configuration block drives its per-channel reset vector (src_cfg_channel_reset, snk_cfg_channel_reset) from SRC/SNK.CHANNEL_RESET.CH_RST and GLOBAL_CTRL.GLOBAL_RST. The top wires each vector to its half of the core (see RAPIDS Core), and it reaches every block that can hold a channel error: a channel reset recovers every error class, including the AXI response and sink packet length errors, and aresetn is needed only at power-up. The top has no port change.

26.4 Differences from RAPIDS Beats

Table 3.12.3: Top Delta

Item	RAPIDS Beats	RAPIDS
Module	rapids_beats_top	rapids_top
Core instance	rapids_core_beats	rapids_core
Register file, APB map, config blocks		Unchanged
Bus meter beat counters	r_rd, r_wr	r_rd_beats, r_wr_beats
Config nets	*_xfer	*_xfer_beats
Port list		Unchanged

The register-file field names RD_XFER_BEATS and WR_XFER_BEATS are the same in both designs. Their meaning is beats in both, since bursts are beat-based.

Last Updated: 2026-09-30

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27 Network Planes Specification

Modules: `rapids_plane_axis_rx.sv`, `rapids_plane_axis_tx.sv`, `rapids_plane_cxs_rx.sv`, `rapids_plane_cxs_tx.sv` **Location:** `projects/components/dma-ip/rapids/rtl/plane/` **Generator:** `projects/components/dma-ip/rapids/bin/gen_plane_tops.py` **Status:** Implemented (rapids TASK-024, TASK-025)

27.1 Overview

A network plane is the endpoint between the core's network-side AXIS stream and the pins: the RX plane feeds the sink half's ingress, the TX plane takes the source half's egress. Each plane carries the monitor-lite for its port. Both RAPIDS trees (byte-granular `rtl/top`, beat-granular `rtl/top_beats`) share the same four planes.

Every plane has the same parameters and the same inner interface, so a top selects AXIS or CXS by module name alone:

- `fub_axis_*`: the core-facing AXIS stream (`tdata`, `tstrb`, `tlast`, `tid` = channel, `tdest`, `tuser`)
- `cfg_mon_*` in, `sts_mon_*` out: the `SNK.WRMON_*` / `SRC.RDMON_*` monitor CSRs, from `rapids_config_block`
- `monbus_*` out: client 1 of the half's 3:1 `monbus_arbiter`, entering the core on `snk_plane_mon_*` / `src_plane_mon_*`

Table 3.13.1: RAPIDS Network Planes

Plane	Contents	Outer pins
`rapids_plane_axis_rx`	`axis4_slave_monlite` (skid stage, `SKID_DEPTH` = 4, + `axis_monitor_lite` tap)	`s_axis_*`
`rapids_plane_axis_tx`	`axis4_master_monlite` (skid stage + tap)	`m_axis_*`
`rapids_plane_cxs_rx`	`cxs_slave_monlite` (`u_endpoint`) + `cxs_to_axis` (`u_cxs_to_axis`)	`s_cxs_*`, `snk_cxs_*` status
`rapids_plane_cxs_tx`	`axis_to_cxs` (`u_axis_to_cxs`) + `cxs_master_monlite`	`m_cxs_*`, `src_cxs_*` status

Plane	Contents	Outer pins
	(u_endpoint)	

The monitor counters of every plane are at `u_snk_plane.u_endpoint` and `u_src_plane.u_endpoint` in the top.

27.2 Generated Tops

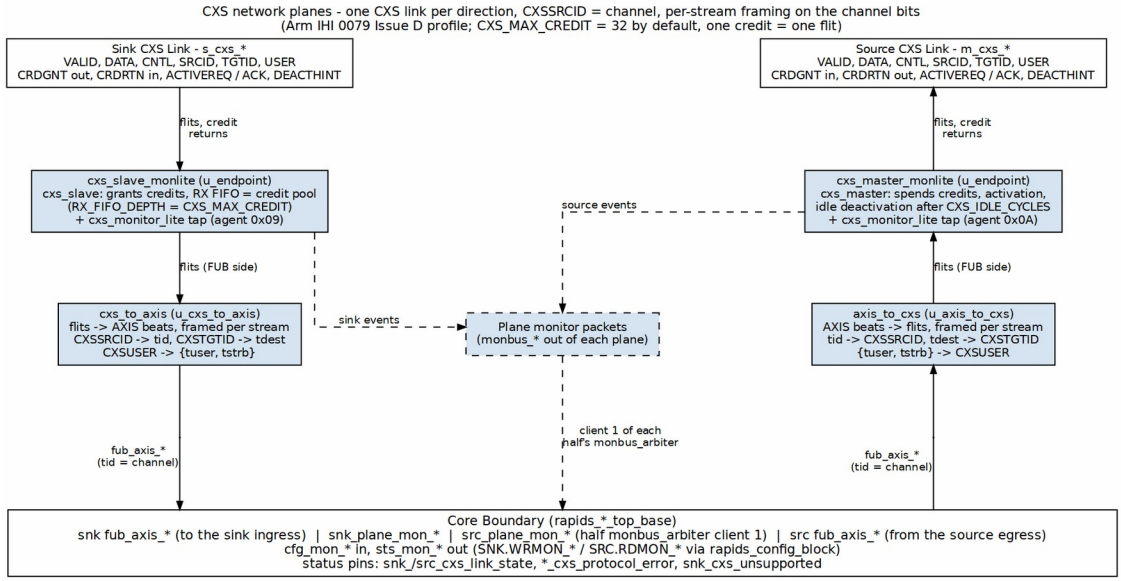
Table 3.13.2: Generated Tops

Top	Tree	Base	Planes
rapids_top	byte	rapids_top_base	AXIS
rapids_cxs_top	byte	rapids_top_base	CXS
rapids_beats_top	beats	rapids_beats_top_base	AXIS
rapids_beats_cxs_top	beats	rapids_beats_top_base	CXS

`gen_plane_tops.py` copies the base's parameter and port text verbatim, then instantiates the base (`u_base`) and the two planes (`u_snk_plane`, `u_src_plane`). The AXIS tops therefore keep exactly the interface they had before the split, and existing instantiations are unchanged. Run the generator with `--check` to fail on a stale top.

27.3 CXS Planes

27.3.1 Figure 3.13.1: CXS Network Planes



CXS network planes

Source: 23_cxs_planes.dot

27.3.2 Receive (sink) plane

`cxs_slave_monlite` is `cxs_slave` with a `cxs_monitor_lite` tap. `cxs_slave` grants credits up to `CXS_MAX_CREDIT`, runs the activation handshake, and holds received flits in a receive FIFO of `RX_FIFO_DEPTH` (default `CXS_MAX_CREDIT`) entries. The FIFO is the credit pool: a credit is only granted for a free entry, so a flit always has space. `cxs_to_axis` turns flits back into AXIS beats: `CXSSRCID` becomes `tid`, `CXSTGTID` becomes `tdest`, and `CXSUSER` is split into `{tuser, tstrb}`. When the core stops taking beats the FIFO fills, and the slave stops granting credits; that is the CXS form of the sink's `tready` backpressure.

`cxs_to_axis` is combinational and stateless, so it can only represent the flits `axis_to_cxs` makes: at most one START and one END per flit, with a packet starting at byte 0. Any other flit (a transmitter that packs two packets into one flit) is still passed on, with `tlast` from END[0], and sets the sticky `snk_cxs_unsupported`.

27.3.3 Transmit (source) plane

`axis_to_cxs` packs AXIS beats into flits (`tid` -> `CXSSRCID`, `tdest` -> `CXSTGTID`, `{tuser, tstrb}` -> `CXSUSER`) and `cxs_master_monlite` sends them, one credit per flit. The master requests activation when it has a flit to send and deactivates the link after `CXS_IDLE_CYCLES` idle cycles.

27.3.4 Per-stream framing

axis_to_cxs and both endpoints' control checkers track an open packet per stream, keyed on the low CXS_STREAM_SRCID_BITS = CIW bits of CXSSRCID (the channel); cxs_to_axis needs no state because each beat carries its own tid. The byte-granular source arbitrates per drain block while a packet spans the whole descriptor, so with two or more channels active the egress switches channel in the middle of a packet. Per-link framing (one open packet per link) rejects that traffic; per-stream framing accepts it, as IHI 0079D 2.5.2 allows.

27.3.5 Parameters

Table 3.13.3: CXS Plane Parameters

Parameter	Default	Description
CXS_MAX_CREDIT	32	Credits granted by the receiver. One credit is one flit; the grant -> flit -> free round trip is about 3 cycles
CXS_RX_FIFO_DEPTH	CXS_MAX_CREDIT	Receive FIFO depth (the credit pool); need not be a power of two
CXS_CREDIT_BEYOND_SPEC	CXS_MAX_CREDIT > 63	Permits more credits than IHI 0079D allows; the far end must match
CXS_IDLE_CYCLES	32	Idle cycles before the transmitter deactivates the link
CXS_IW, CXS_DESTW, CXS_UW, CXS_CNTL_PW	derived	Port widths; do not override

DATA_WIDTH must be at most 1024: CXSUSER is at most 128 bits and carries tstrb (DATA_WIDTH/8 bits). tuser is dropped from CXSUSER when DATA_WIDTH/8 + AXIS_USER_WIDTH > 128.

27.3.6 Monitoring

The CXS planes drive cxs_monitor_lite from the same cfg_mon_* CSRs as the AXIS planes and use the same agent IDs (sink 0x09, source 0x0A). Packets are tagged PROTOCOL_CXS; the top's monitor group filters them through its AXIS mask slot. A PACKET_END completion carries the channel in event_data[55:48]; channel_id holds {PRCLTYPE[2:0], TGTID[5:0]}. The lite's output queue is 4 entries deep; when events outrun it the lite drops and counts them (an Error / EVENT_DROPPED packet), so a check on completions counts delivered plus dropped.

27.4 Proposed Changes

A proposal (2026-10-16, not implemented) puts a tile packet header at the front of every packet: destination and source coordinates, payload type (`cfg` / `prog` / `mem`), length and target address. It is described once in the compute-engine tile's `TILE_SPEC.md` Section 11; CXS carries it as the first flit. The link signals, credits and per-stream framing on this page do not change, because `CXSSRCID` keeps the channel.

27.5 Verification

Table 3.13.4: CXS Top Test Cells

Suite	Cells
`dv/tests/top/ test_rapids_cxs_top.py`	source, sink, source_bytes, sink_bytes, source_concurrent, sink_concurrent, cxs_mon, ext_addressing
`dv/tests/top_beats/ test_rapids_beats_cxs_top.py`	source, sink, source_concurrent, sink_concurrent, cxs_mon, ext_addressing

The concurrent cells start four channels with one `KICK_ENABLE` write, so channels interleave on the link. Every cell also requires a clean link: no BFM or wire-monitor violation and the DUT's protocol-error flags low. The core testbenches drive packets on `plane_mon_*` and require them on the MonBus output.

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28 AXI4 Interface Specification

Module: `axi_read_engine.sv` (source AXI read master), `axi_write_engine.sv` (sink AXI write master) **Location:** `projects/components/dma-ip/rapids/rtl/fub/` **Status:** Implemented **Last Updated:** 2026-09-30

28.1 Overview

RAPIDS drives two AXI4 masters: a read master on the source half and a write master on the sink half. The channel signals, the outstanding-transaction limits and the response handling are those of RAPIDS Beats and are specified in AXI4 Interface Specification (shared with RAPIDS Beats). This chapter records only what the byte-granular design adds.

The AXI side stays beat-oriented. Every burst starts at a beat-aligned address and moves whole beats. A descriptor that starts or ends inside a beat is handled by the byte lanes of the data path, not by narrow AXI transfers. Both engines are specified in AXI Read Engine and AXI Write Engine.

28.2 Byte Address to Beat Mapping

The scheduler splits a byte address into a beat address and a byte offset. `OFF_W` is `$clog2(DATA_WIDTH/8)`, which is 5 for the 256-bit board build.

Table 4.1.1: Address Split

Quantity	Value
Offset	`addr[OFF_W-1:0]`
First beat address	`addr` with the low `OFF_W` bits cleared
Beats for a descriptor	`(offset + length + lanes - 1) >> OFF_W`, zero for zero length

The engines receive the working byte address and the remaining beat count. `ARADDR` and `AWADDR` are driven with the low `OFF_W` bits at zero, so every burst is beat-aligned.

28.3 Burst Rules

Table 4.1.2: Burst Rules

Rule	Read master	Write master
Burst start	beat-aligned	beat-aligned
Burst size	full bus width	full bus width
4 KB boundary	no burst crosses one	no burst crosses one
Burst cap	smaller of the configured burst and the beats left to the 4 KB boundary	same

The number of beats left to the boundary is computed from the beat-aligned address, so a byte offset never lets a burst run past it.

28.4 Write Strobes

The write master carries a byte enable with every beat. The data path stores the strobe next to the data in the sink SRAM, and the write engine drives it onto `WSTRB`. Interior beats of a packet are fully strobed. The first beat and the last beat of a descriptor that start or end inside a beat carry a partial strobe, so only the bytes of the descriptor are written and neighbouring bytes in memory are untouched.

The read master has no byte enables. It always reads whole beats and the source data path discards the bytes outside the descriptor.

Table 4.1.3: Byte-Granular Write Signal

Signal	Width	Meaning
`m_axi_wstrb`	`DATA_WIDTH/8`	Byte lanes written by the beat, from the stored strobe.

28.5 Limitations

- TYPE=EXT descriptors remain beat-aligned in address and length. This is a permanent limitation.
- A descriptor never issues a narrow AXI transfer. Sub-beat behavior is entirely in the strobes.

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29 AXI-Stream Interface Specification

Module: `src_data_path_axis.sv` (source master), `snk_data_path_axis.sv` (sink slave)
Location: `projects/components/dma-ip/rapids/rtl/macro/` **Status:** Implemented **Last Updated:** 2026-09-30

29.1 Overview

RAPIDS has one AXI-Stream master on the source half and one AXI-Stream slave on the sink half. Both carry packed bytes. Unlike RAPIDS Beats, the stream is no longer a train of full beats: each descriptor is one packet, the last beat of a packet may be partial, and `TSTRB` says which byte lanes are valid. The handshake, the width parameters and the monitor tap are those of RAPIDS Beats, see AXI-Stream Interface Specification (shared with RAPIDS Beats). The two data paths are specified in Source Data Path AXIS and Sink Data Path AXIS.

29.2 Packet Contract

Table 4.2.1: Packet Contract

Property	Rule
Packet	One descriptor is one packet, ending with `TLAST`.
Packing	Bytes are packed from lane 0 upward. The descriptor start offset in memory does not appear on the stream.
`TSTRB`	Contiguous from lane 0. All ones on every beat but the last.
Partial beat	Only the `TLAST` beat may be partial, carrying the remaining bytes.
Packet length	Equals the descriptor length in bytes.
Zero length	A zero-length descriptor produces no packet.

29.3 Source Master Signals

Table 4.2.2: Source Master Signals

Signal	Value
`m_axis_tdata`	Re-packed bytes from the drain shifter.
`m_axis_tstrb`	Low `min(bytes_left, DATA_WIDTH/8)` lanes set.
`m_axis_tlast`	Set when the remaining byte count equals the bytes in this beat.
`m_axis_tid`	Channel number.
`m_axis_tdest`	Channel number.
`m_axis_tuser`	Zero.

The source path pops memory beats and re-packs them, so a stream beat is not tied one to one to a memory beat. The first pop of an offset descriptor only primes the hold register and produces no output unless the packet fits in that single beat. When the last memory beat leaves bytes in the hold register, one extra flush beat carries them.

29.4 Sink Slave Signals

Table 4.2.3: Sink Slave Signals

Signal	Use
`s_axis_tdata`	Payload, shifted into the per-channel hold by the descriptor offset.
`s_axis_tstrb`	Valid byte lanes. Only the `TLAST` beat may be partial.
`s_axis_tid`	Channel select, low `CIW` bits.
`s_axis_tdest`, `s_axis_tuser`	Unused.
`s_axis_tlast`	Ends the packet.
`s_axis_tready`	Deasserted while the output register is busy, a flush beat

Signal	Use
	is pending, or the selected channel has no packet record.

The sink stalls a packet until the scheduler has issued the record of its descriptor. A packet for a channel whose record queue is empty is not accepted, so no byte enters the SRAM before the destination offset is known. The queue holds four records per channel.

The spill of a shifted beat is held per channel. After TLAST, if the hold register still holds bytes, TREADY drops for one flush beat that writes them to the SRAM.

29.5 Length Check

At TLAST the sink compares the bytes received with the length in the head record of the channel. A mismatch sets the sticky per-channel `r_pkt_len_error` bit, which is ORed with the write engine error into `sched_wr_error`. Nothing is padded or dropped silently. The bytes received are counted from the lanes set in TSTRB on each beat.

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30 MonBus Interface Specification

Last Updated: 2025-01-10

30.1 Overview

The MonBus (Monitor Bus) is a 64-bit packet-based interface for reporting internal events, status, and performance data. All RAPIDS subsystems generate MonBus packets that are aggregated and output through a unified interface.

30.2 MonBus Signal Interface

30.2.1 Signal Table

Table 4.3.1: MonBus Interface Signals

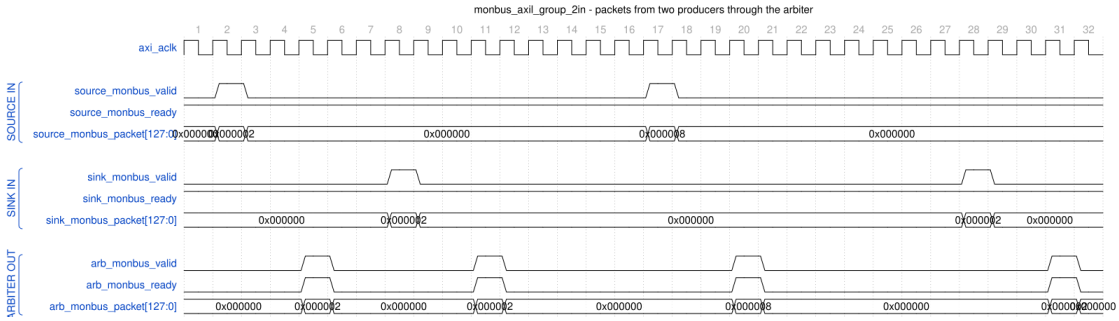
Signal	Direction	Width	Description
`monbus_pkt_valid`	output	1	Packet valid
`monbus_pkt_ready`	input	1	Consumer ready
`monbus_pkt_data`	output	64	Packet data

30.2.2 Handshaking Protocol

Standard valid/ready handshaking: - Data transfers when `valid & ready` both high - Producer holds `valid` until `ready` seen - Consumer asserts `ready` when able to accept

30.2.3 Timing Diagram

30.2.4 Figure 4.3.1: MonBus Packet Transfer Timing



monbus_axil_group_2in - packets from two producers through the arbiter

Source: monbus_group_packet_transfer.json, captured from `dv/tests/macro/test_monbus_axil_group.py` (`basic_flow`, error FIFO 64 x 32, `TEST_LEVEL=full`) with `WAVES=1`. The 128-bit packets are shown as their top 24 bits (`[127:104]`: packet type nibble, then the protocol field at `[108:105]`, so `0x000002` is AXIS and `0x000008` is CORE).

Reading it: a MonBus link is a single-beat valid/ready handshake with the packet presented alongside `valid`. The group holds `ready` high on both inputs, so the source and sink packets are accepted the cycle they are offered, and each one reappears on the arbiter output three cycles later as another one-cycle `valid`/`ready` pair. Between packets the links idle low; nothing is held across transfers.

30.3 MonBus Packet Format

30.3.1 64-bit Packet Structure

```
63      56 55      48 47      40 39      32 31                0
+--------+---------+---------+---------+--------------------+
| Type   | Unit ID | Agent ID| Channel | Payload (32 bits) |
+--------+---------+---------+---------+--------------------+
```

30.3.2 Figure 4.3.2: MonBus Packet Format

Table 4.3.2: MonBus Packet Fields

Field	Bits	Description
Type	[63:56]	Packet type code
Unit ID	[55:48]	Reporting unit identifier

Field	Bits	Description
Agent ID	[47:40]	Source agent within unit
Channel	[39:32]	Channel number (0-7)
Payload	[31:0]	Type-specific payload

30.4 Packet Types

30.4.1 Type Code Definitions

Table 4.3.3: MonBus Packet Types

Type Code	Name	Description
0x01	DESC_FETCH	Descriptor fetch started
0x02	DESC_COMPLETE	Descriptor fetch completed
0x03	SCHED_START	Scheduler started transfer
0x04	SCHED_DONE	Scheduler transfer complete
0x10	AXI_RD_START	AXI read burst started
0x11	AXI_RD_DONE	AXI read burst completed
0x12	AXI_WR_START	AXI write burst started
0x13	AXI_WR_DONE	AXI write burst completed
0x20	SRAM_FILL	SRAM fill event
0x21	SRAM_DRAIN	SRAM drain event
0x30	ERROR	Error event
0x31	TIMEOUT	Timeout event

Type Code	Name	Description
0xFF	HEARTBEAT	Periodic heartbeat

30.4.2 Payload Formats by Type

30.4.2.1 DESC_FETCH / DESC_COMPLETE (0x01, 0x02)

Table 4.3.4: Descriptor Packet Payload

Bits	Description
[31:16]	Descriptor address [47:32]
[15:0]	Descriptor address [31:16]

30.4.2.2 SCHED_START / SCHED_DONE (0x03, 0x04)

Table 4.3.5: Scheduler Packet Payload

Bits	Description
[31:16]	Transfer ID
[15:0]	Beat count

30.4.2.3 AXI_RD/WR_START/DONE (0x10-0x13)

Table 4.3.6: AXI Event Packet Payload

Bits	Description
[31:24]	AXI ID
[23:16]	Burst length
[15:0]	Reserved

30.4.2.4 ERROR (0x30)

Table 4.3.7: Error Packet Payload

Bits	Description
[31:24]	Error code
[23:16]	Error source
[15:0]	Error details

30.5 MonBus Source Aggregation

30.5.1 RAPIDS Core MonBus Sources

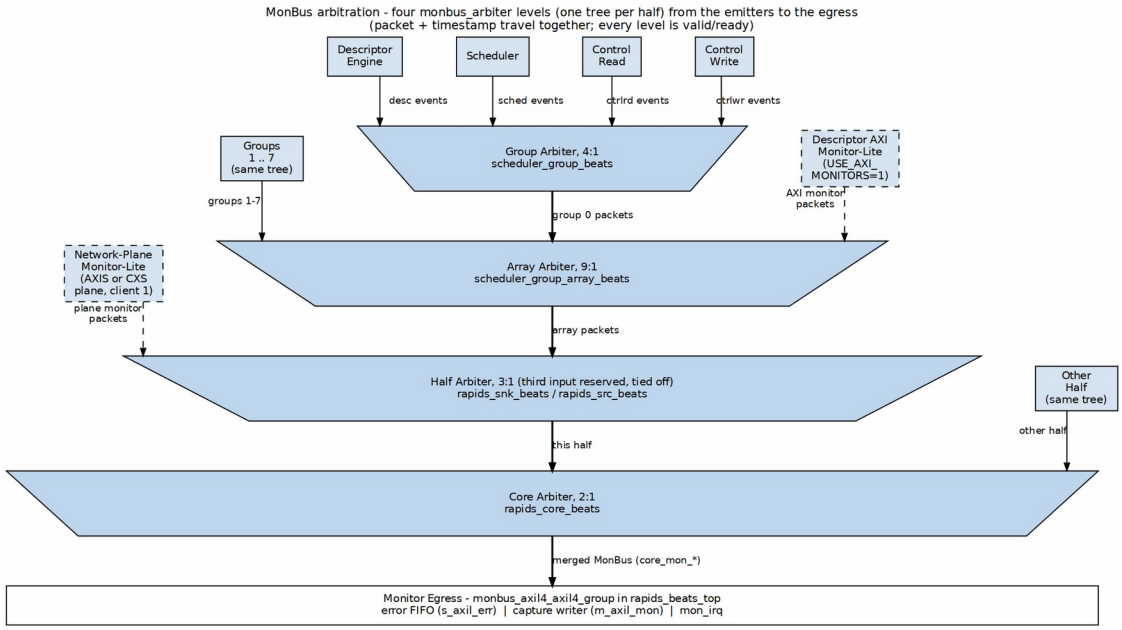
Table 4.3.8: MonBus Source Assignment

Source ID	Agent	Description
0-7	Scheduler[0:7]	Per-channel scheduler events
8-15	DescEngine[0:7]	Per-channel descriptor events
16-23	SnkSRAM[0:7]	Sink SRAM controller events
24-31	SrcSRAM[0:7]	Source SRAM controller events
32	AXI Write Engine	Write transaction events
33	AXI Read Engine	Read transaction events
34	Error Handler	Error events

Scheduler and DescEngine COMPLETION packets are subject to the half's `SCHED_CONFIG.COMPL_EN` (reset 1): the scheduler group drops them ahead of its arbiter when the bit is clear and acknowledges the emitter, so the data path is unaffected. Their ERROR packets are always forwarded (rapids ISSUE-005).

30.5.2 Aggregation Hierarchy

30.5.3 Figure 4.3.3: MonBus Aggregation Tree



MonBus aggregation tree

Source: 21_monbus_aggregation.dot

30.6 Integration Guidelines

30.6.1 Consumer Requirements

MonBus consumers must: 1. Provide sufficient ready cycles to prevent packet loss 2. Handle bursty traffic (multiple sources may have packets simultaneously) 3. Process or buffer packets at wire rate

30.6.2 Recommended Consumer Patterns

```
// Pattern 1: Direct FIFO buffering
gaxi_fifo_sync #(
    .DATA_WIDTH(64),
    .DEPTH(256)
) u_gaxi_fifo_sync (
    .axi_aclk               (axi_aclk),
    .axi_aresetn            (axi_aresetn),
    .wr_valid               (wr_valid),
    .wr_ready               (wr_ready),
```

```
    .wr_data                (wr_data),
    .rd_ready               (rd_ready),
    .count                  (count),
    .rd_valid               (rd_valid),
    .rd_data                (rd_data)
);
```

30.7 Timing Requirements

Table 4.3.9: MonBus Timing Parameters

Parameter	Value	Description
Min Ready	1 cycle	Must assert ready within N cycles
Max Latency	Configurable	Before packet dropped
Packet Rate	Variable	Depends on activity

Last Updated: 2025-01-10

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31 Packet Record Interface Specification

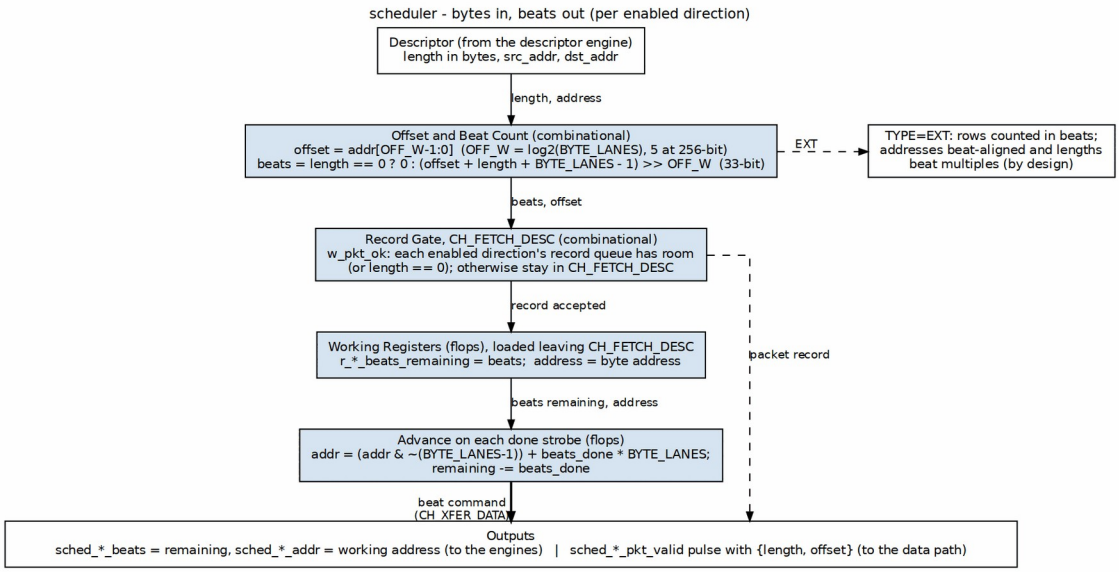
Module: `scheduler.sv` (producer), `src_data_path_axis.sv` and `snk_data_path_axis.sv` (consumers) **Location:** `projects/components/dma-ip/rapids/rtl/fub/` and `rtl/macro/`
Status: Implemented **Last Updated:** 2026-09-30

31.1 Overview

The packet record interface is new in the byte-granular design. It tells a data path how many bytes a descriptor covers and where in its first beat the data starts. RAPIDS Beats has no equivalent, because every transfer there is a whole number of aligned beats.

The scheduler is the producer. It emits one record per data descriptor. The source or sink data path is the consumer and holds the records in a small queue per channel. The record interface is internal to each half: it does not appear on the `rapids_core` or `rapids_top` ports.

31.1.1 Figure 4.4.1: Scheduler Byte Math



Scheduler byte math

Source: assets/graphviz/03_scheduler_byte_math.dot

31.2 Signals

There is one set per direction. The source half uses `sched_rd_pkt_*` and the sink half uses `sched_wr_pkt_*`. Each signal is a vector with one entry per channel.

Table 4.4.1: Packet Record Signals

Signal	Direction (scheduler view)	Width per channel	Description
`sched_*_pkt_valid`	output	1	One-cycle pulse: a record is presented.
`sched_*_pkt_ready`	input	1	The consumer queue has room.
`sched_*_pkt_bytes`	output	32	Descriptor length in bytes.
`sched_*_pkt_offset`	output	`OFF_W`	Address offset inside the first beat.

`OFF_W` is `$clog2(DATA_WIDTH/8)`, 5 for a 256-bit build.

31.3 Protocol

The scheduler raises `valid` for one cycle, in the cycle it leaves the descriptor fetch state. It does not leave that state until every enabled direction can take the record, so `ready` is a pre-condition of the pulse, not a wait state after it. Both directions of a channel see exactly one pulse per data descriptor.

A consumer accepts a record when `valid` and its queue is not full. It drives `ready` low when the queue is full, which is what holds the scheduler in its fetch state.

Table 4.4.2: Record Rules

Rule	Detail
Records per descriptor	One, for data descriptors of nonzero length.
Zero-length descriptors	No record.
Non-data descriptors	No record.
Order	Records are consumed in the order pushed, per channel.

Rule	Detail
Queue depth	Four records per channel in each data path.
Pop	The record pops on the last beat of its packet.

The offset is taken from the source address in the read direction and from the destination address in the write direction. The length is the same for both. When both directions are enabled, each data path receives its own offset.

31.4 Unused Direction

A scheduler group array instantiated for only one direction ties the other direction off inside the half. The unused `valid`, `bytes` and `offset` outputs are left open and the unused `ready` inputs are held at all ones, so the scheduler never waits on a queue that does not exist.

31.5 Consumer Behavior

- The source data path uses the record to compute the number of memory beats, to decide when a pop emits a stream beat, to set `TSTRB` and `TLAST`, and to retire the record.
- The sink data path uses the record to decide how far to shift each incoming beat, to check the received length at `TLAST`, and it holds off `TREADY` for a channel until the record exists.

Last Updated: 2026-09-30

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32 CXS Interface Specification

Applies to: `rapids_beats_cxs_top`, `rapids_cxs_top` **Protocol:** Arm IHI 0079 (CXS) Issue D, one fixed profile **Last Updated:** 2026-10-16

32.1 Overview

The CXS tops replace the two AXIS network ports with one CXS link per direction: `s_cxs_*` (a CXS receiver) into the sink half and `m_cxs_*` (a CXS transmitter) out of the source half. One link carries every channel; the channel rides in `CXSSRCID`. The planes that implement the links are described in the Network Planes chapter (Section 3.13).

32.2 Link Profile

Table 4.4.1: CXS Link Profile

Property	Value
Issue	D (the issue with SRCID, TGTID and USER)
`CXSMAXPKTPERFLIT`	2
Packet start alignment	16 bytes; RAPIDS always starts a packet at byte 0
Packet mapping	One AXIS packet (`tlast`-delimited) is one CXS packet
`CXSSRCID`	The channel (AXIS `tid`)
`CXSTGTID`	AXIS `tdest`
`CXSUSER`	`{tuser, tstrb}`; `tuser` dropped when `DATA_WIDTH/8 + AXIS_USER_WIDTH > 128`
Stream framing	Per stream on the channel bits of `CXSSRCID`
Link control	Explicit credit return, activation, deactivation
Parity	Off

32.3 Signal Table

Table 4.4.2: CXS Interface Signals

Signal	Sink link	Source link	Width	Description
{s,m}_cxs_valid	in	out	1	Flit valid
{s,m}_cxs_data	in	out	DATA_WIDTH	Flit data
{s,m}_cxs_cntl	in	out	CXS_CNTL_PW	START / END / pointers
{s,m}_cxs_srcid	in	out	AXIS_ID_WIDTH	Channel
{s,m}_cxs_tgtid	in	out	AXIS_DEST_WIDTH	tdest
{s,m}_cxs_user	in	out	CXS_UW	{tuser, tstrb}
{s,m}_cxs_crdgnt	out	in	1	Credit grant (receiver to transmitter)
{s,m}_cxs_crdrtn	in	out	1	Credit return (transmitter to receiver)
{s,m}_cxs_activereq	in	out	1	Transmitter requests the link
{s,m}_cxs_activeack	out	in	1	Receiver acknowledges
{s,m}_cxs_deacthint	out	in	1	Receiver asks the transmitter to deactivate
snk_cxs_link_state, src_cxs_link_state	out	out	2	{ACTIVEREQ, ACTIVEACK}
snk_cxs_protocol_error,	out	out	1	Sticky protocol violation

Signal	Sink link	Source link	Width	Description
src_cxs_protocol_error				
snk_cxs_unsupported	out	-	1	Sticky: a flit AXIS framing cannot carry

32.4 Credit Flow

One credit permits one flit (IHI 0079D 2.1). The receiver grants up to CXS_MAX_CREDIT (default 32) credits, one per free entry in its receive FIFO, so every flit sent against a credit has space. The transmitter spends one credit per flit and returns the credits it still holds on deactivation. Credits are a single pool for the link; a grant carries no ID and cannot be reserved for a channel.

The grant -> flit -> free round trip between the RAPIDS endpoints is about 3 cycles, so 32 credits sustains one flit per cycle with margin for pipeline stages or a clock crossing on the link. More than 63 credits is beyond the spec and requires CXS_CREDIT_BEYOND_SPEC on both ends.

32.5 Link Activation

Table 4.4.3: Link Activation Sequence

Step	Transmitter (m_cxs_*)	Receiver (s_cxs_*)
1	Raises ACTIVEREQ when it has a flit to send (and only once the previous activation's ACTIVEACK has fallen: 4-phase)	-
2	-	Raises ACTIVEACK; starts granting credits
3	Sends flits, one credit each	Frees an entry and regrants as the core drains
4	After CXS_IDLE_CYCLES idle cycles in RUN (or on DEACTHINT) drops ACTIVEREQ, then returns every held credit on CRDRTN, one per cycle	-
5	-	Drops ACTIVEACK once its credits are back

32.6 Backpressure

On the sink link, a full receive FIFO means no further grants: the transmitter stalls for lack of credit. This is the CXS form of the AXIS sink's `s_axis_tready = 0`, including the byte-granular sink's hold until a channel's descriptor has been fetched. On the source link, the plane stalls the core's egress (`fub_axis_tready = 0`) while it has no credit.

32.7 Proposed Changes

A proposal (2026-10-16, not implemented) puts a tile packet header at the front of every packet: destination and source coordinates, payload type (`cfg` / `prog` / `mem`), length and target address. It is described once in the compute-engine tile's `TILE_SPEC.md` Section 11; CXS carries it as the first flit. The link signals, credits and per-stream framing on this page do not change, because `CXSSRCID` keeps the channel.

33 RAPIDS Beats Architecture MAS

Version: 0.7 **Date:** 2026-07-13 **Purpose:** Module Architecture Specification for RAPIDS Phase 1 “Beats” Architecture

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33.1 Document Organization

Note: All chapters linked below for automated document generation.

33.1.1 Front Matter

- Document Information

33.1.2 Chapter 1: Overview

- Architecture Overview
- Top-Level Port List
- Clocks and Reset

33.1.3 Chapter 2: FUB (Functional Unit Blocks)

Control Path: - Scheduler - Descriptor Engine

AXI Engines: - AXI Read Engine - AXI Write Engine

Flow Control: - Beats Alloc Control - Beats Drain Control - Beats Latency Bridge

Control Engines: - Control-Read Engine - Control-Write Engine

33.1.4 Chapter 3: Macro (Integration Blocks)

Scheduler Integration: - Beats Scheduler Group - Beats Scheduler Group Array

Sink Data Path (Network to Memory): - Sink Data Path - Sink Data Path AXIS - Sink SRAM Controller

Source Data Path (Memory to Network): - Source Data Path - Source Data Path AXIS - Source SRAM Controller

Top-Level Integration: - RAPIDS Core Beats - RAPIDS Registers - RAPIDS Config Block - RAPIDS Beats Top - Network Planes

33.1.5 Chapter 4: Interfaces

- Interface Overview

- AXI4 Interface Specification
- AXIS Interface Specification
- MonBus Interface Specification
- CXS Interface Specification

33.2 Quick Reference

33.2.1 FUB Modules (fub_beats/)

FUB Module Summary

Module	File	Purpose	Status
scheduler_beats	`scheduler_beats.sv`	Transfer coordinator, descriptor processing	Implemented
descriptor_engine_beats	`descriptor_engine_beats.sv`	Descriptor fetch/parse (256-bit)	Implemented
axi_read_engine_beats	`axi_read_engine_beats.sv`	AXI read master, streaming pipeline	Implemented
axi_write_engine_beats	`axi_write_engine_beats.sv`	AXI write master, streaming pipeline	Implemented
alloc_ctrl_beats	`alloc_ctrl_beats.sv`	Space allocation tracking (virtual FIFO); kept and tested, no longer in the SRAM path since `bdf4e0dff`	Implemented
drain_ctrl_beats	`drain_ctrl_beats.sv`	Data availability tracking (virtual FIFO); kept and tested, no longer in the SRAM path	Implemented
latency_bridge_beats	`latency_bridge_beats.sv`	Registered-FIFO to valid/ready skid bridge; kept and tested, no longer in the SRAM path	Implemented

Module	File	Purpose	Status
ctrlrd_engine	`ctrlrd_engine.sv`	Control-read consumer gate (poll-until-match, retry budget)	Implemented
ctrlwr_engine	`ctrlwr_engine.sv`	Control-write producer doorbell (single-beat write)	Implemented

33.2.2 Macro Modules (macro_beats/)

Macro Module Summary

Module	File	Purpose	Status
scheduler_group_beats	`scheduler_group_beats.sv`	Scheduler + Descriptor Engine wrapper	Implemented
scheduler_group_array_beats	`scheduler_group_array_beats.sv`	8-channel scheduler array with arbitration	Implemented
snk_data_path_beats	`snk_data_path_beats.sv`	Network-to-memory path integration	Implemented
snk_data_path_axis_beats	`snk_data_path_axis_beats.sv`	AXIS variant of sink path	Implemented
src_data_path_beats	`src_data_path_beats.sv`	Memory-to-network path integration	Implemented
src_data_path_axis_beats	`src_data_path_axis_beats.sv`	AXIS variant of source path	Implemented
snk_sram_controller_beats	`snk_sram_controller_beats.sv`	Sink SRAM: naming wrapper around STREAM's `sram_controller` (one instance, per-channel FIFOs inside it)	Implemented
src_sram_controller_beats	`src_sram_controller_beat`	Source SRAM: naming wrapper	Implemented

Module	File	Purpose	Status
	s.sv	around STREAM's sram_controller	
rapids_src_beats / rapids_snk_beats	rapids_src_beats.sv, rapids_snk_beats.sv	One half each: scheduler array + data path + monbus merge	Implemented
rapids_core_beats	rapids_core_beats.sv	Both halves + the core monbus arbiter	Implemented
rapids_regs	regs/generated/rtl/rapids_regs.sv (generated from rtl/macro_beats/rapids_regs.rdl)	PeakRDL register block (base + MON regfile @ 0x1000)	Implemented
rapids_config_block	rapids_config_block.sv	Maps register hwif_out to core/monitor cfg_*	Implemented
rapids_beats_top	rapids_beats_top.sv	Top-level: APB slave, kickoff, monitors, MonBus AXI-Lite group	Implemented

33.3 Architecture Comparison: RAPIDS vs STREAM

The "beats" architecture is a Phase 1 implementation of RAPIDS that shares concepts with STREAM but targets network-to-memory (and vice versa) transfers rather than memory-to-memory.

RAPIDS Beats vs STREAM Comparison

Feature	STREAM	RAPIDS Beats
Primary Use	Memory-to-memory DMA	Network-to-memory accelerator
Data Paths	Single	Separate sink/source paths

Feature	STREAM	RAPIDS Beats
	bidirectional	
Network Interface	None	AXIS master/slave
SRAM Buffering	Shared buffer	Separate sink/source buffers
Descriptor Format	256-bit	256-bit (compatible)
Channel Count	8	8
Flow Control	alloc_ctrl/ drain_ctrl	STREAM `stream_alloc_ctrl`/`stream_drain_ctrl` inside the shared `sram_controller` (`alloc_ctrl_beats`/`drain_ctrl_beats` kept, out of the path)

33.4 Clock and Reset Summary

33.4.1 Clock Domains

Clock Domains

Clock	Frequency	Usage
`aclk`	100-500 MHz	Primary - all RAPIDS logic, AXI/AXIS interfaces

33.4.2 Reset Signals

Reset Signals

Reset	Polarity	Type	Usage
`aresetn`	Active-low	Async assert, sync deassert	Primary - all RAPIDS logic

See: Clocks and Reset for complete timing specifications

33.5 Interface Summary

33.5.1 External Interfaces

External Interfaces

Interface	Type	Width	Purpose
AXI4 (Descriptor)	Master	256-bit	Descriptor fetch
AXI4 (Sink Write)	Master	512-bit (param)	Sink data write to memory
AXI4 (Source Read)	Master	512-bit (param)	Source data read from memory
AXIS (Sink)	Slave	512-bit (param)	Network data ingress
AXIS (Source)	Master	512-bit (param)	Network data egress
MonBus	Master	64-bit	Monitor packet output

33.5.2 Internal Buses

Internal Buses

Interface	Width	Purpose
MonBus	64-bit	Internal monitoring bus
Descriptor Bus	256-bit	Descriptor distribution
SRAM Data Bus	512-bit	SRAM read/write data

33.6 Related Documentation

- **PRD.md** - Product requirements and overview
- **CLAUDE.md** - AI development guide
- **RAPIDS Beats HAS** - Architecture specification (the original pre-beats rapids_spec/ tree was retired)

33.7 Specification Conventions

33.7.1 Signal Naming

- **Clock:** `aclk`
- **Reset:** `aresetn` (active-low)
- **Valid/Ready:** Standard AXI/AXIS handshake
- **Registers:** `r_` prefix (e.g., `r_state`, `r_counter`)
- **Wires:** `w_` prefix (e.g., `w_next_state`, `w_grant`)

33.7.2 Parameter Naming

- **Uppercase:** `NUM_CHANNELS`, `DATA_WIDTH`, `ADDR_WIDTH`
- **Derived:** `CHAN_WIDTH = $clog2(NUM_CHANNELS)`

33.7.3 Figure and Table Numbering

- **Figures:** `Figure X.Y.Z: Title` where X=chapter, Y=section, Z=figure number
- **Tables:** Caption line `: Table Title` after table markdown

Last Updated: 2026-07-13 **Maintained By:** RAPIDS Architecture Team